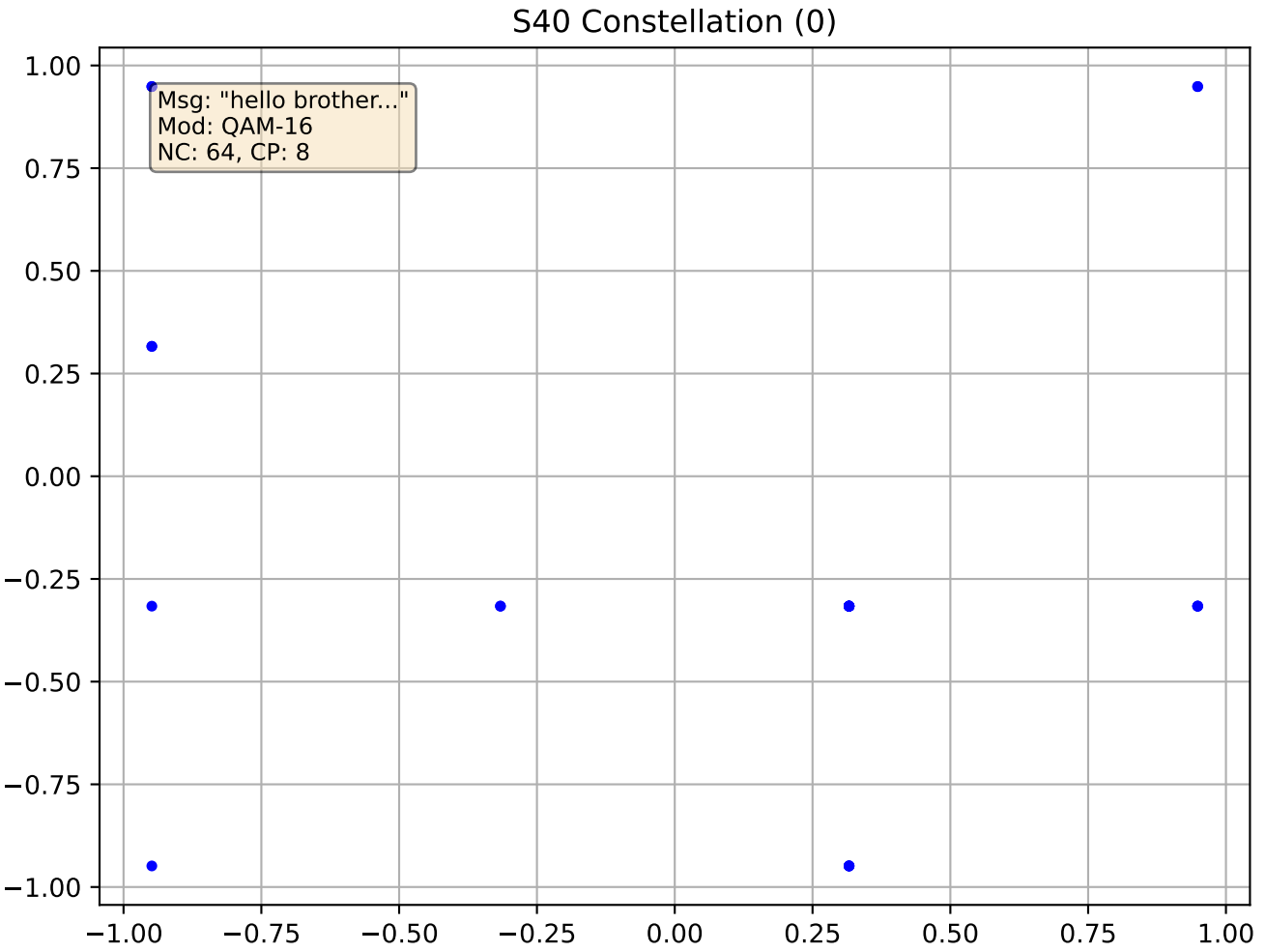
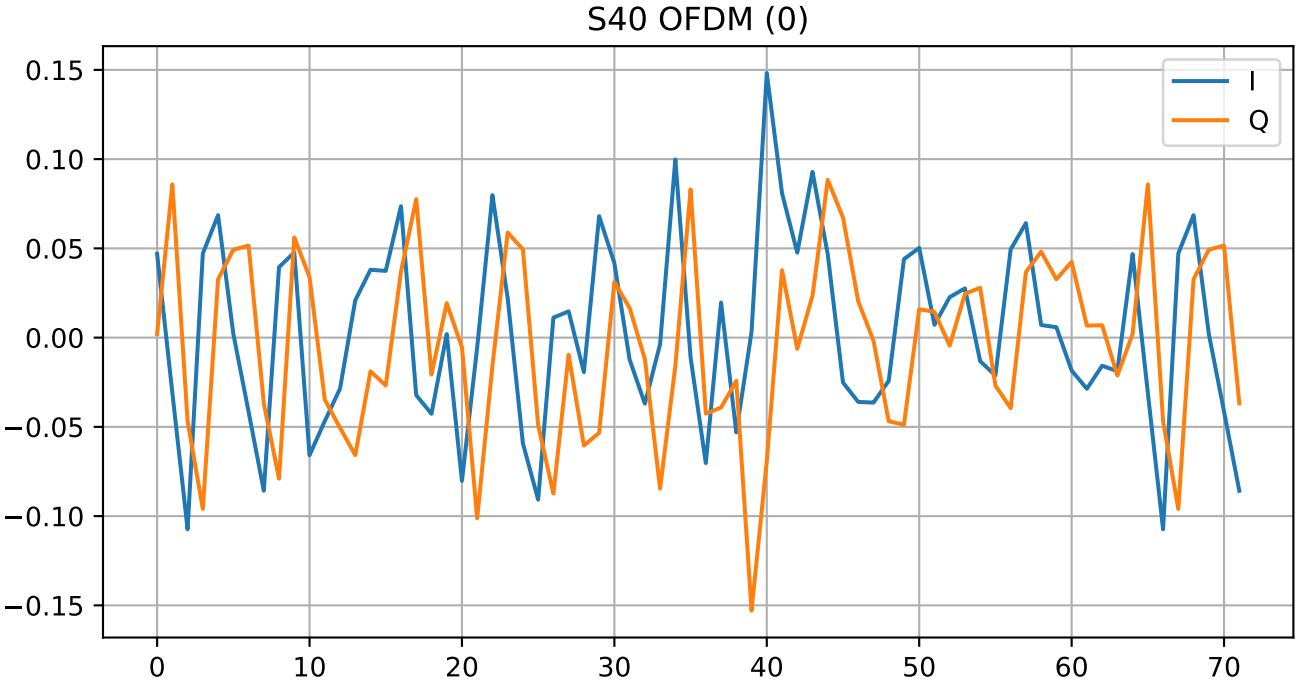
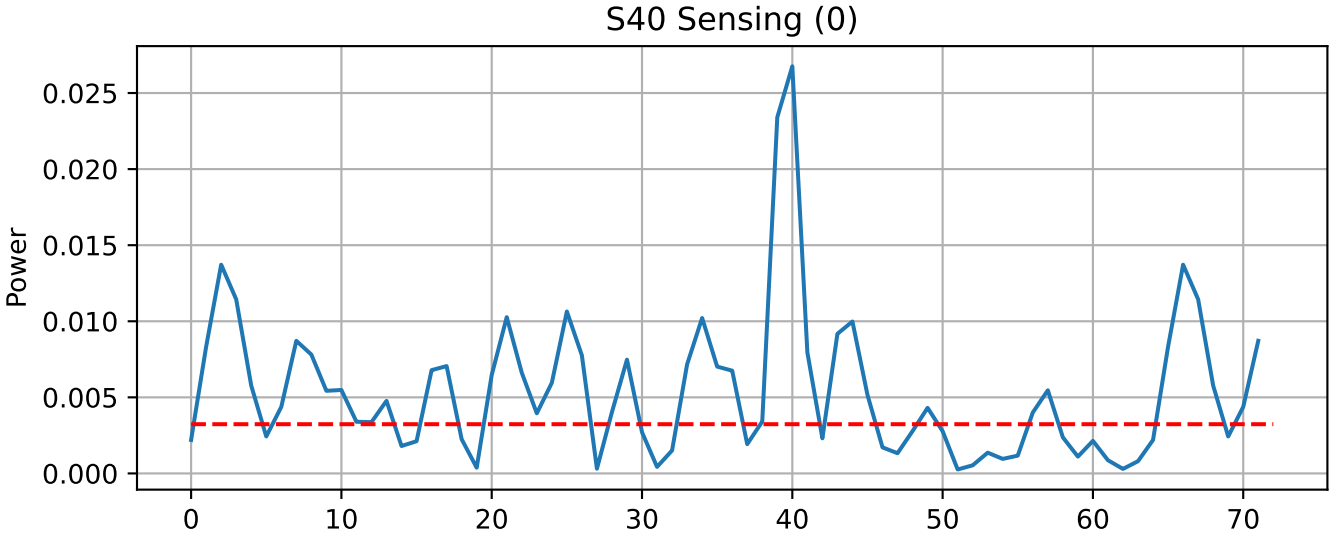
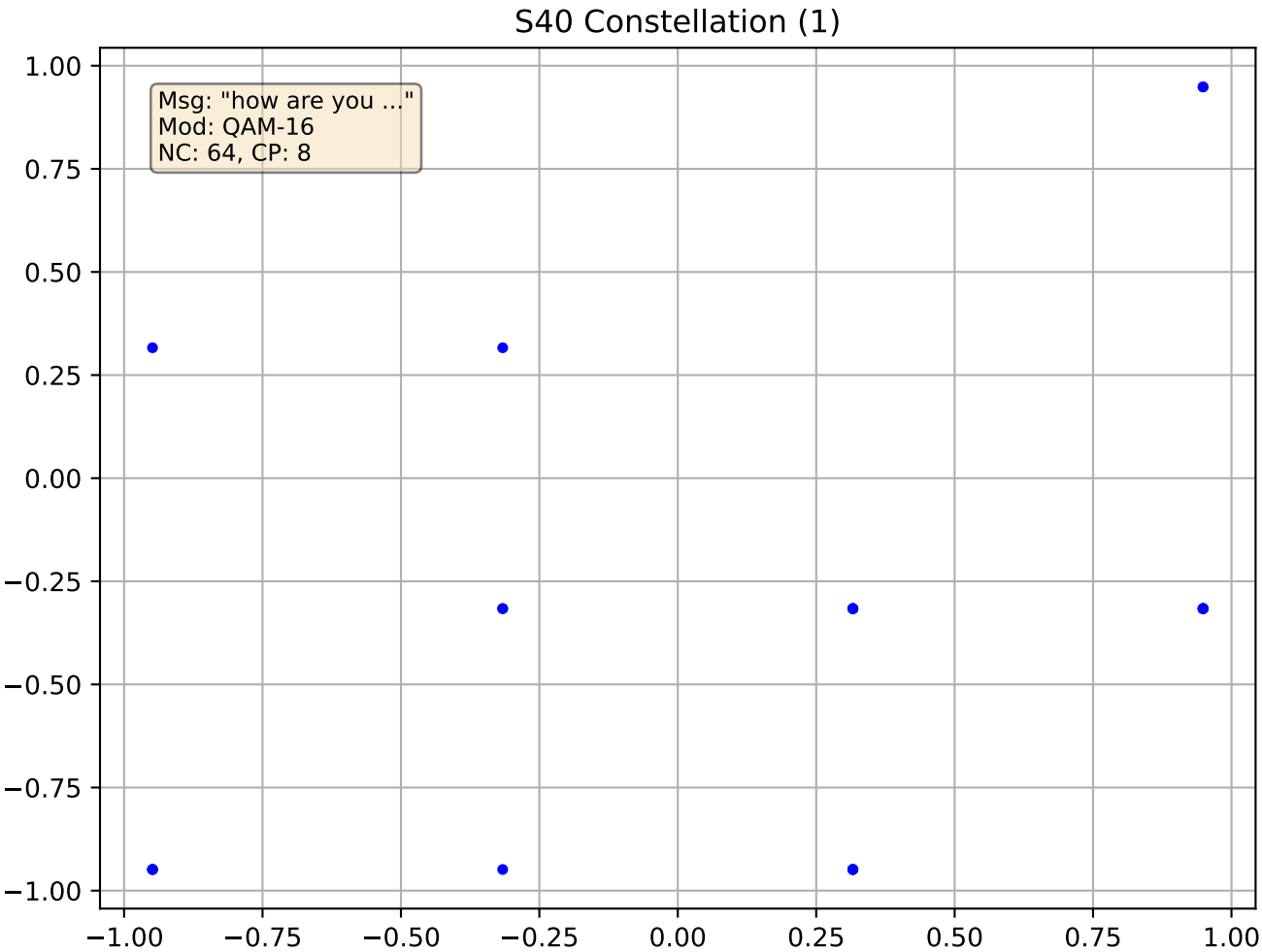
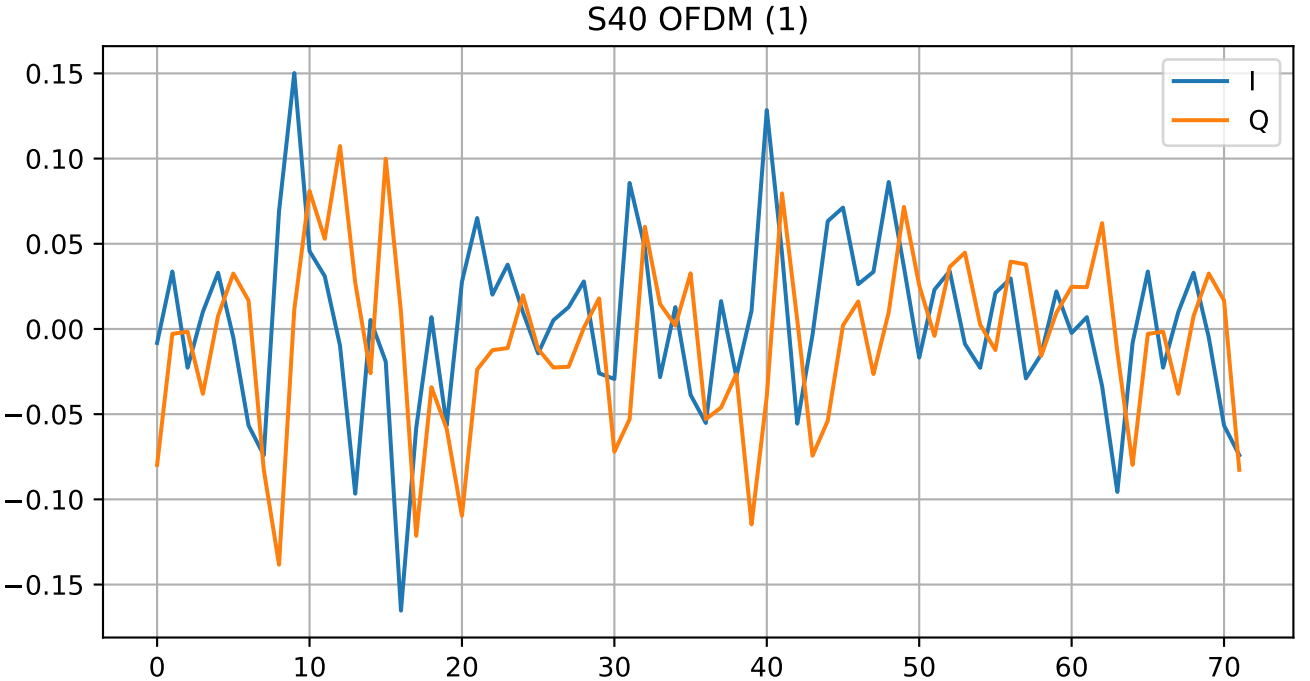
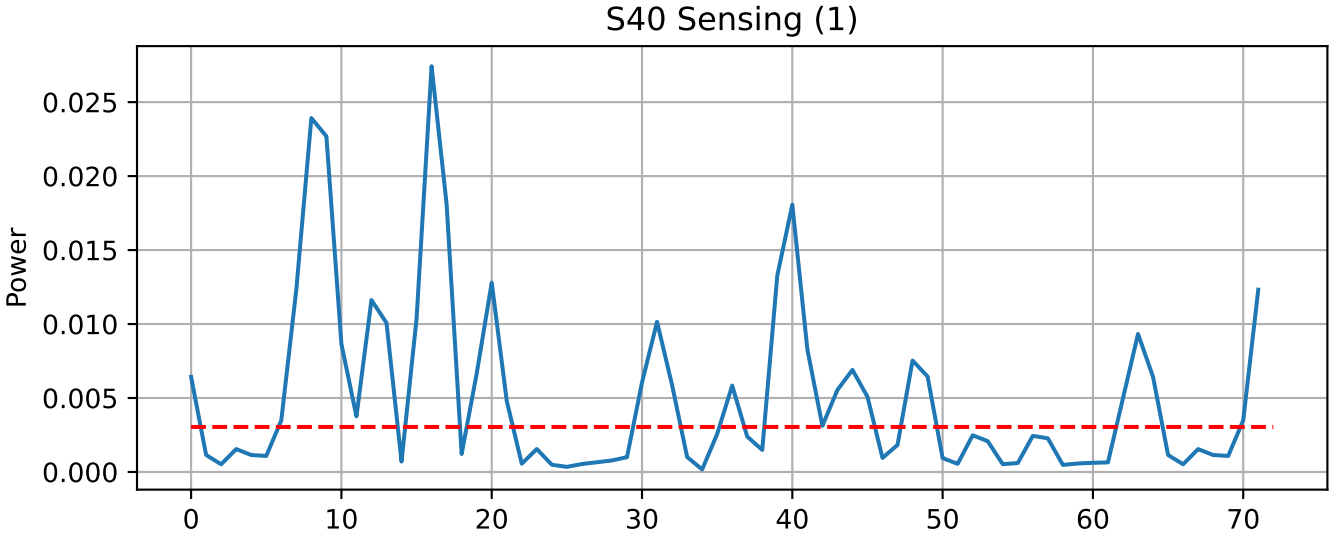

S40 Constellation (0)
Msg: "hello brother..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

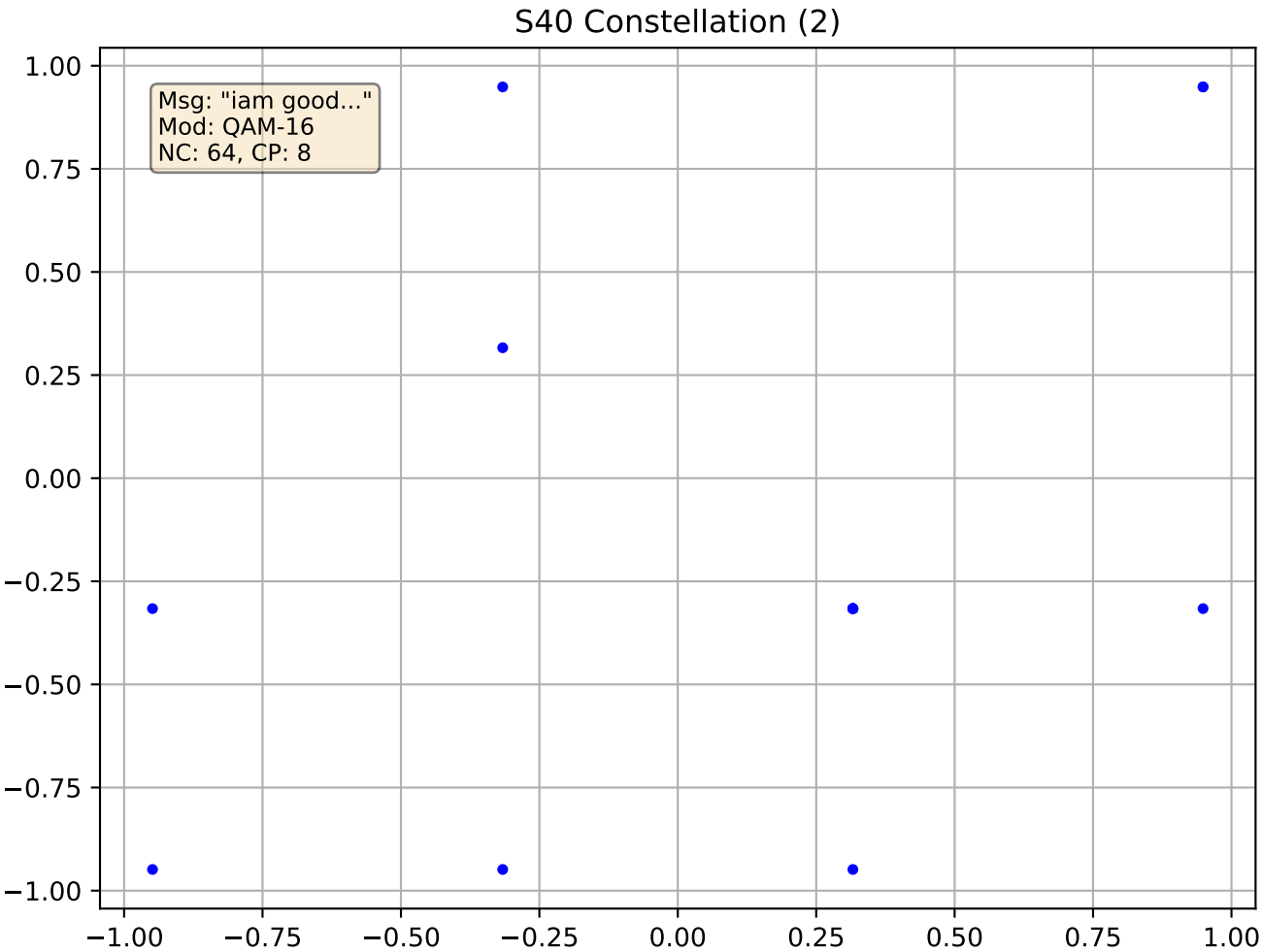

S40 OFDM (0)
I
Q
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
50
60
70

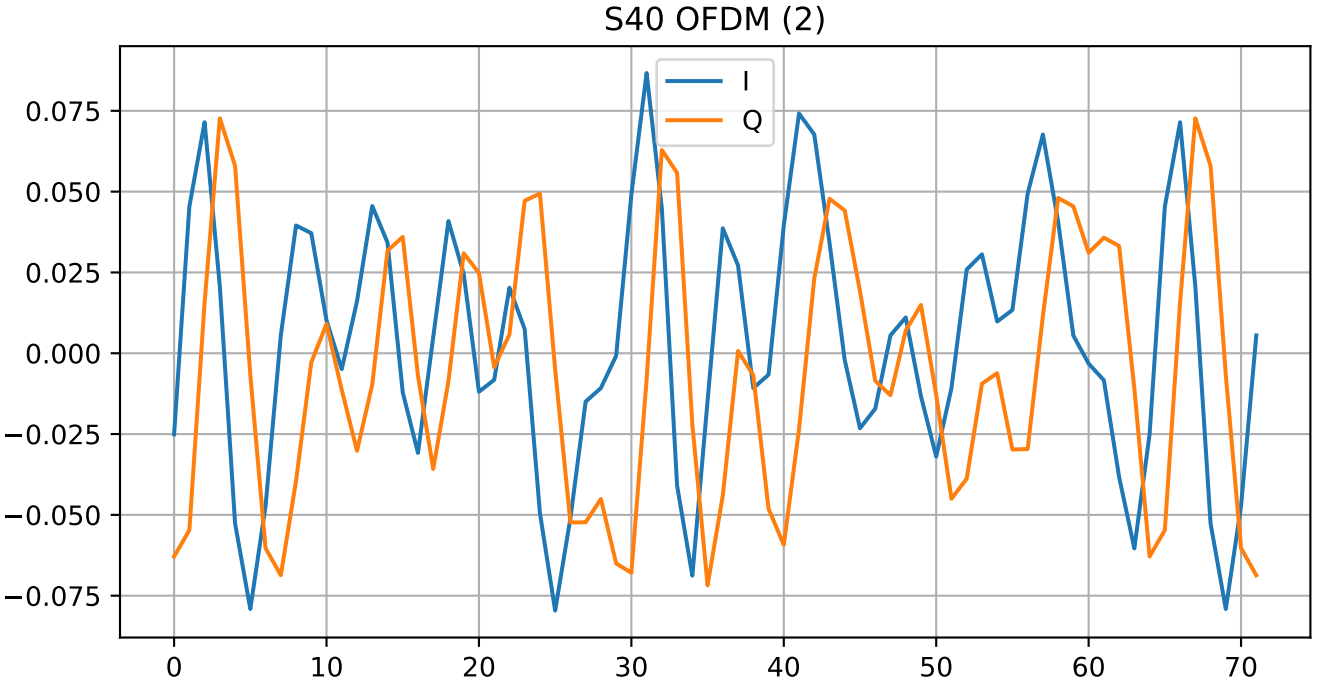

S40 Sensing (0)
Power
0.025
0.020
0.015
0.010
0.005
0.000
0
10
20
30
40
50
60
70

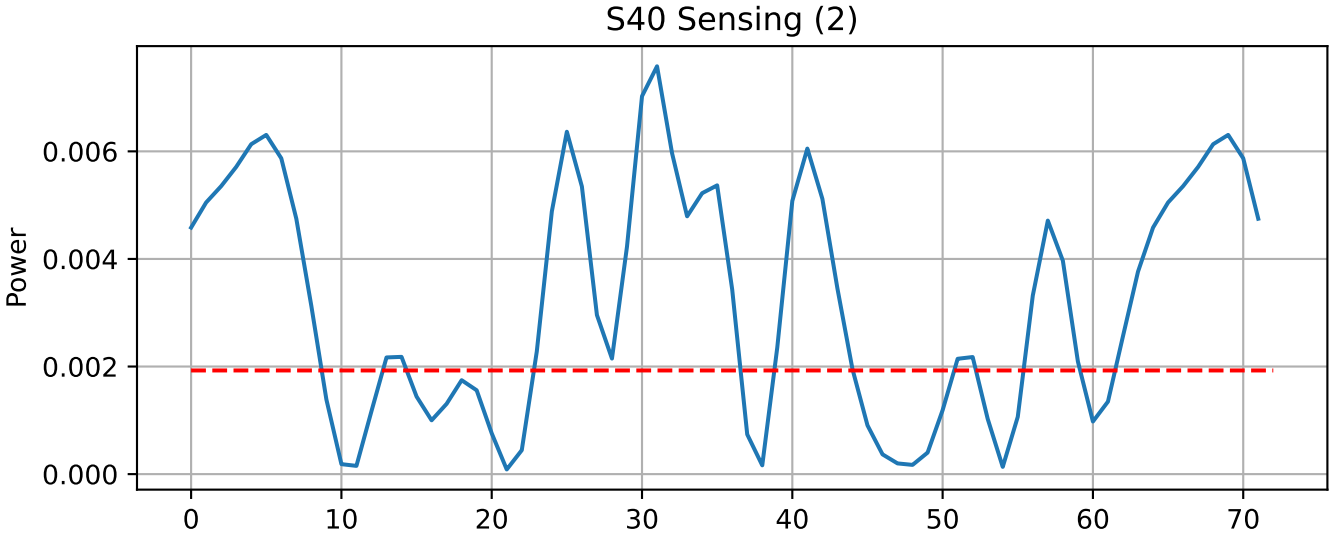

S40 Constellation (1)
Msg: "how are you ..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

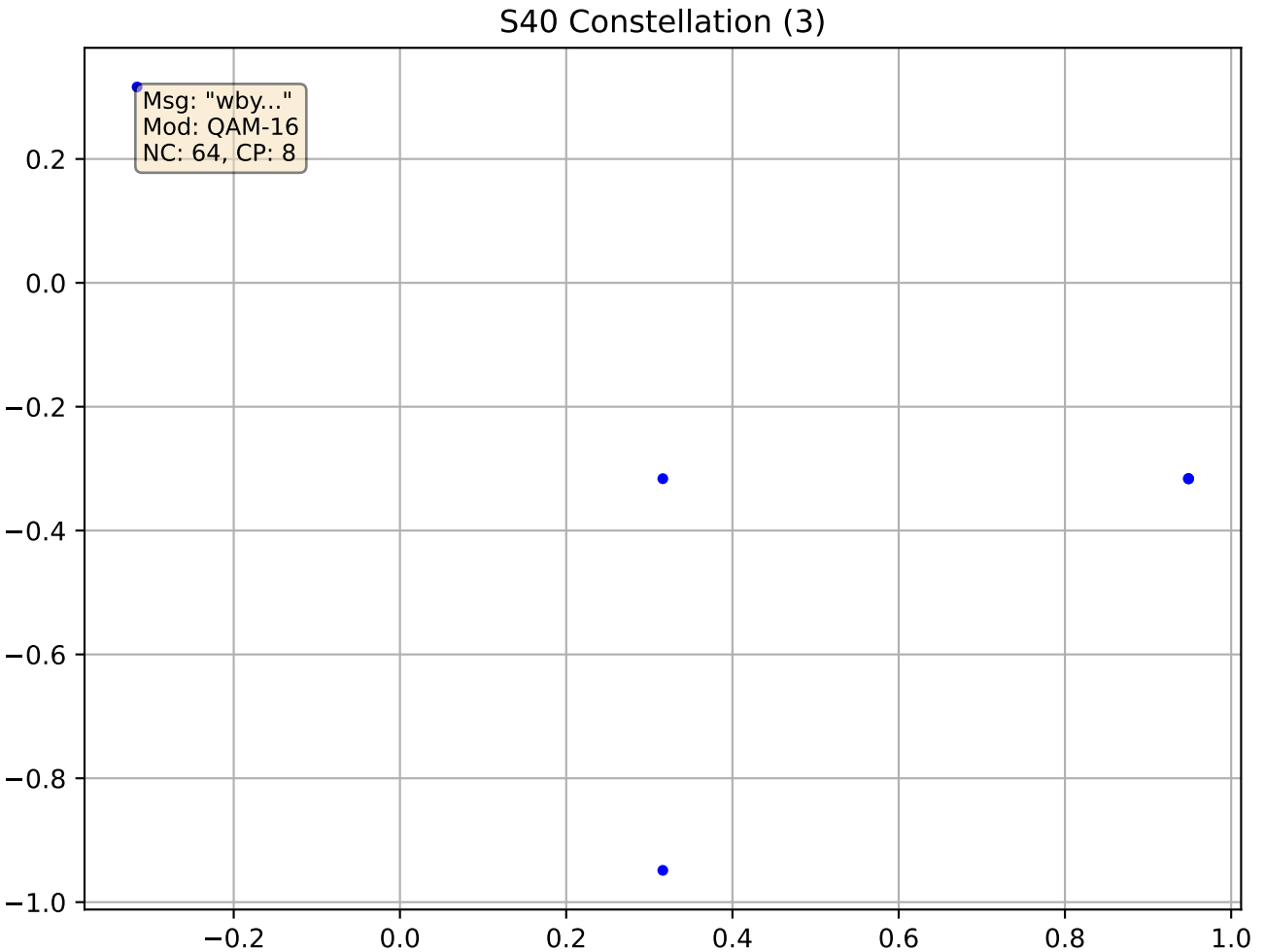

S40 OFDM (1)
I
Q
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
50
60
70

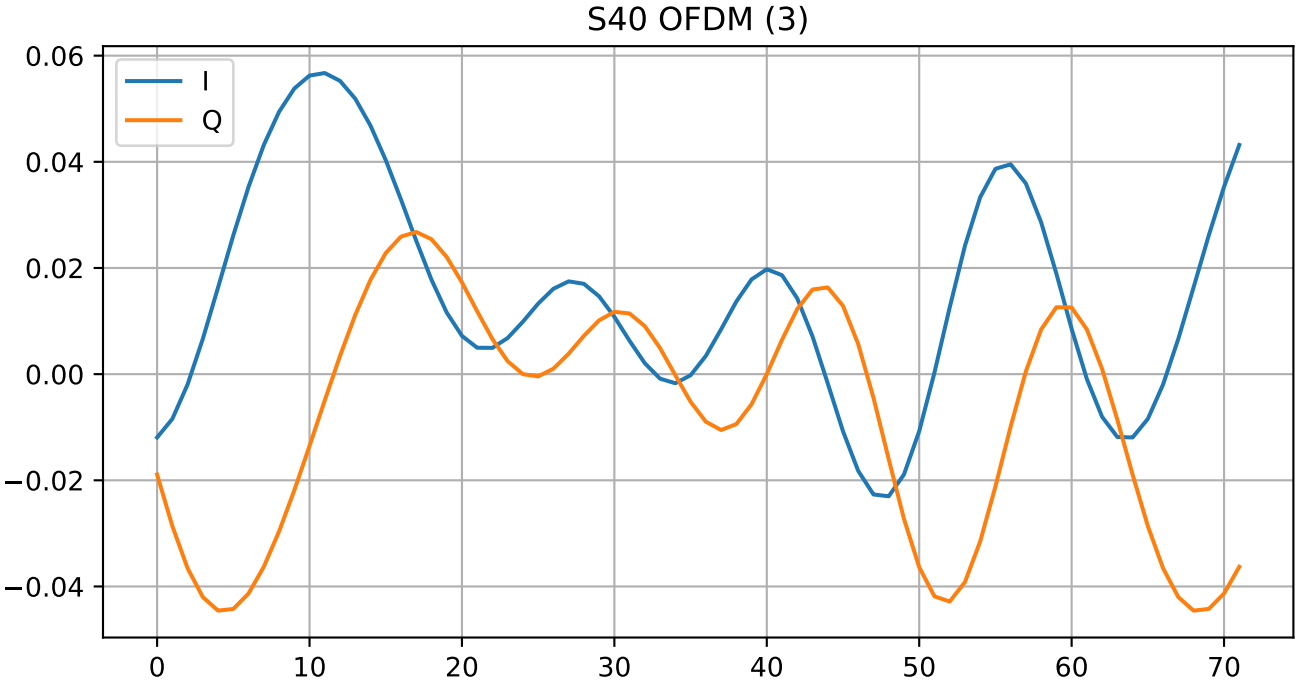

S40 Sensing (1)
Power
0.025
0.020
0.015
0.010
0.005
0.000
0
10
20
30
40
50
60
70

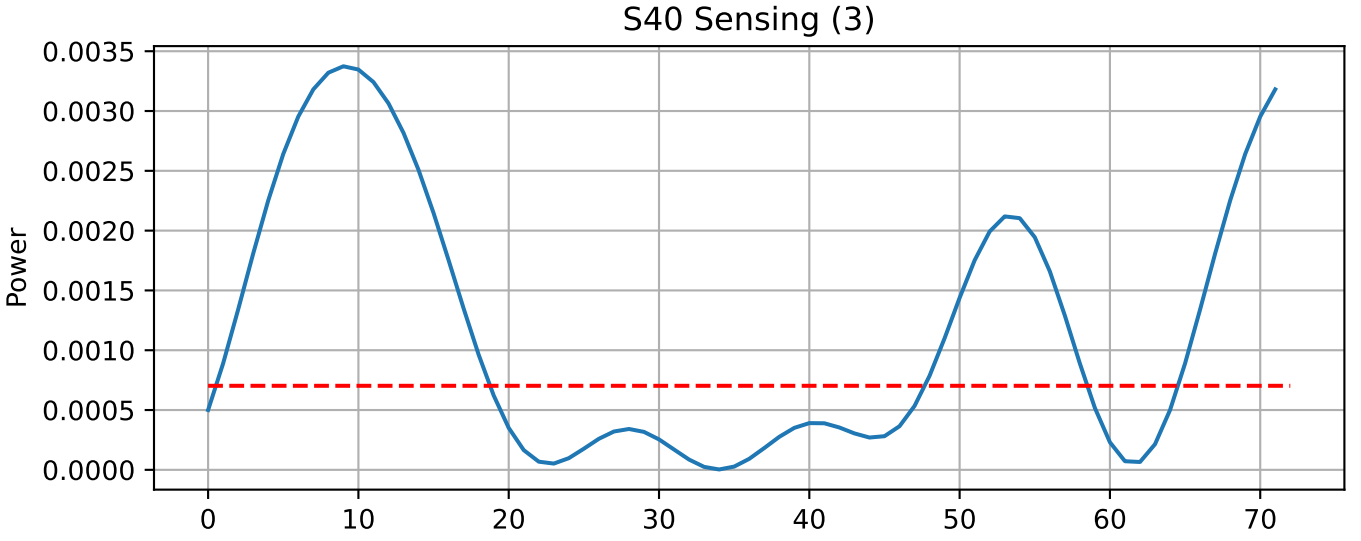

S40 Constellation (2)
Msg: "iam good..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

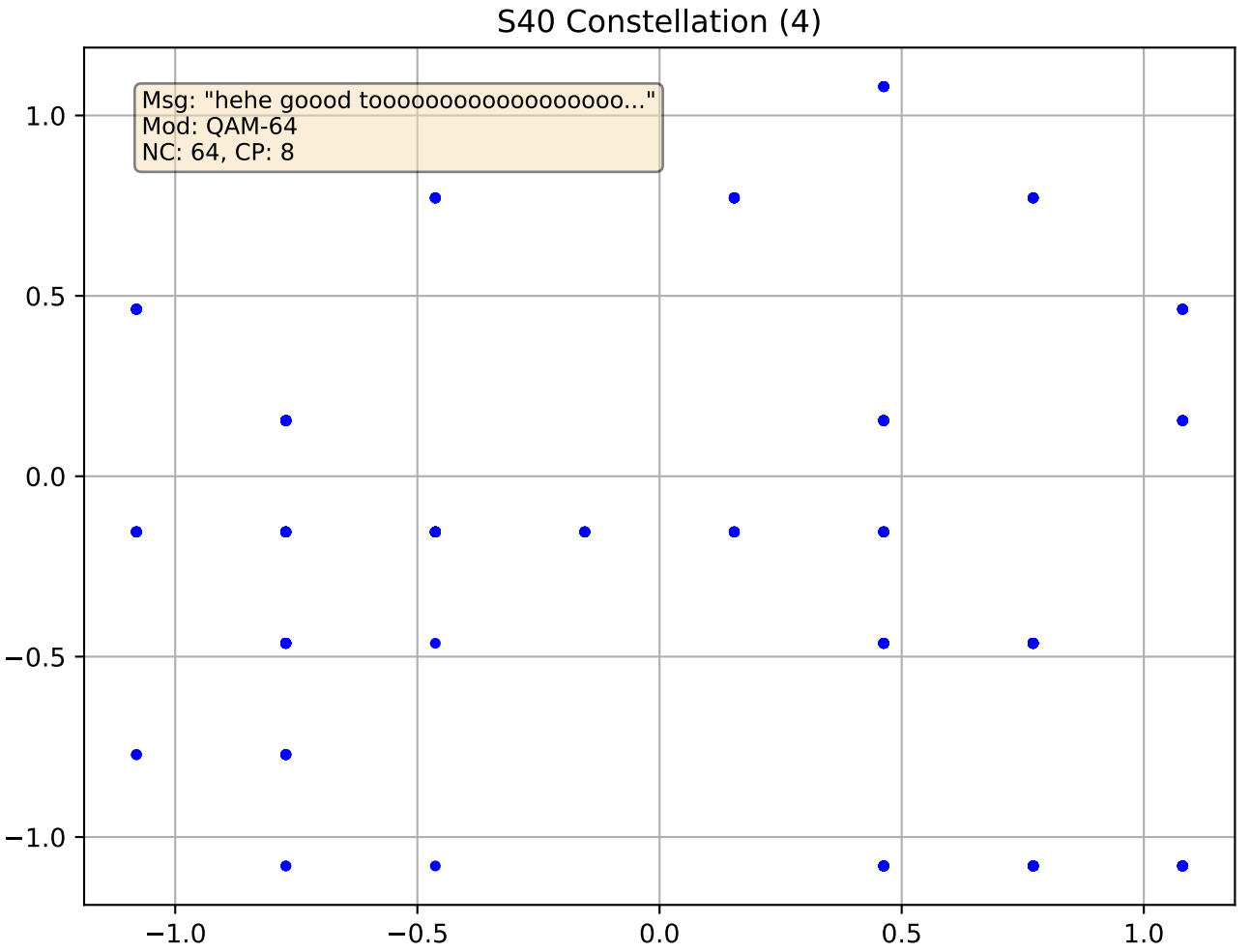

S40 OFDM (2)
I
Q
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
0
10
20
30
40
50
60
70

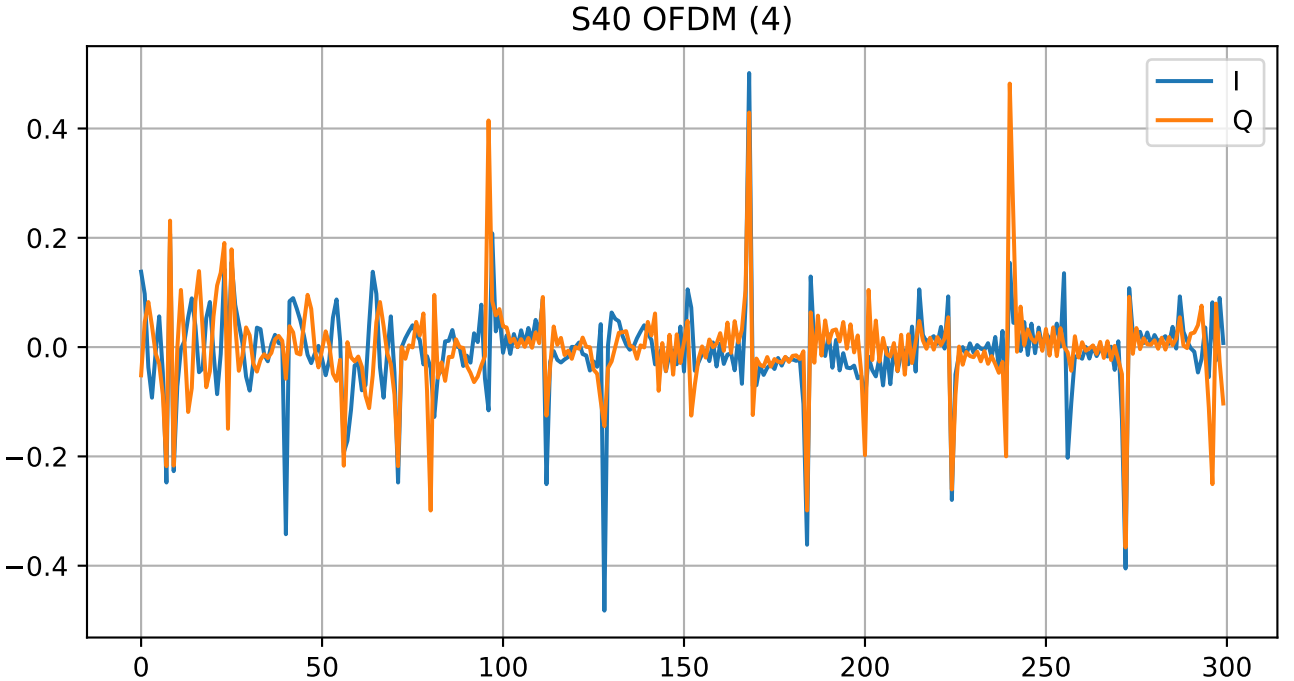

S40 Sensing (2)
Power
0.000
0.002
0.004
0.006
0
10
20
30
40
50
60
70

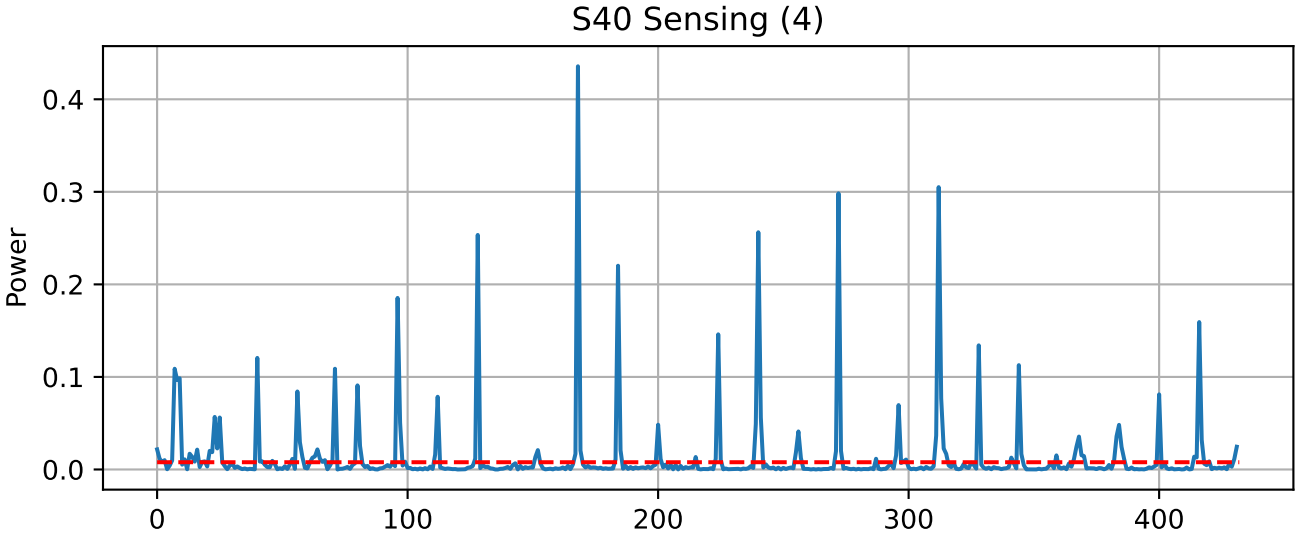

S40 Constellation (3)
Msg: "wby..."
Mod: QAM-16
NC: 64, CP: 8
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0

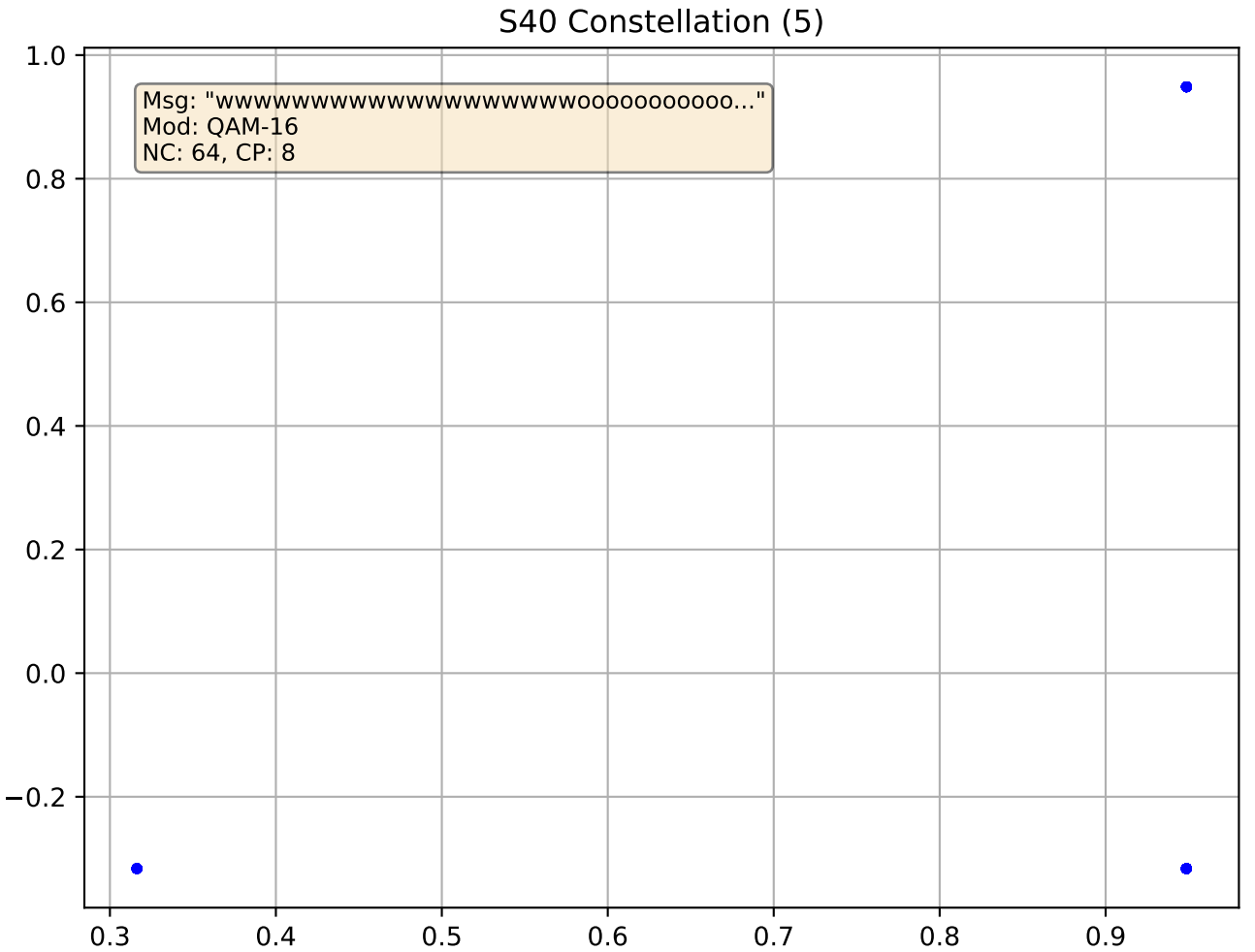

S40 OFDM (3)
I
Q
0.06
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
50
60
70

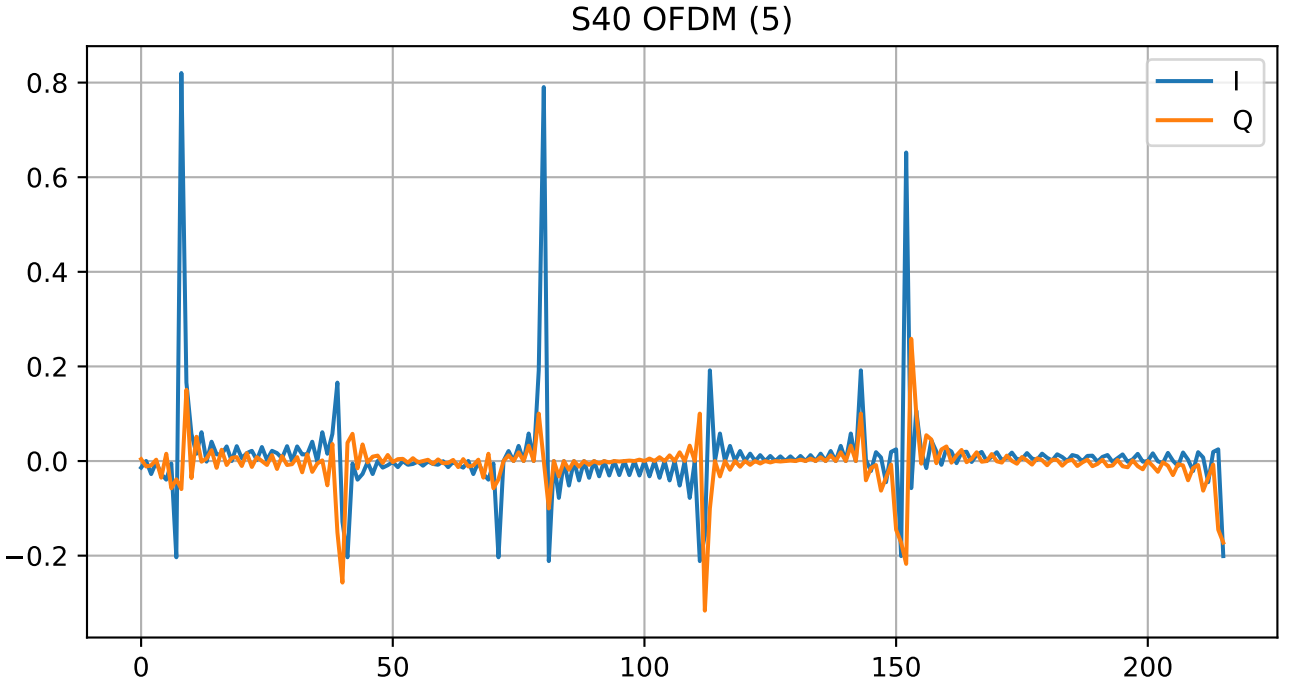

S40 Sensing (3)
Power
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
10
20
30
40
50
60
70

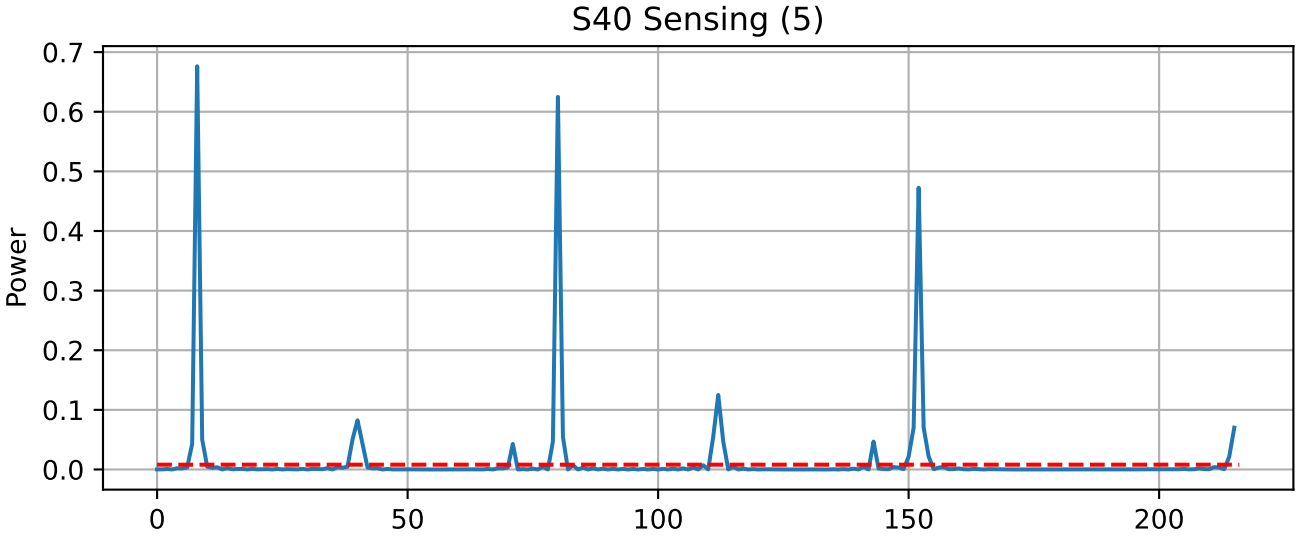

S40 Constellation (4)
Msg: "hehe goood tooooooooooooooooo..."
Mod: QAM-64
NC: 64, CP: 8
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0

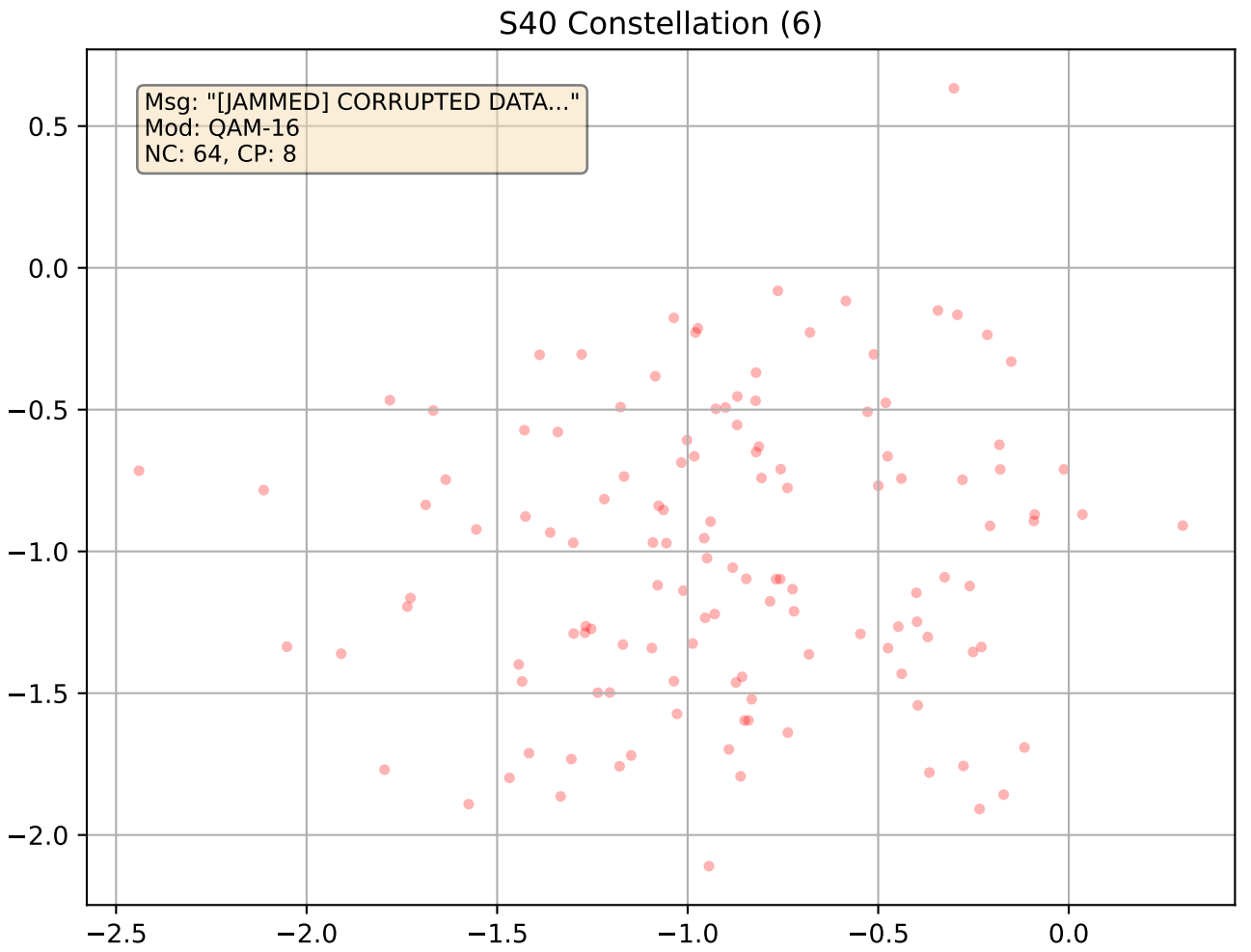

S40 OFDM (4)
I
Q
0.4
0.2
0.0
−0.2
−0.4
0
50
100
150
200
250
300

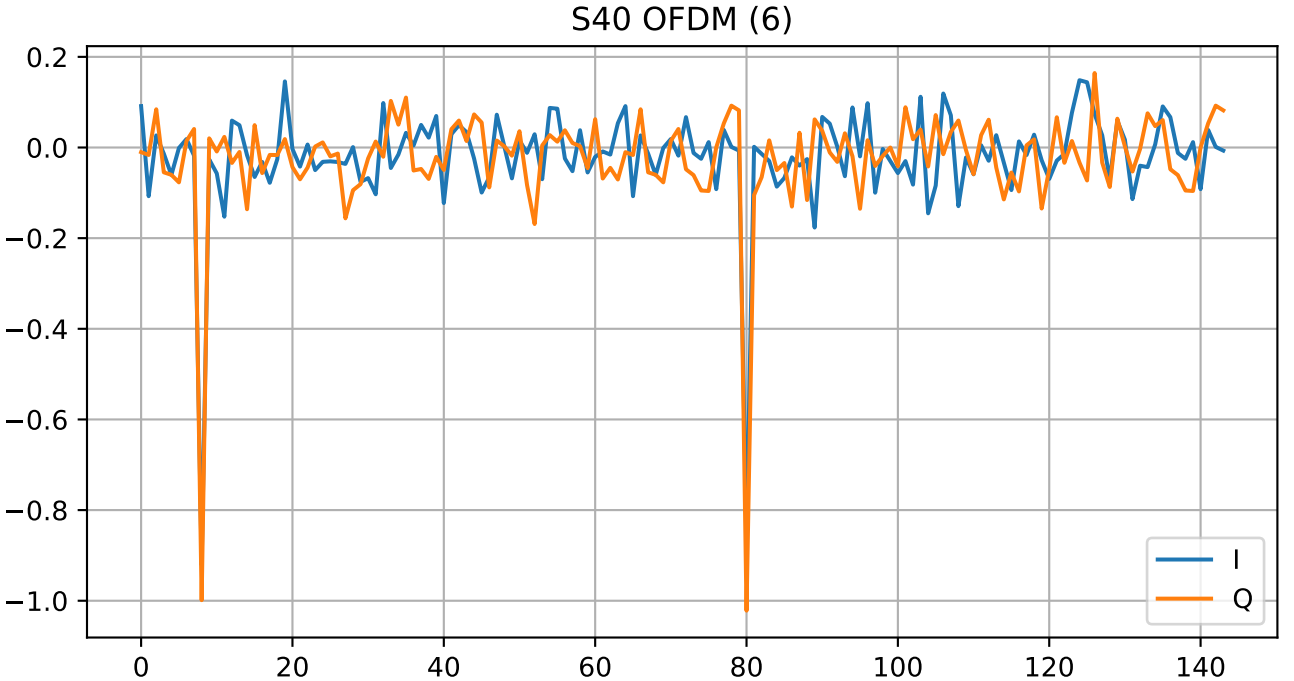

S40 Sensing (4)
Power
0.0
0.1
0.2
0.3
0.4
0
100
200
300
400

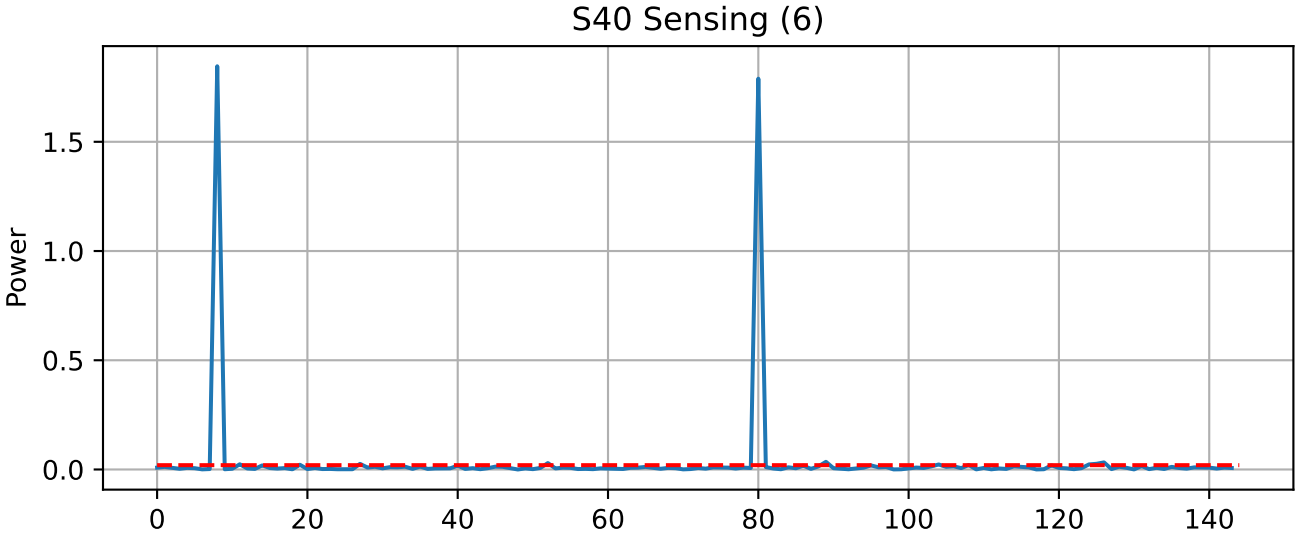

S40 Constellation (5)
Msg: "wwwwwwwwwwwwwwwwwwwwoooooooooo..."
Mod: QAM-16
NC: 64, CP: 8
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9

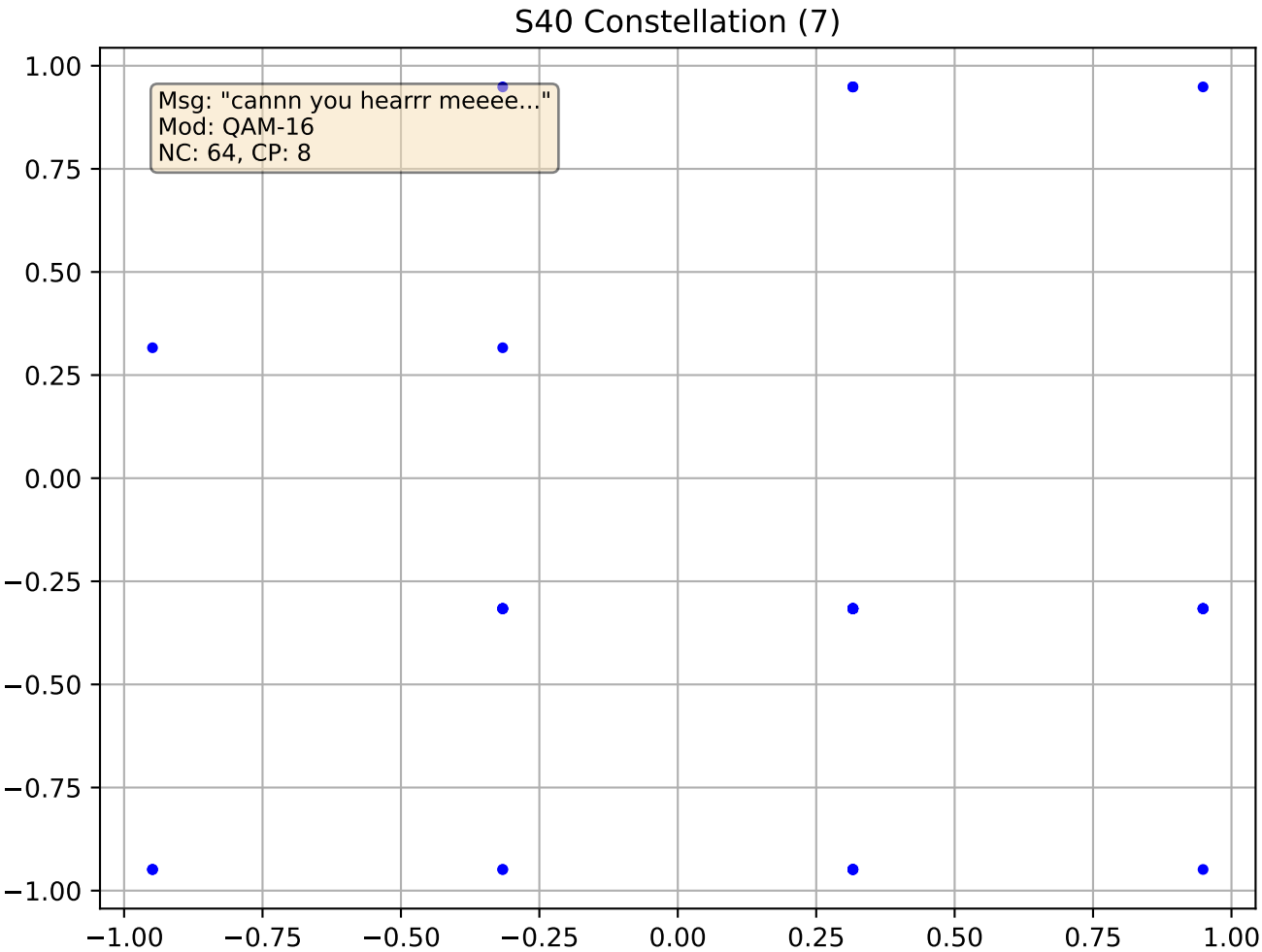

S40 OFDM (5)
I
Q
0.8
0.6
0.4
0.2
0.0
−0.2
0
50
100
150
200

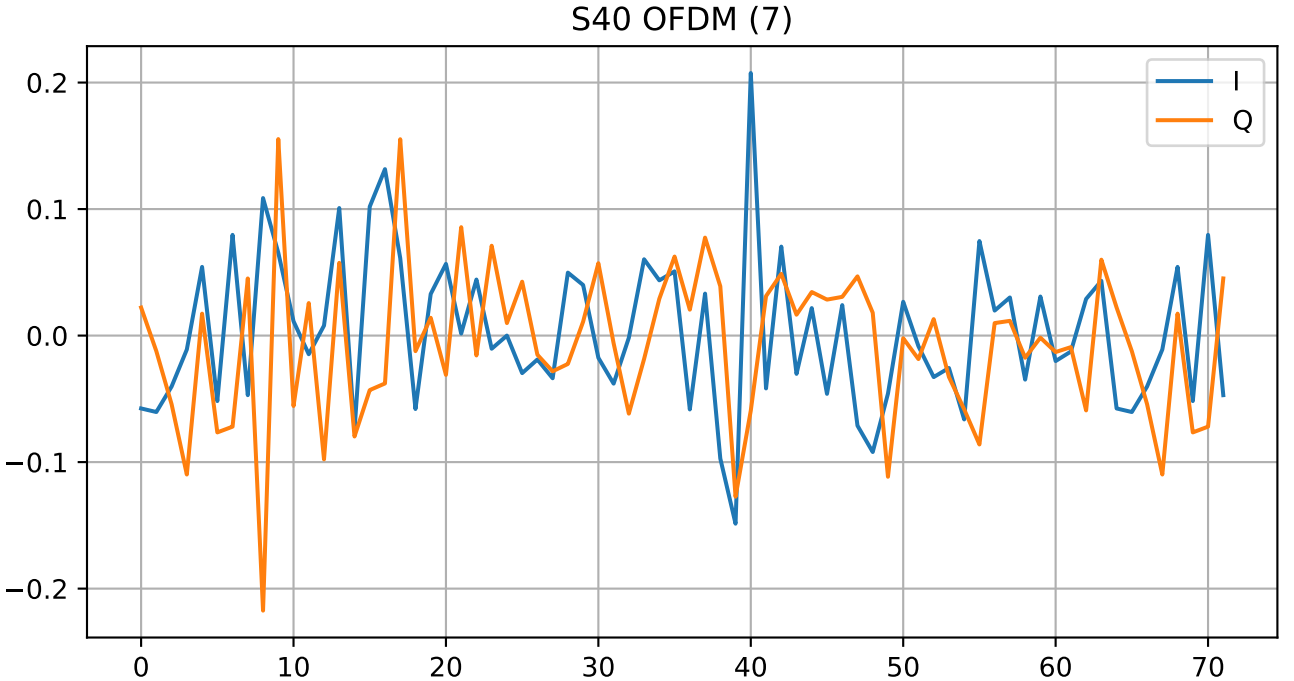

S40 Sensing (5)
Power
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
50
100
150
200

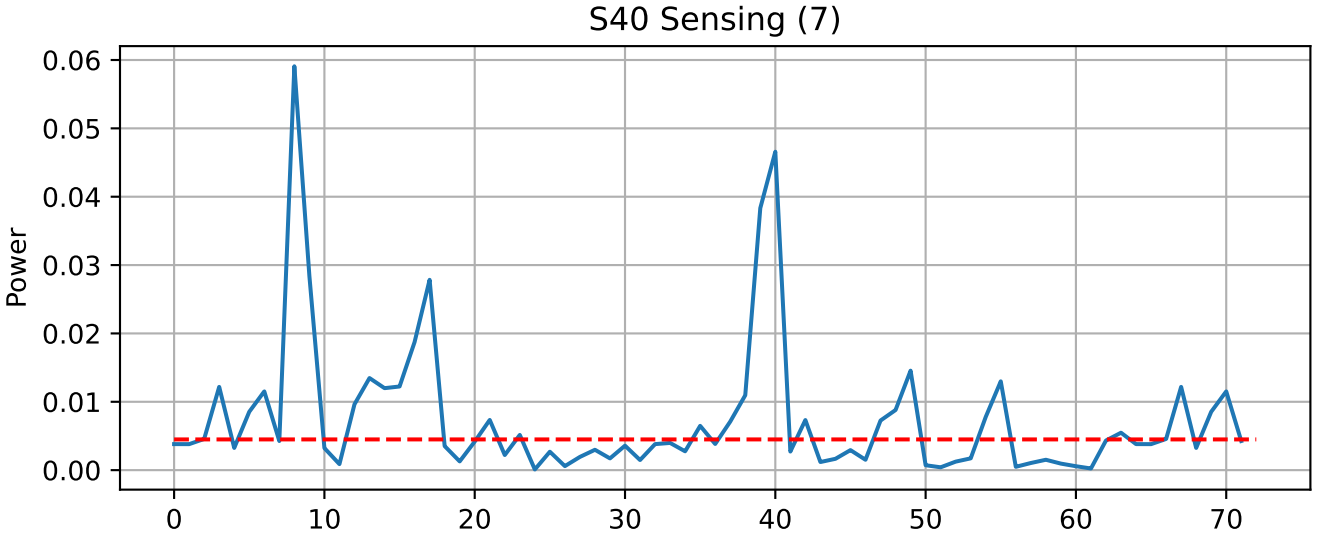

S40 Constellation (6)
Msg: "[JAMMED] CORRUPTED DATA..."
Mod: QAM-16
NC: 64, CP: 8
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0

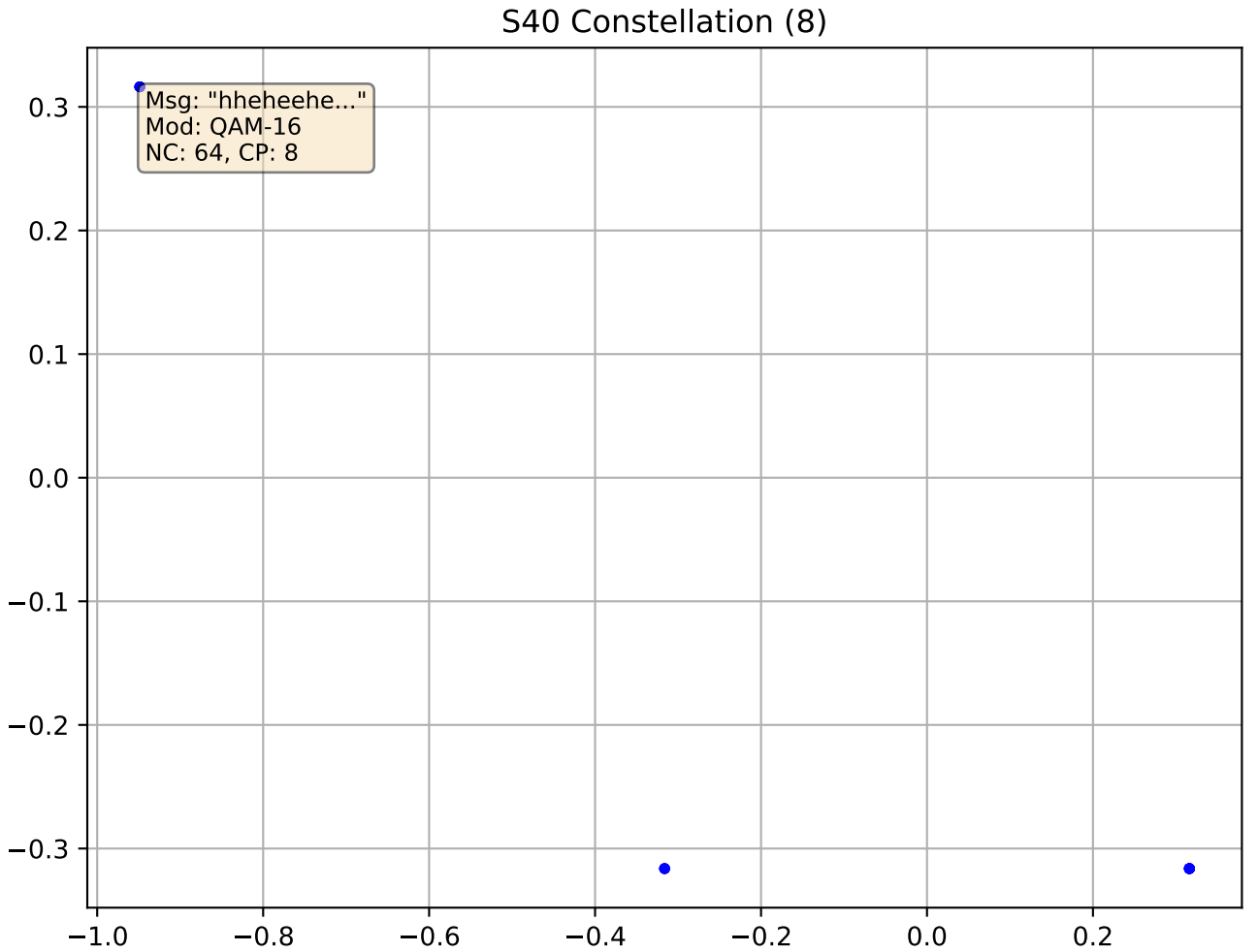

S40 OFDM (6)
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
20
40
60
80
100
120
140
I
Q

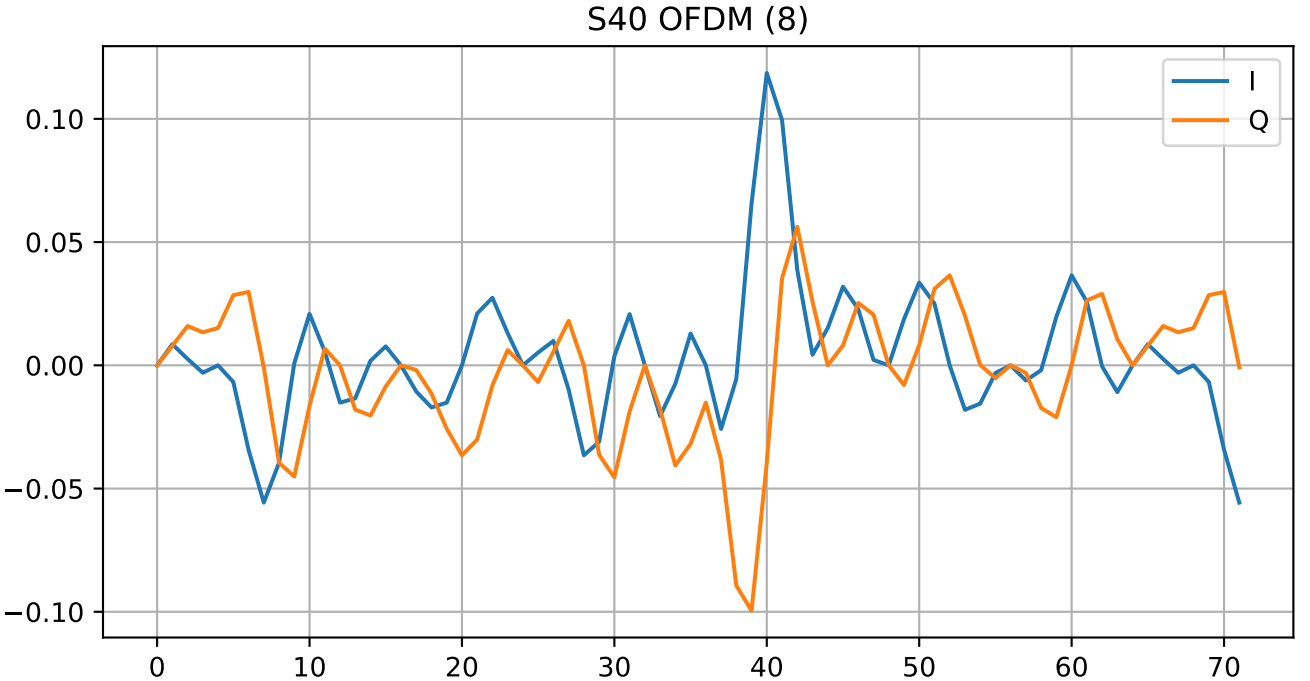

S40 Sensing (6)
Power
1.5
1.0
0.5
0.0
0
20
40
60
80
100
120
140

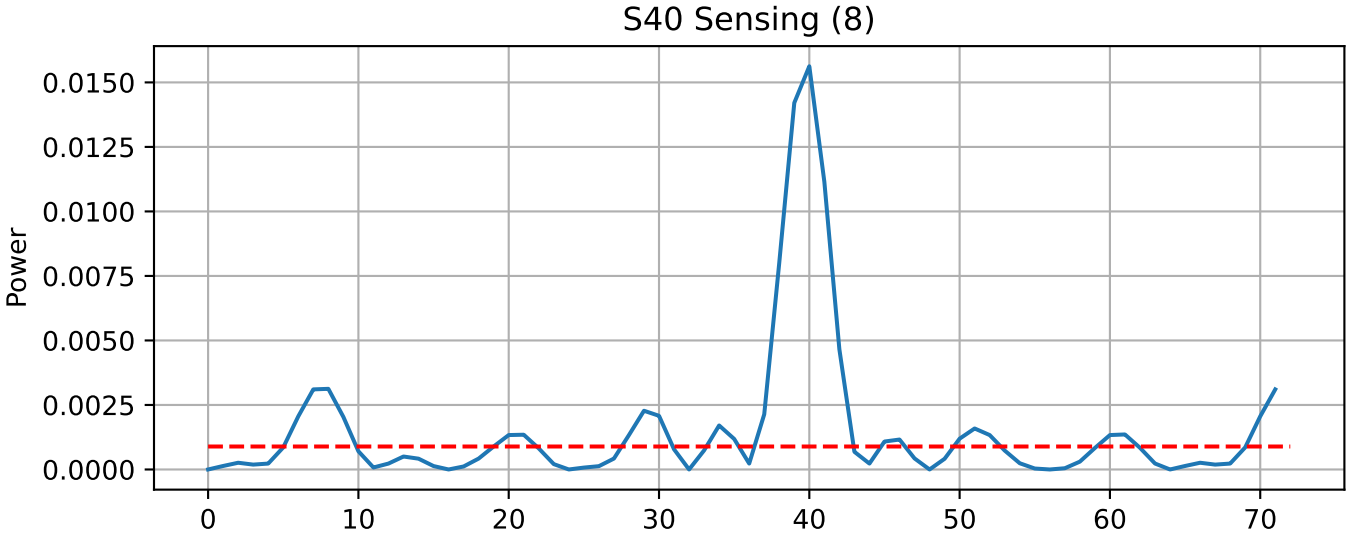

S40 Constellation (7)
Msg: "cannn you hearrr meeee..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

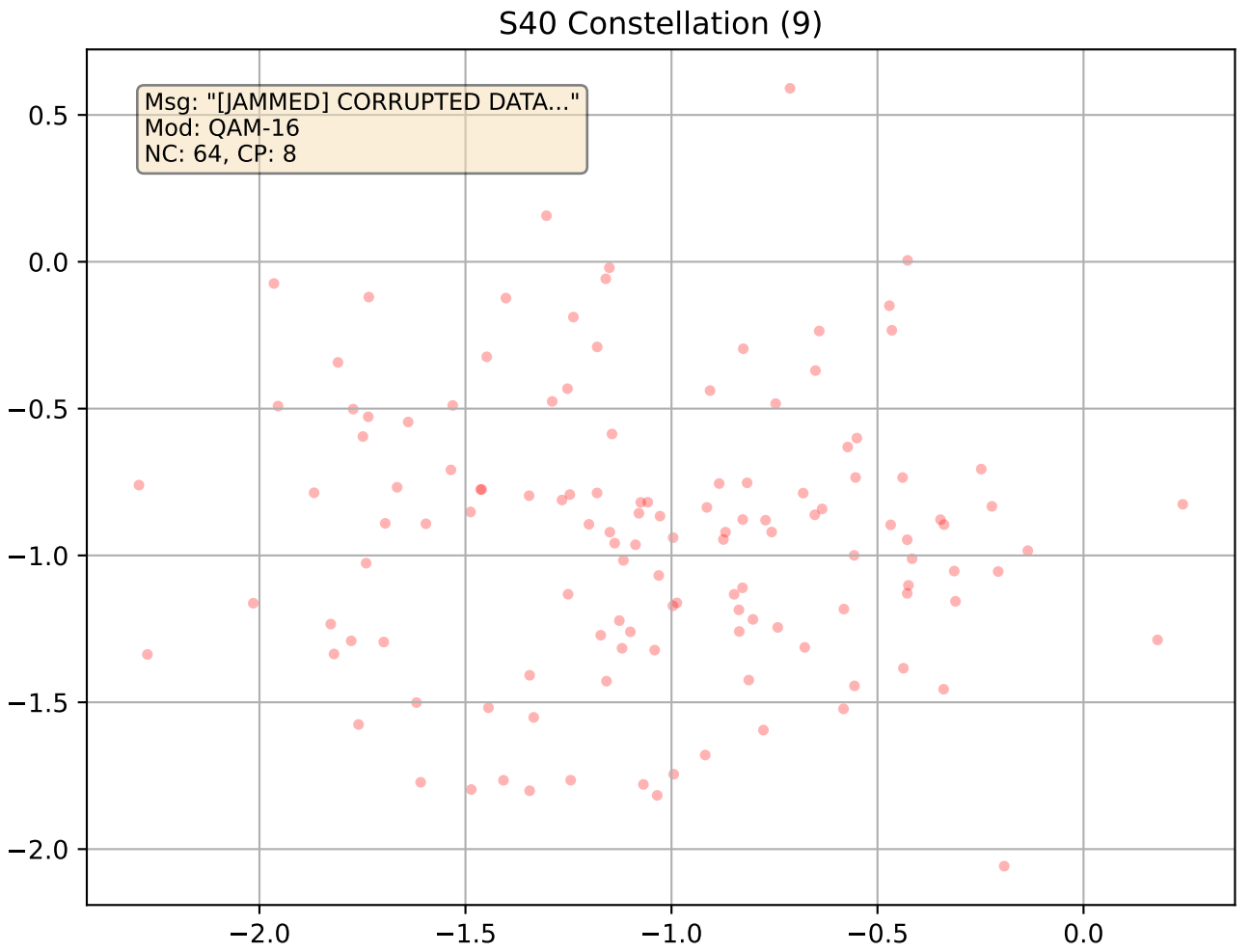

S40 OFDM (7)
I
Q
0.2
0.1
0.0
−0.1
−0.2
0
10
20
30
40
50
60
70

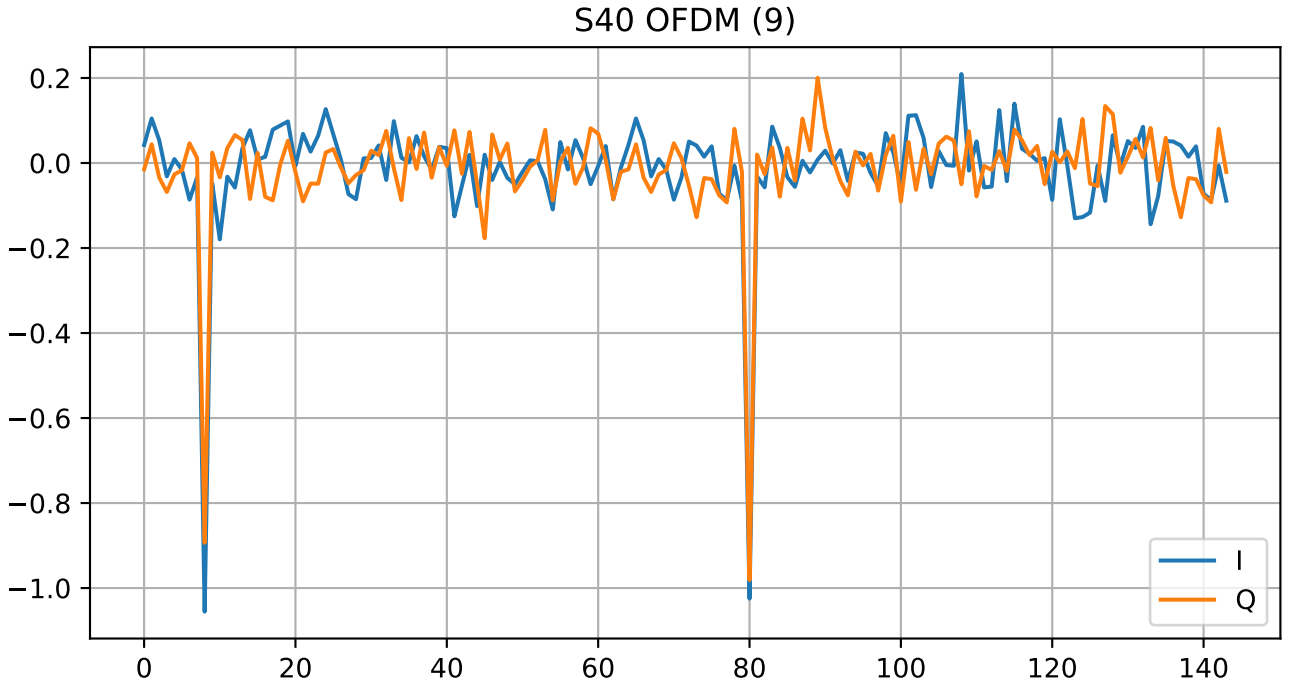

S40 Sensing (7)
Power
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0
10
20
30
40
50
60
70

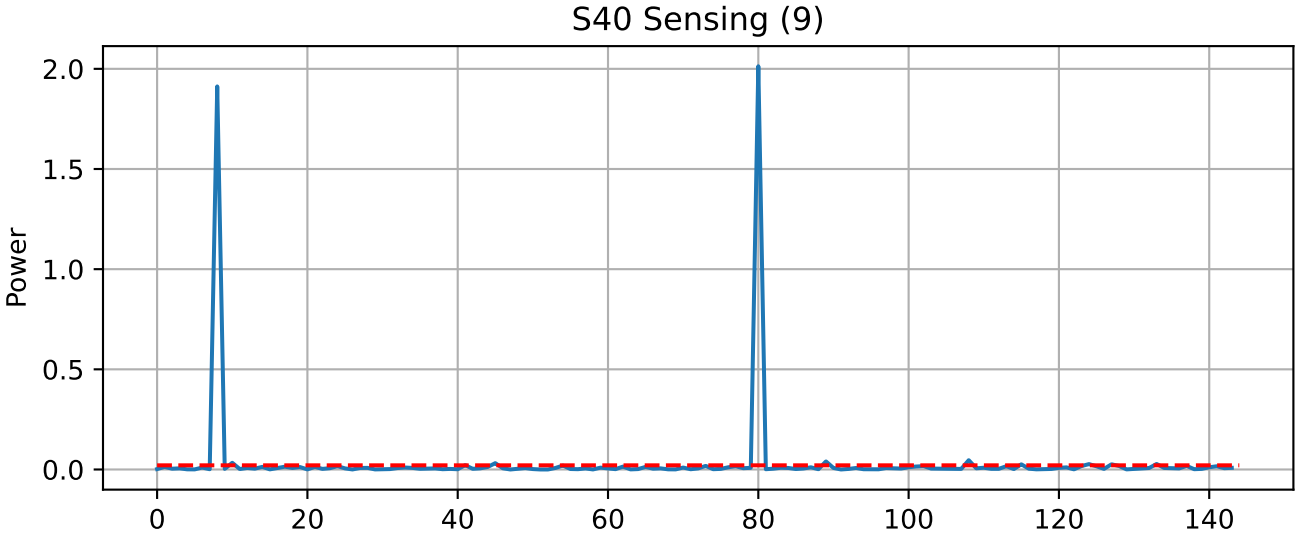

S40 Constellation (8)
Msg: "hheheehe..."
Mod: QAM-16
NC: 64, CP: 8
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2

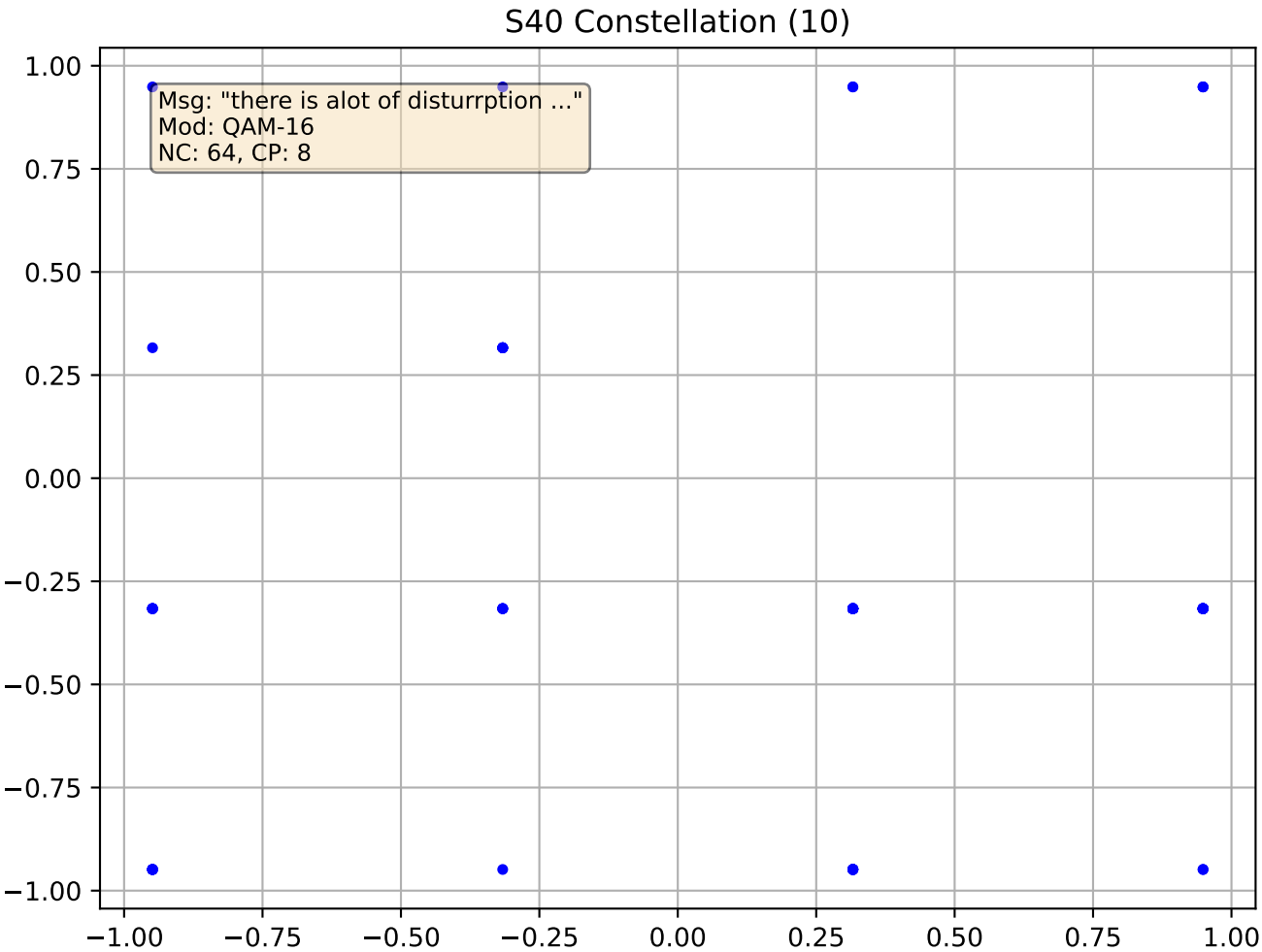

S40 OFDM (8)
I
Q
0.10
0.05
0.00
−0.05
−0.10
0
10
20
30
40
50
60
70

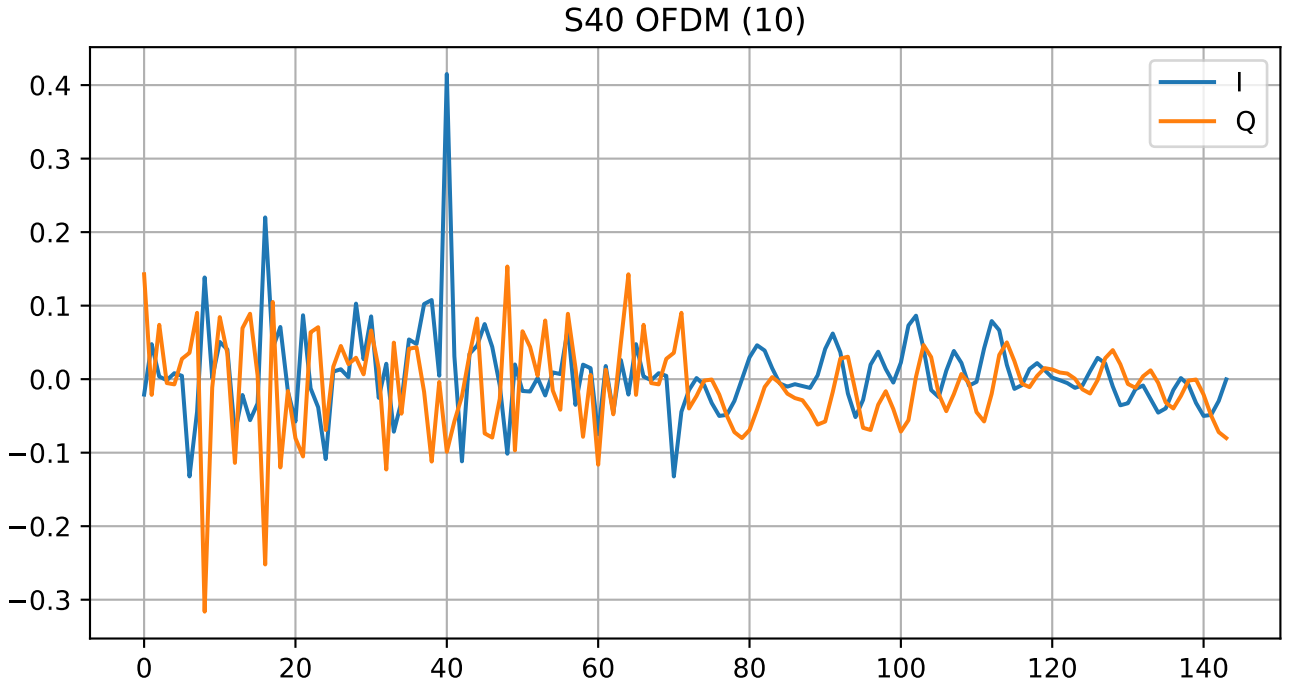

S40 Sensing (8)
Power
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
0
10
20
30
40
50
60
70

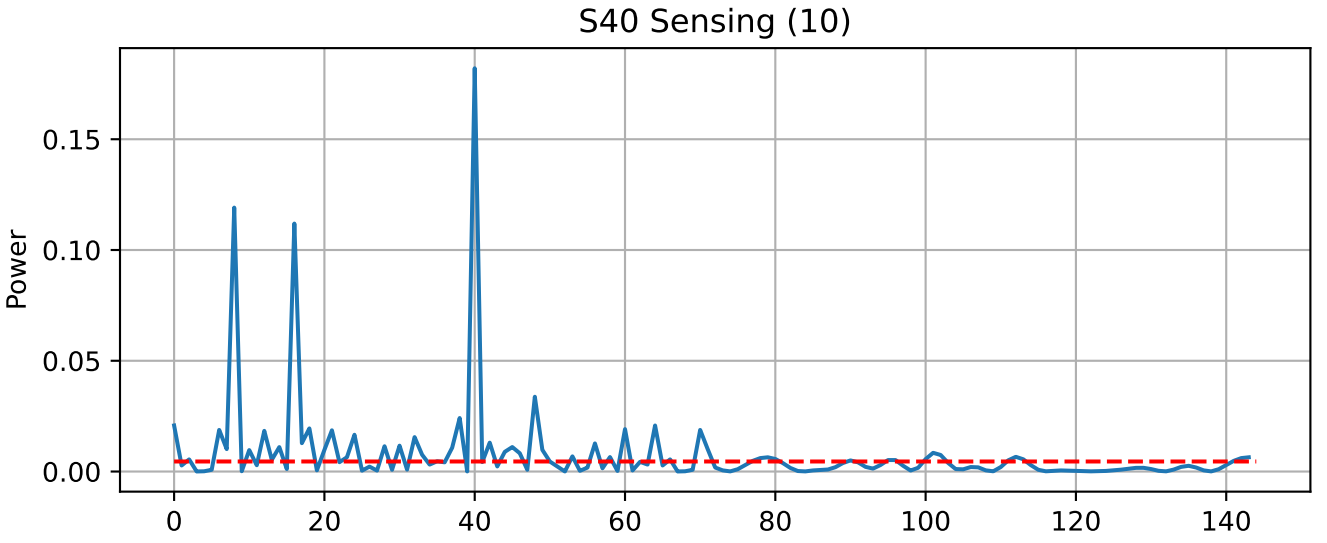

S40 Constellation (9)
Msg: "[JAMMED] CORRUPTED DATA..."
Mod: QAM-16
NC: 64, CP: 8
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.0
−1.5
−1.0
−0.5
0.0

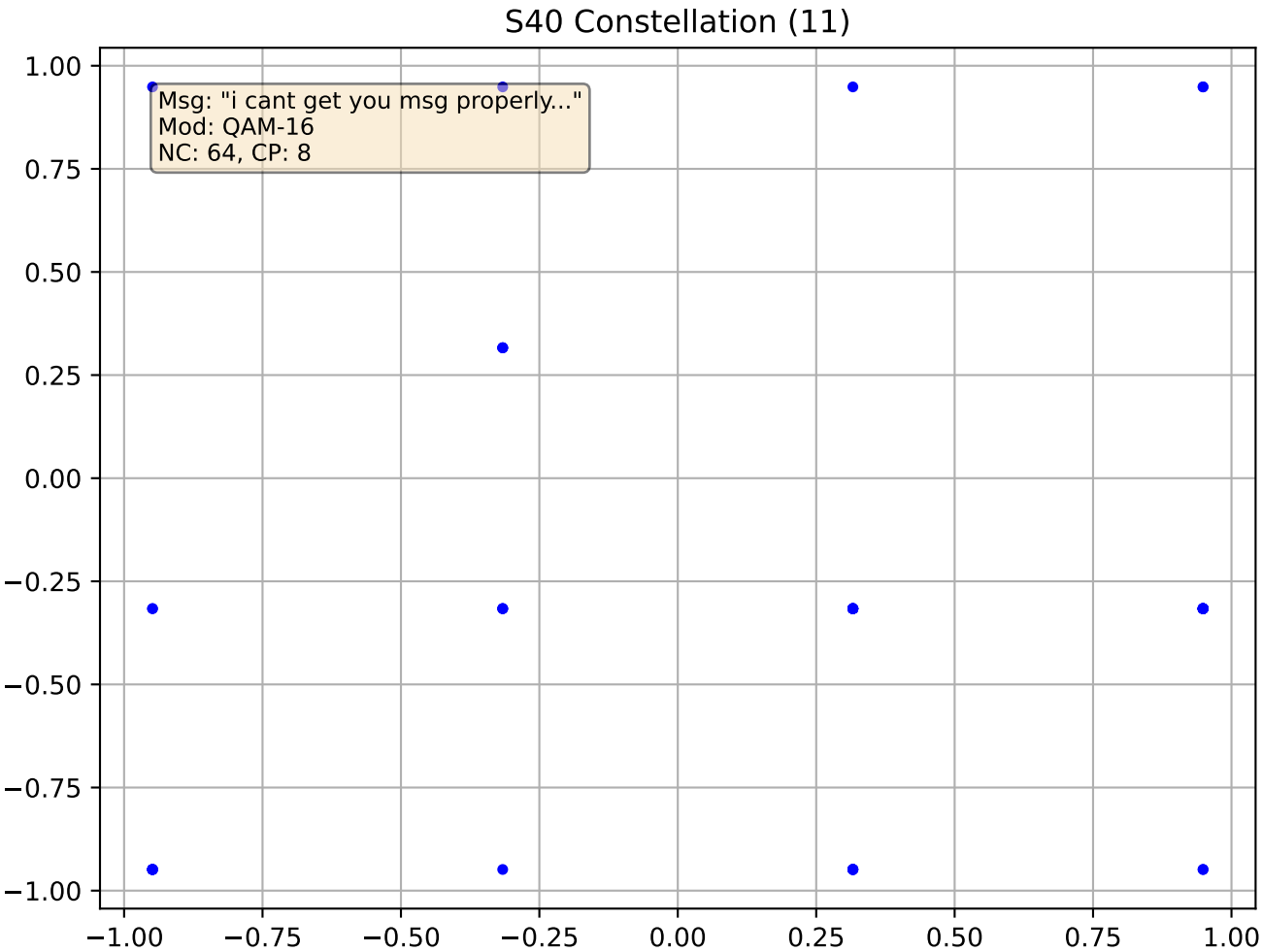

S40 OFDM (9)
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
20
40
60
80
100
120
140
I
Q

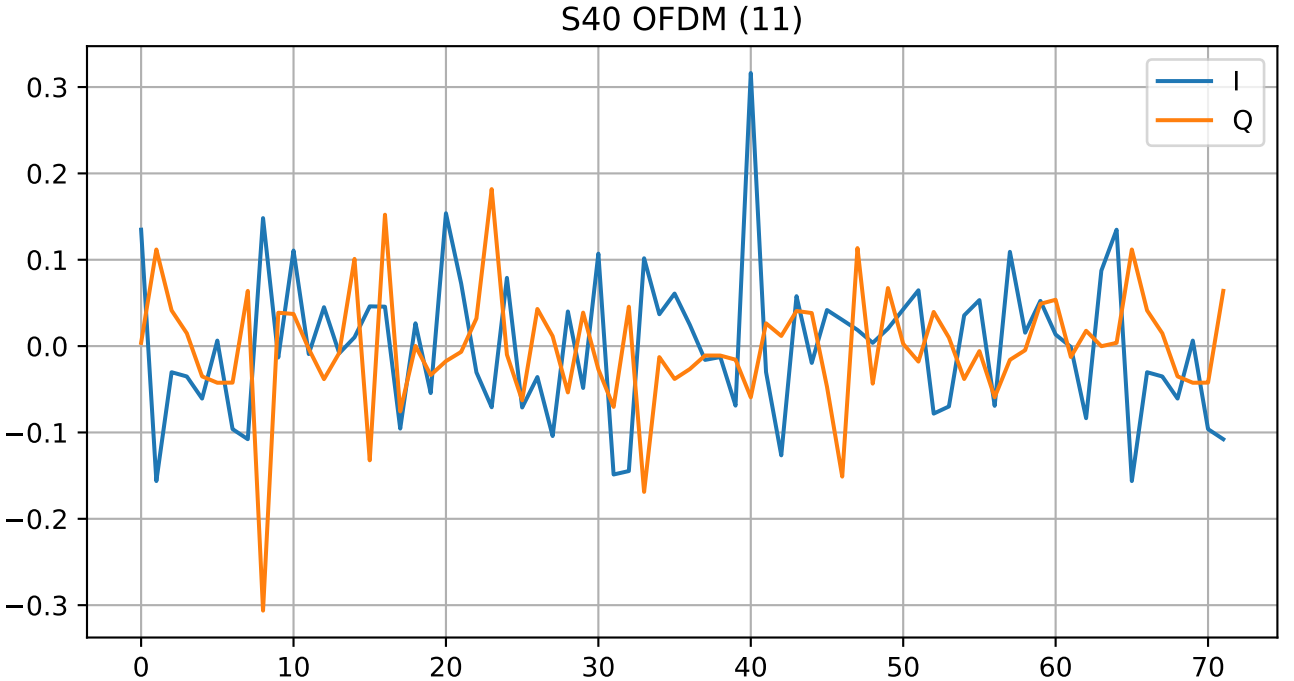

S40 Sensing (9)
Power
2.0
1.5
1.0
0.5
0.0
0
20
40
60
80
100
120
140

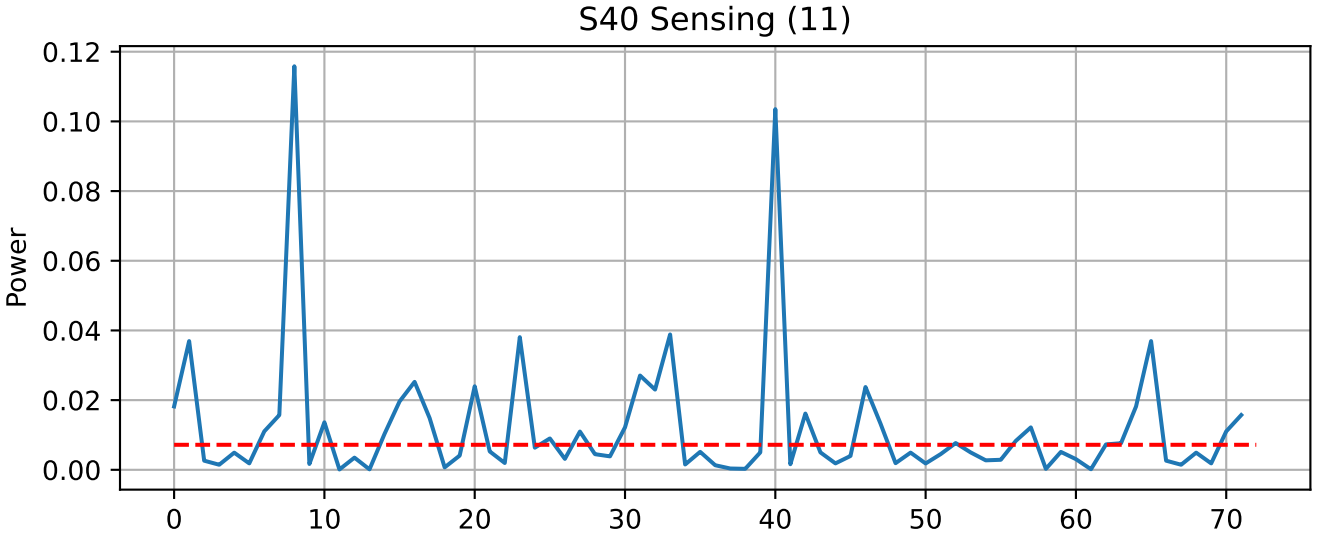

S40 Constellation (10)
Msg: "there is alot of disturrption ..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

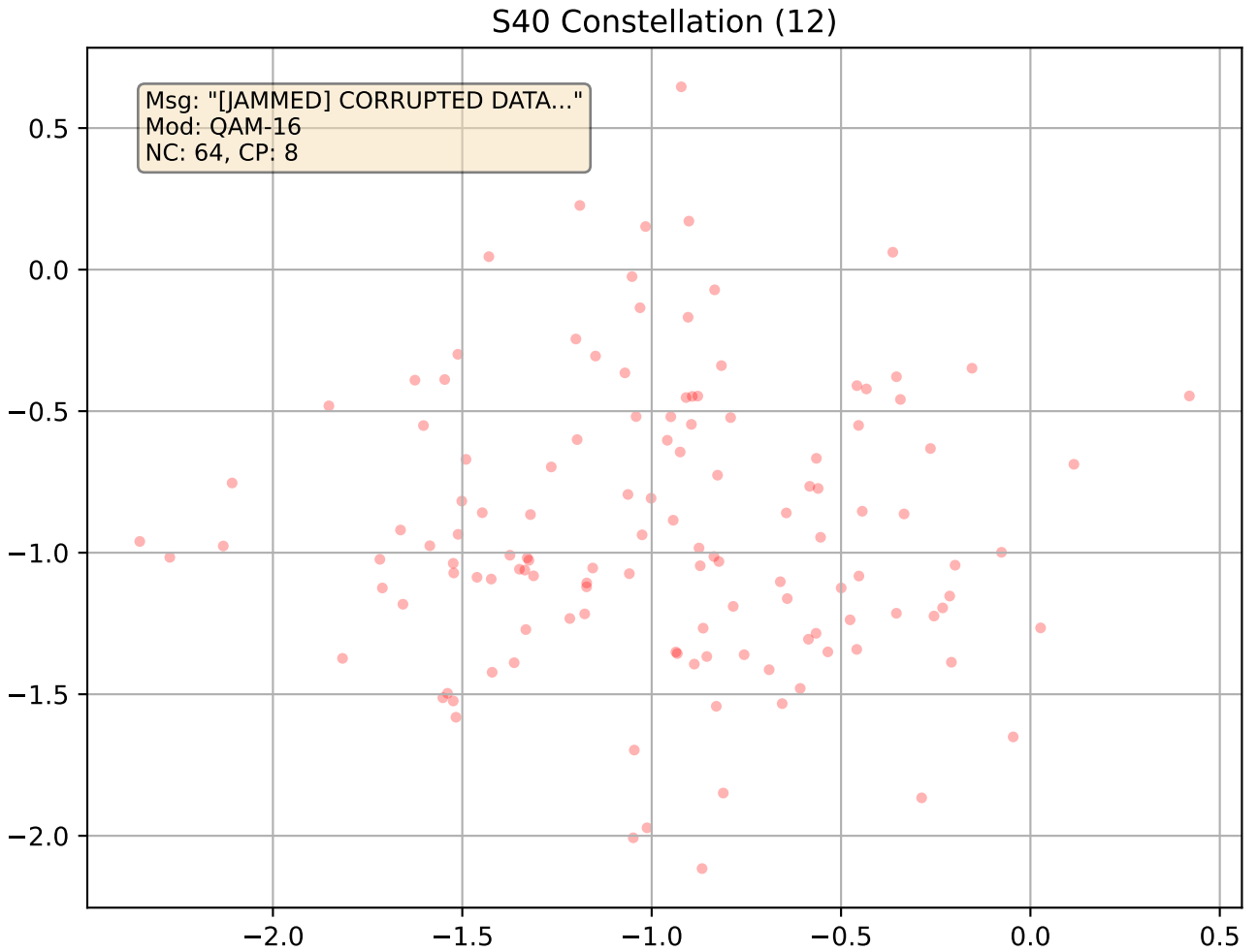

S40 OFDM (10)
I
Q
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
20
40
60
80
100
120
140

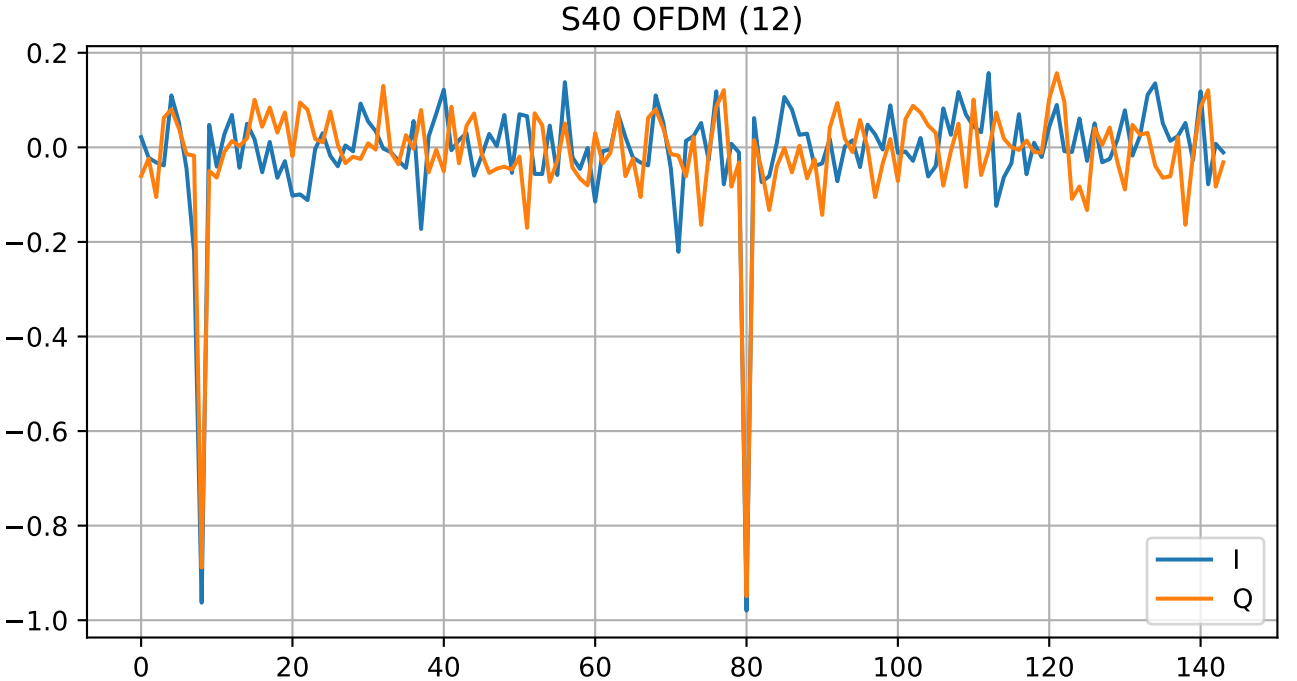

S40 Sensing (10)
Power
0.15
0.10
0.05
0.00
0
20
40
60
80
100
120
140

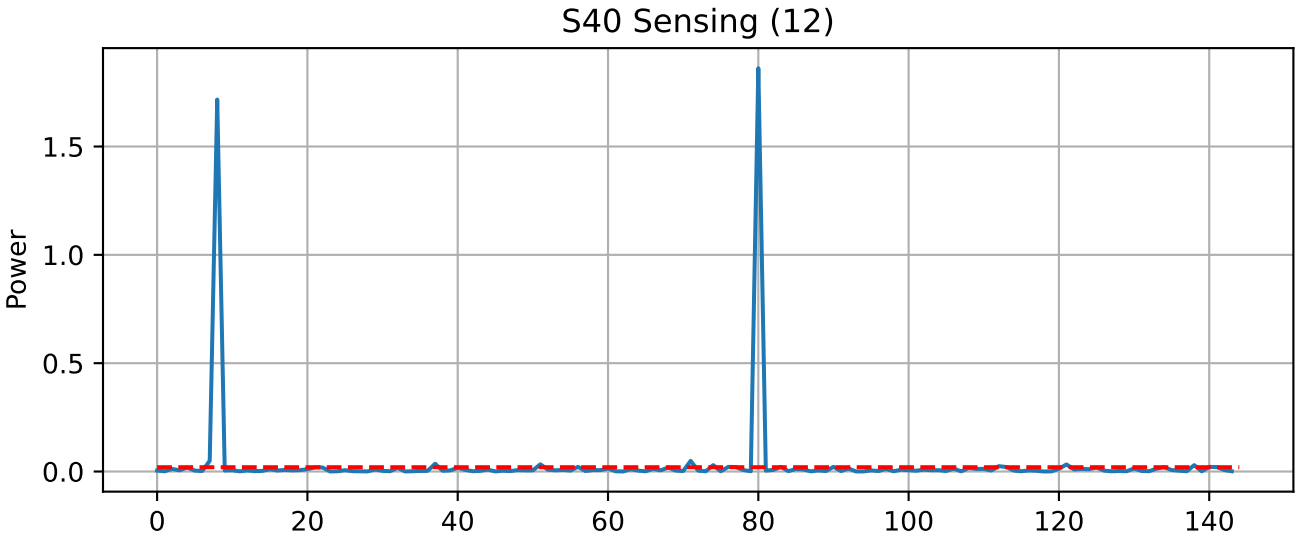

S40 Constellation (11)
Msg: "i cant get you msg properly..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

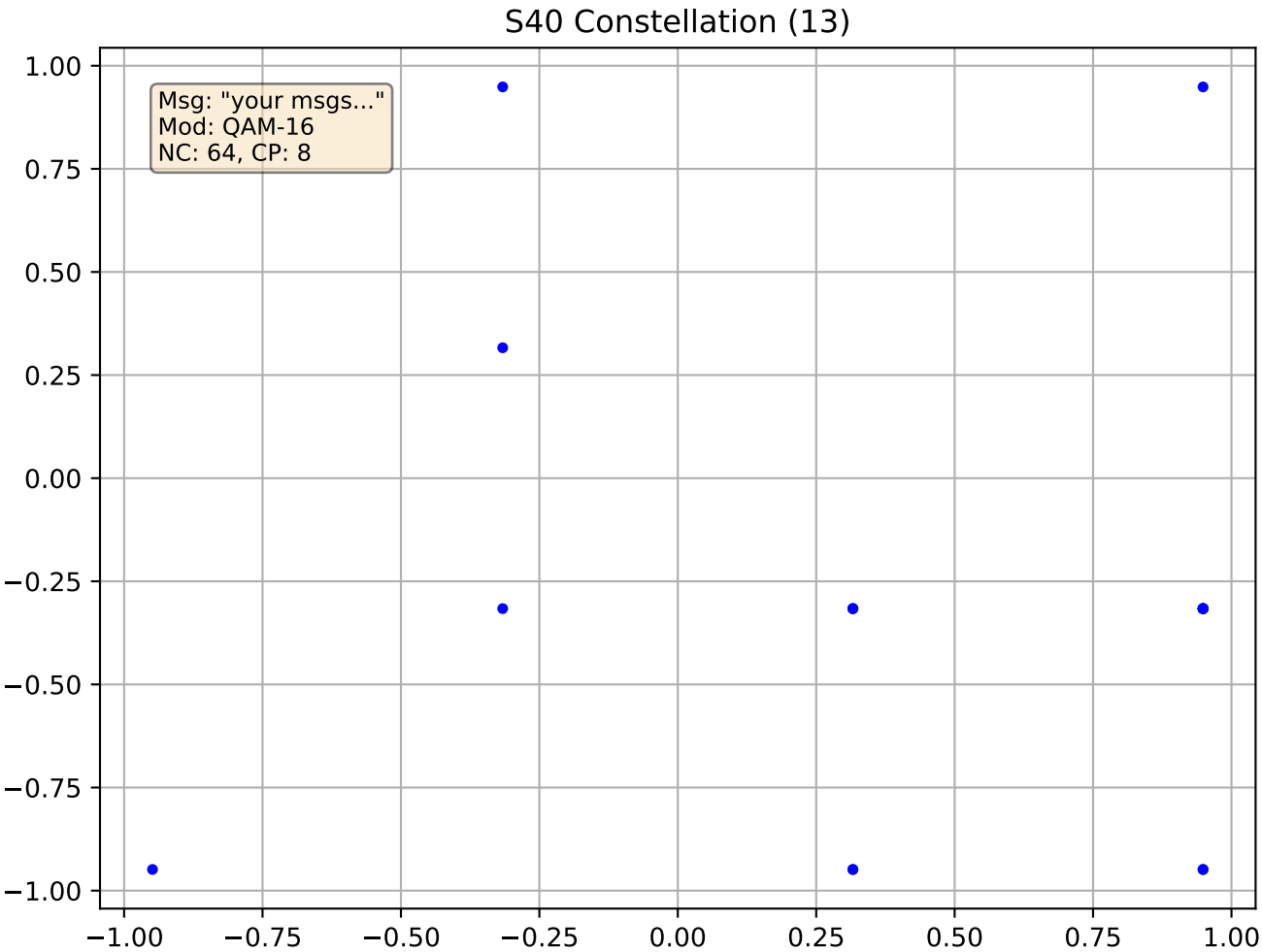

S40 OFDM (11)
I
Q
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
10
20
30
40
50
60
70

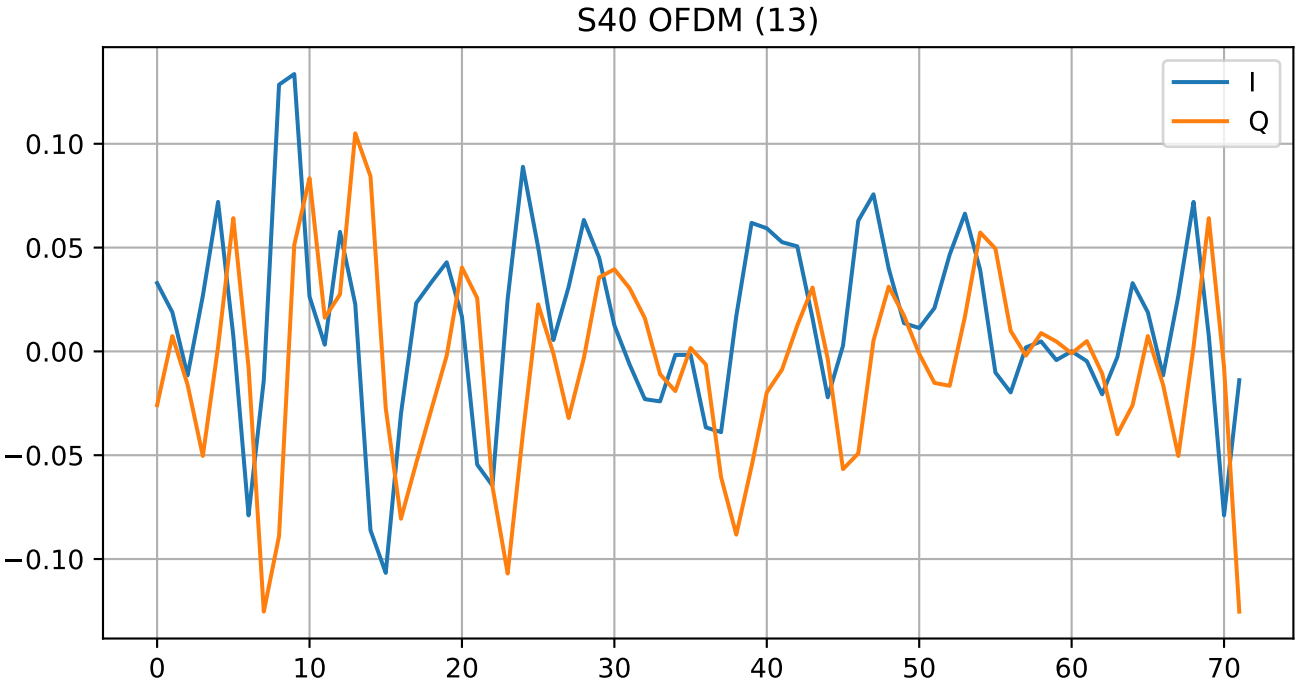

S40 Sensing (11)
Power
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0
10
20
30
40
50
60
70

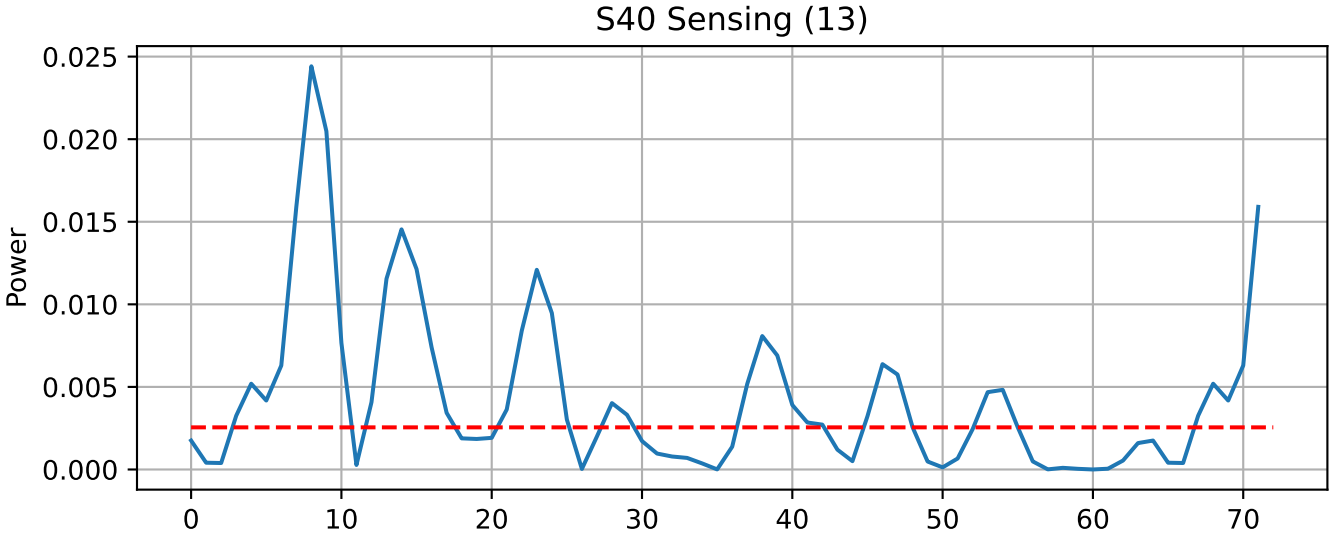

S40 Constellation (12)
Msg: "[JAMMED] CORRUPTED DATA..."
Mod: QAM-16
NC: 64, CP: 8
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5

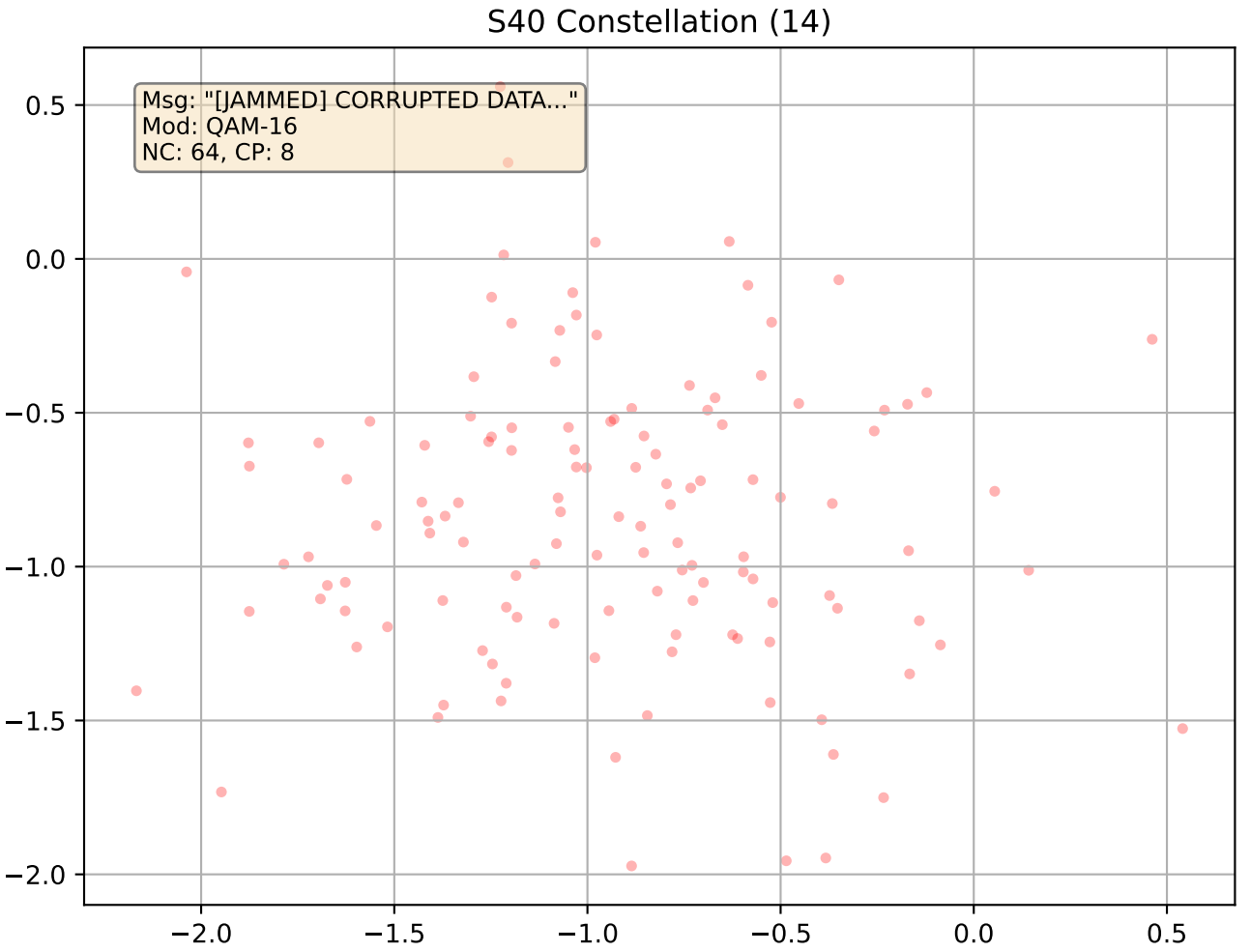

S40 OFDM (12)
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
20
40
60
80
100
120
140
I
Q

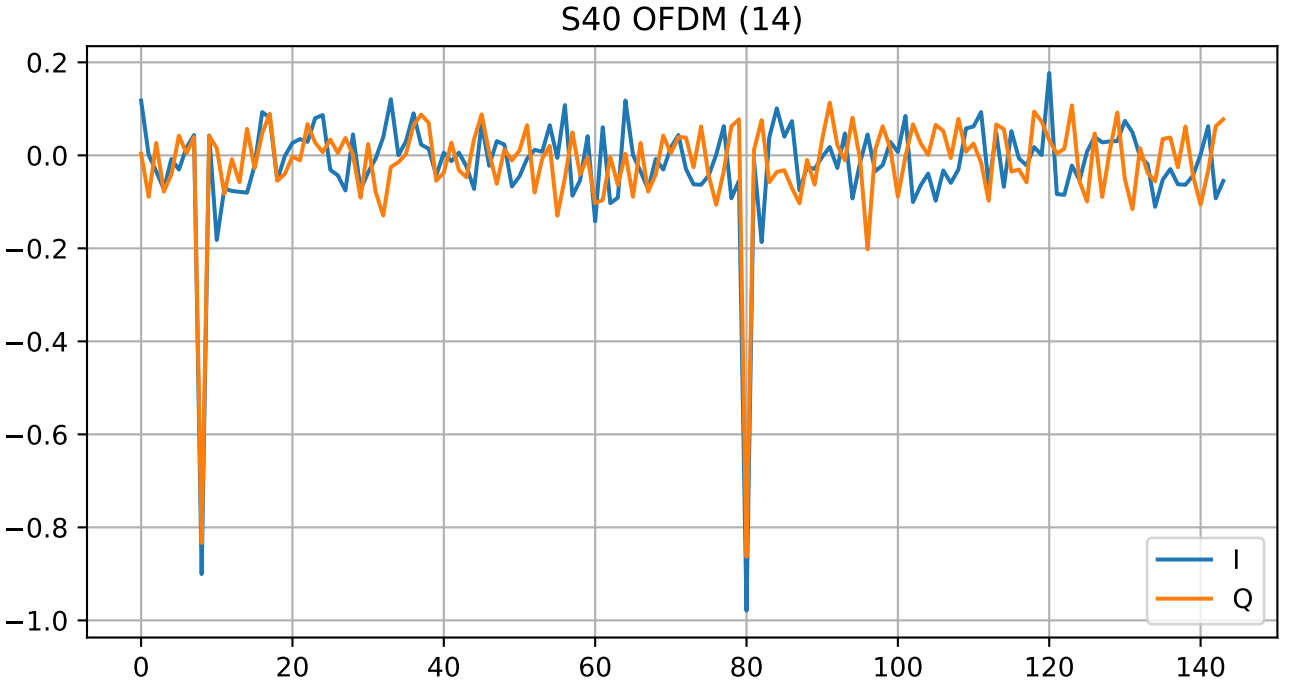

S40 Sensing (12)
Power
1.5
1.0
0.5
0.0
0
20
40
60
80
100
120
140

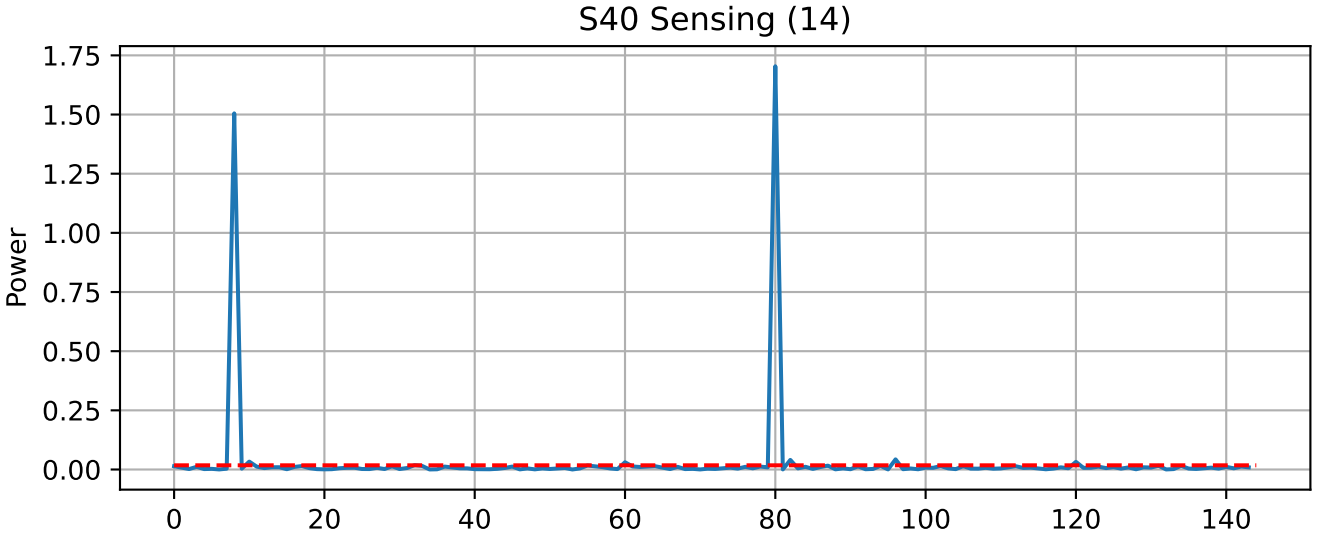

S40 Constellation (13)
Msg: "your msgs..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

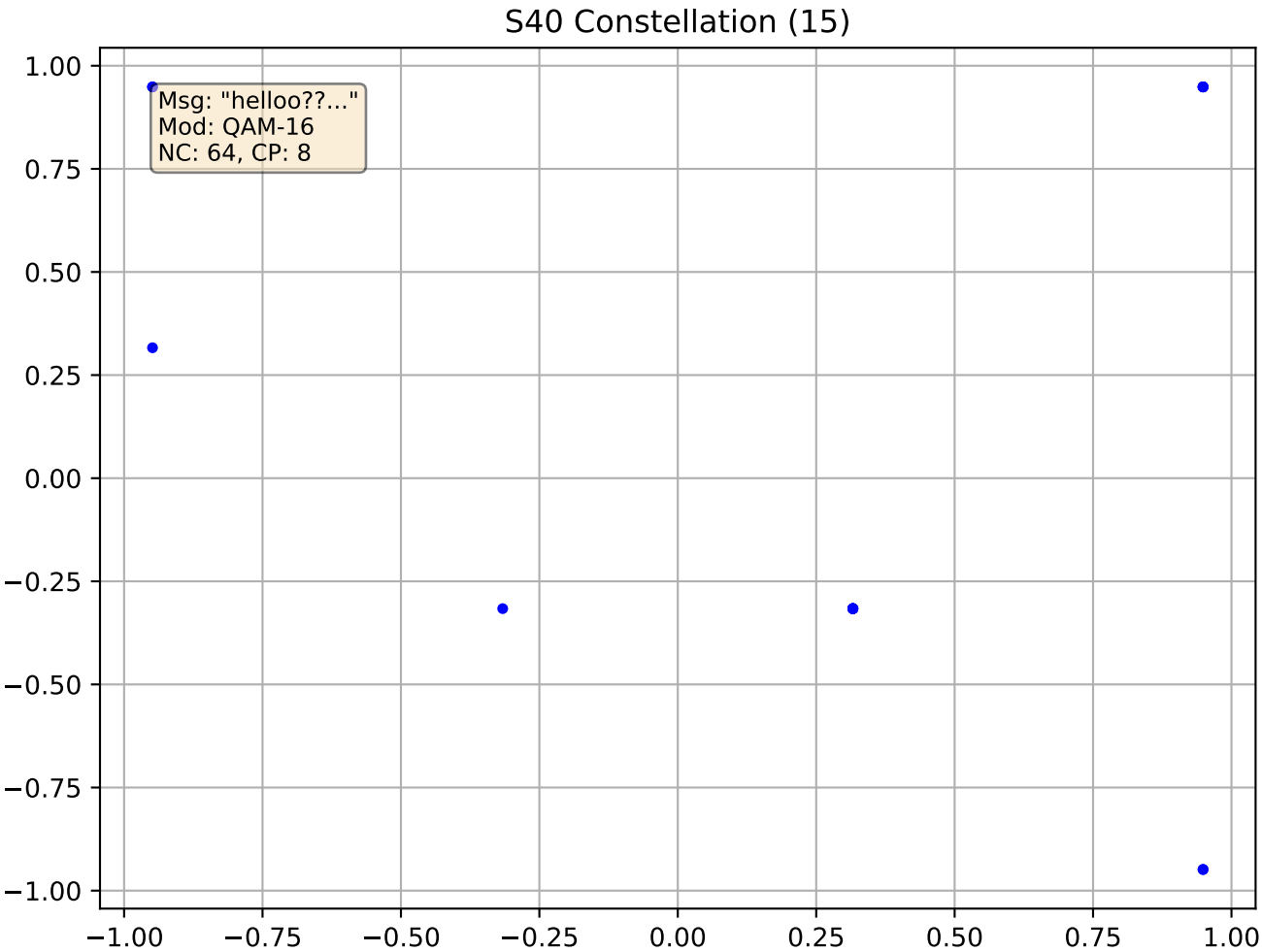

S40 OFDM (13)
I
Q
0.10
0.05
0.00
−0.05
−0.10
0
10
20
30
40
50
60
70

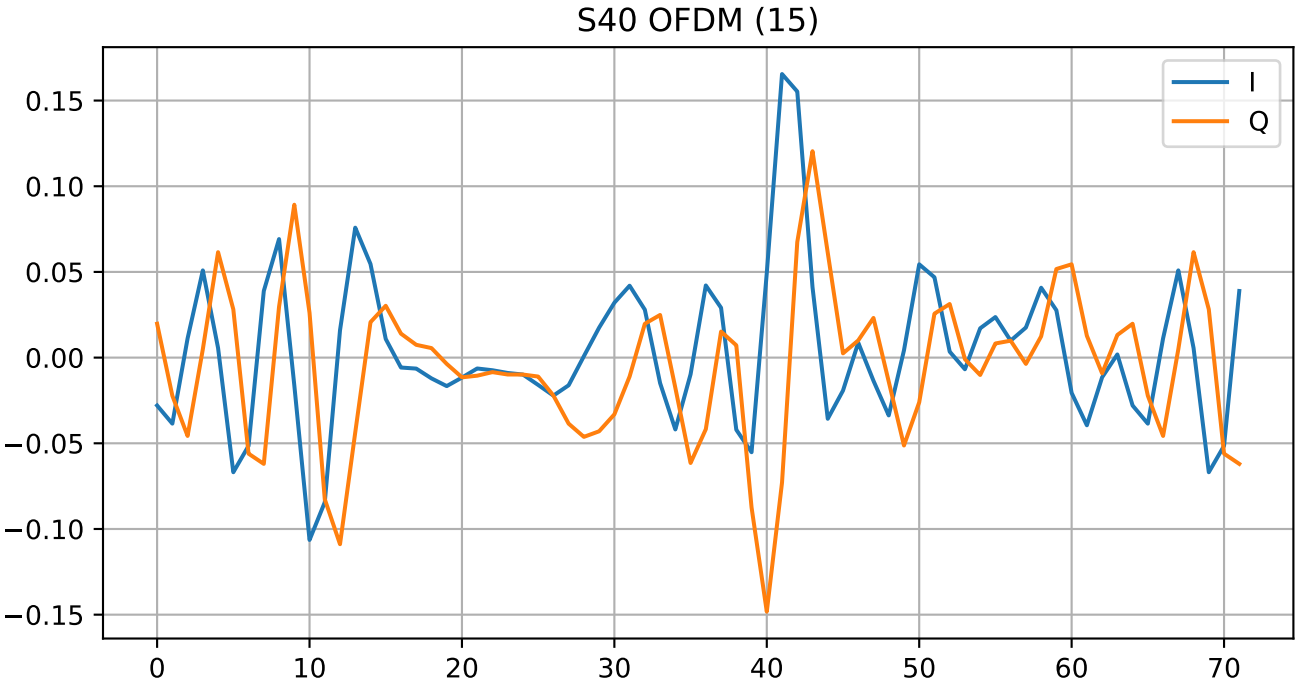

S40 Sensing (13)
Power
0.025
0.020
0.015
0.010
0.005
0.000
0
10
20
30
40
50
60
70

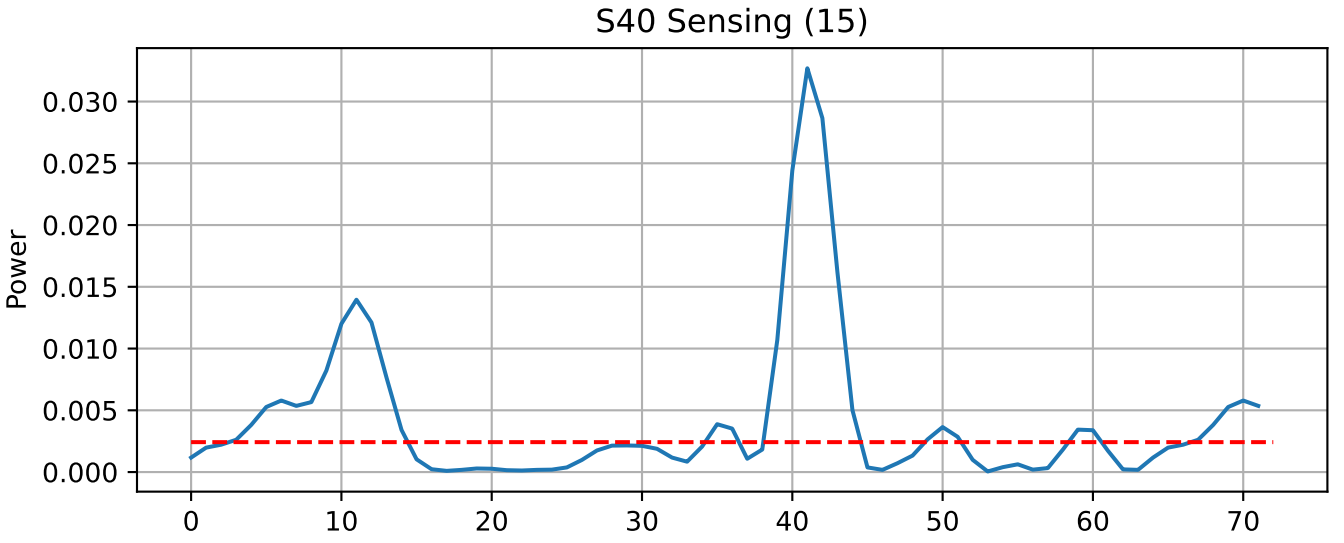

S40 Constellation (14)
Msg: "[JAMMED] CORRUPTED DATA..."
Mod: QAM-16
NC: 64, CP: 8
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5

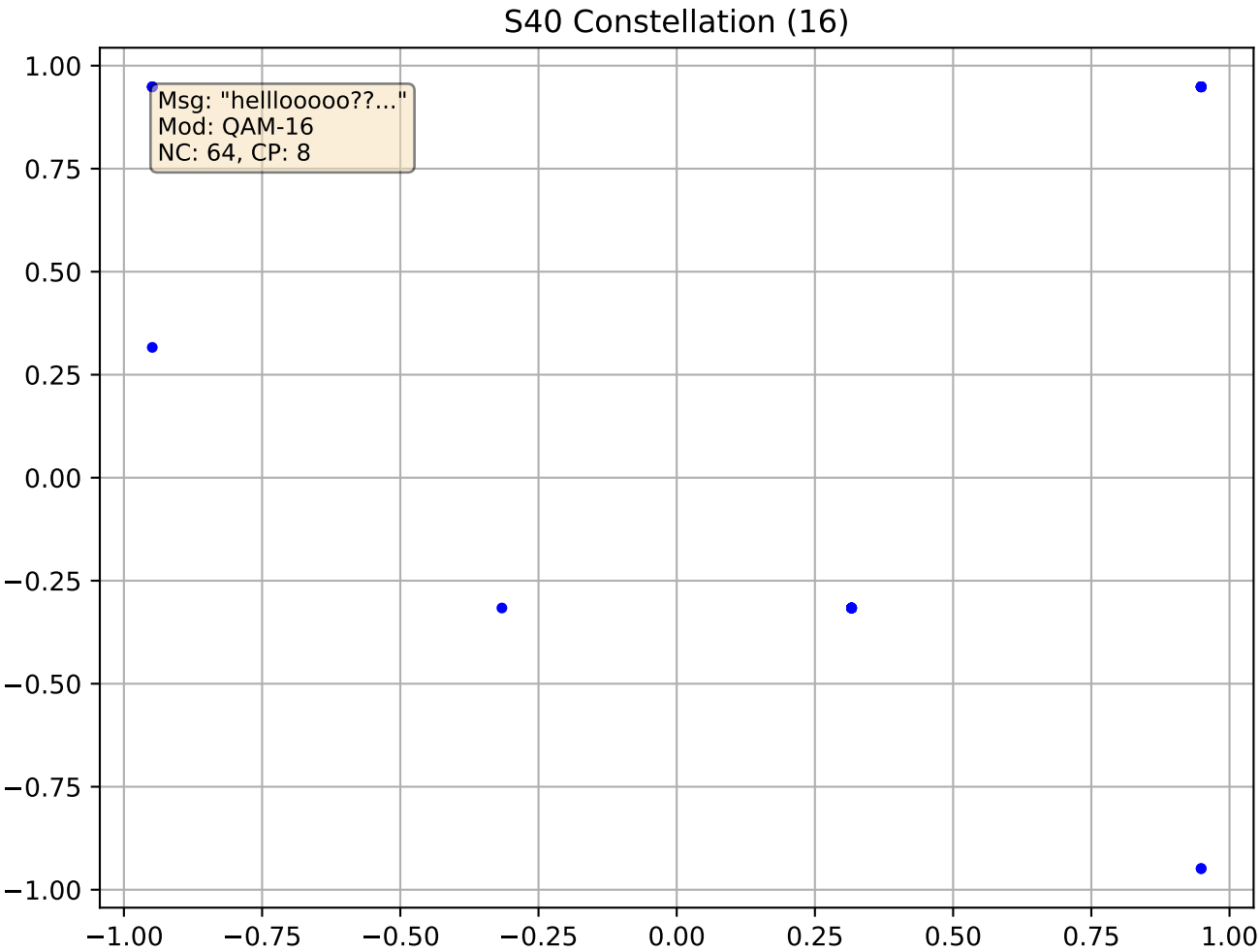

S40 OFDM (14)
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
20
40
60
80
100
120
140
I
Q

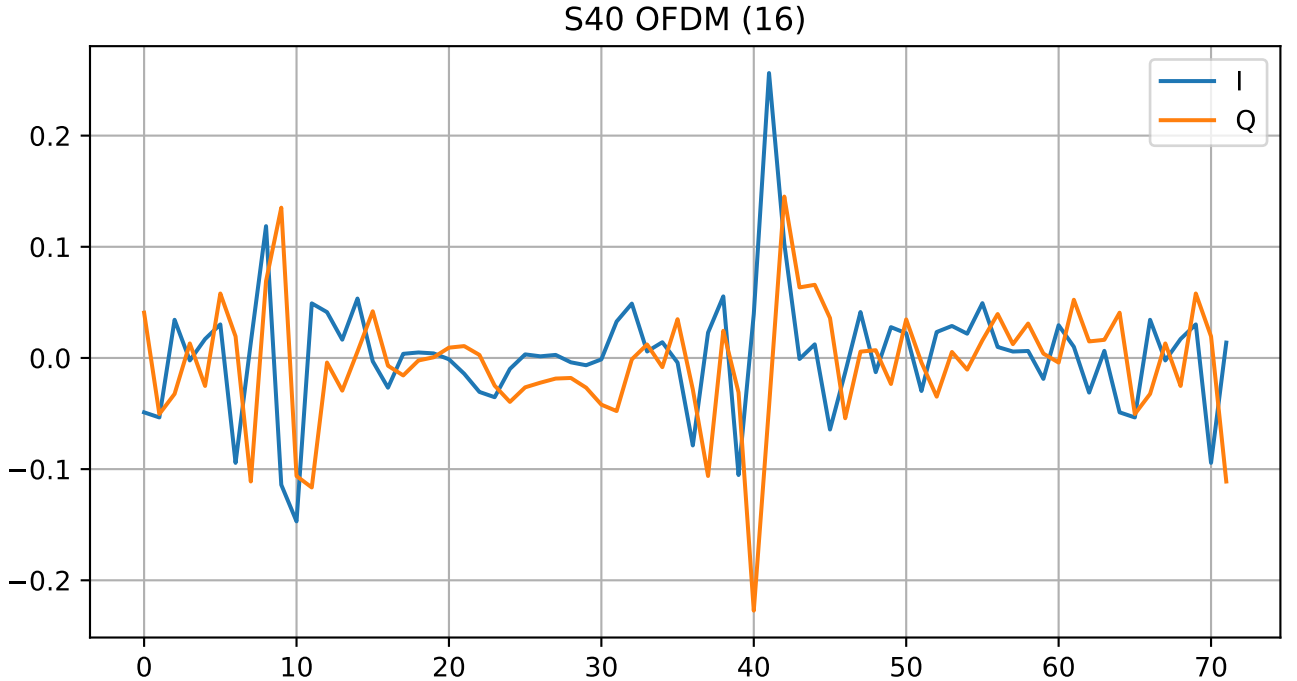

S40 Sensing (14)
Power
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
20
40
60
80
100
120
140

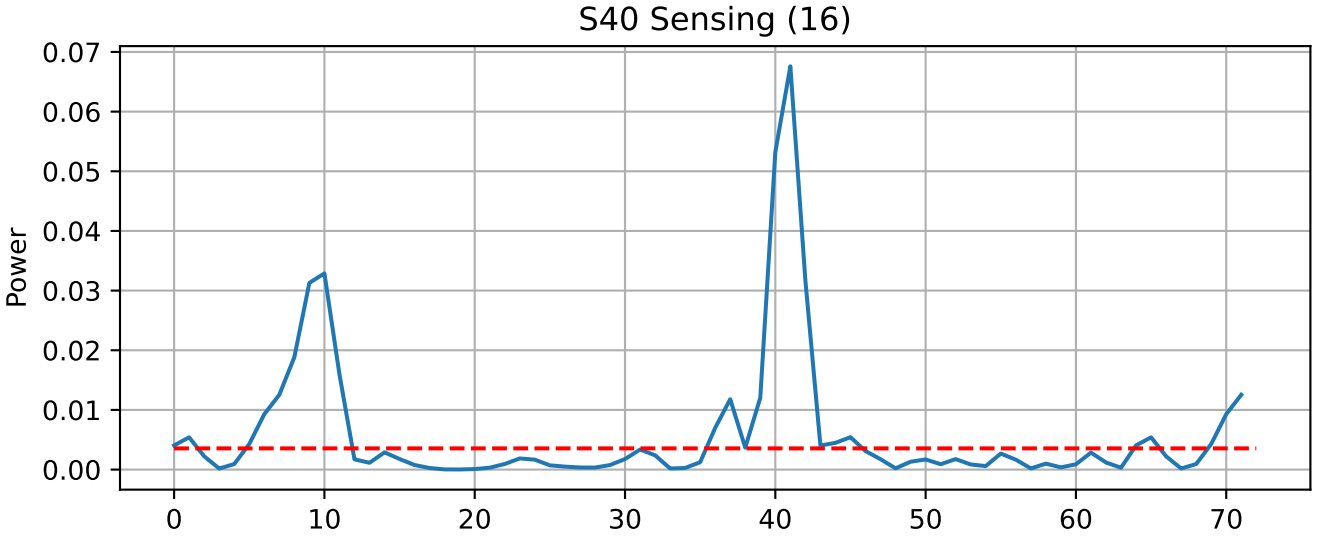

S40 Constellation (15)
Msg: "helloo??..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

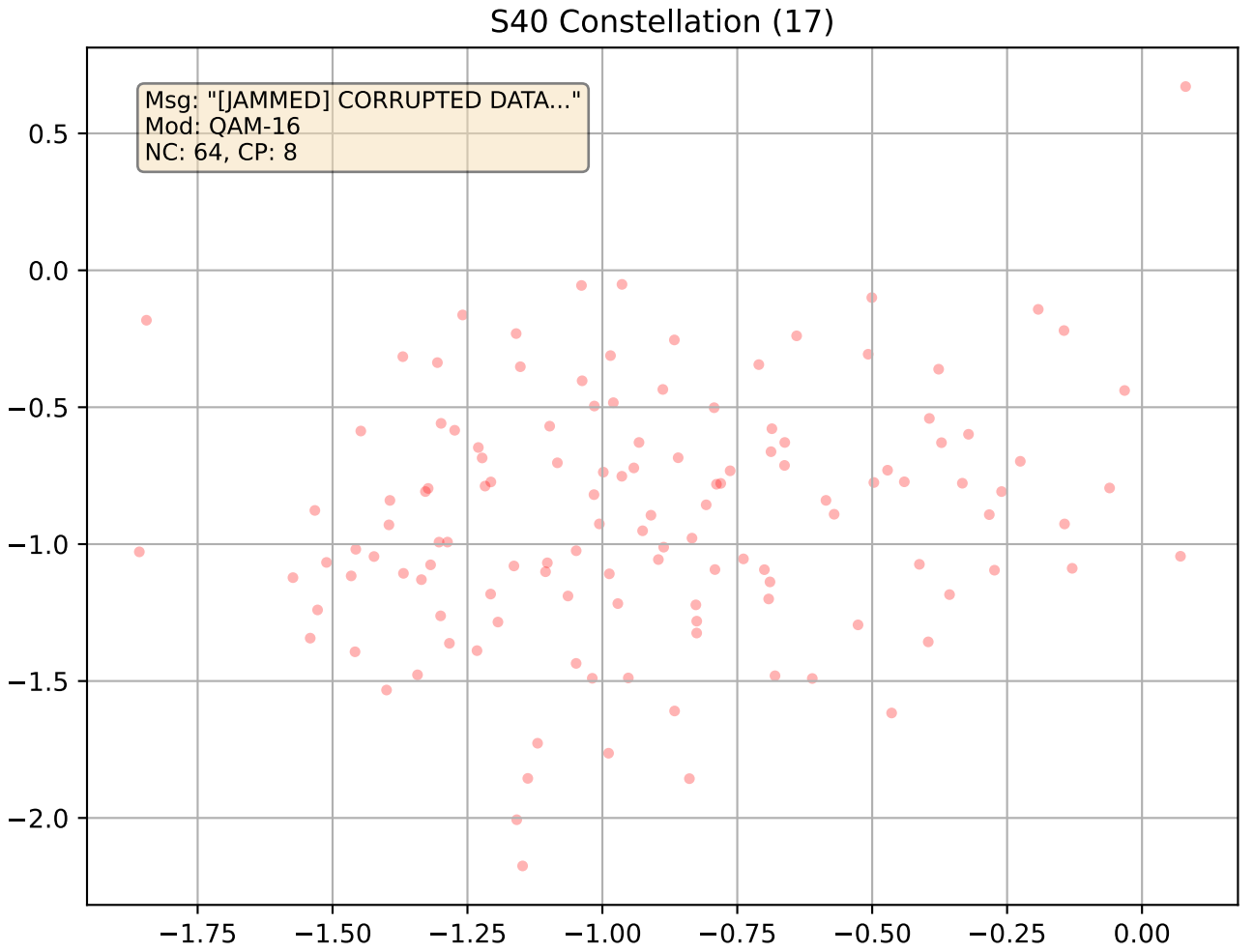

S40 OFDM (15)
I
Q
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
50
60
70

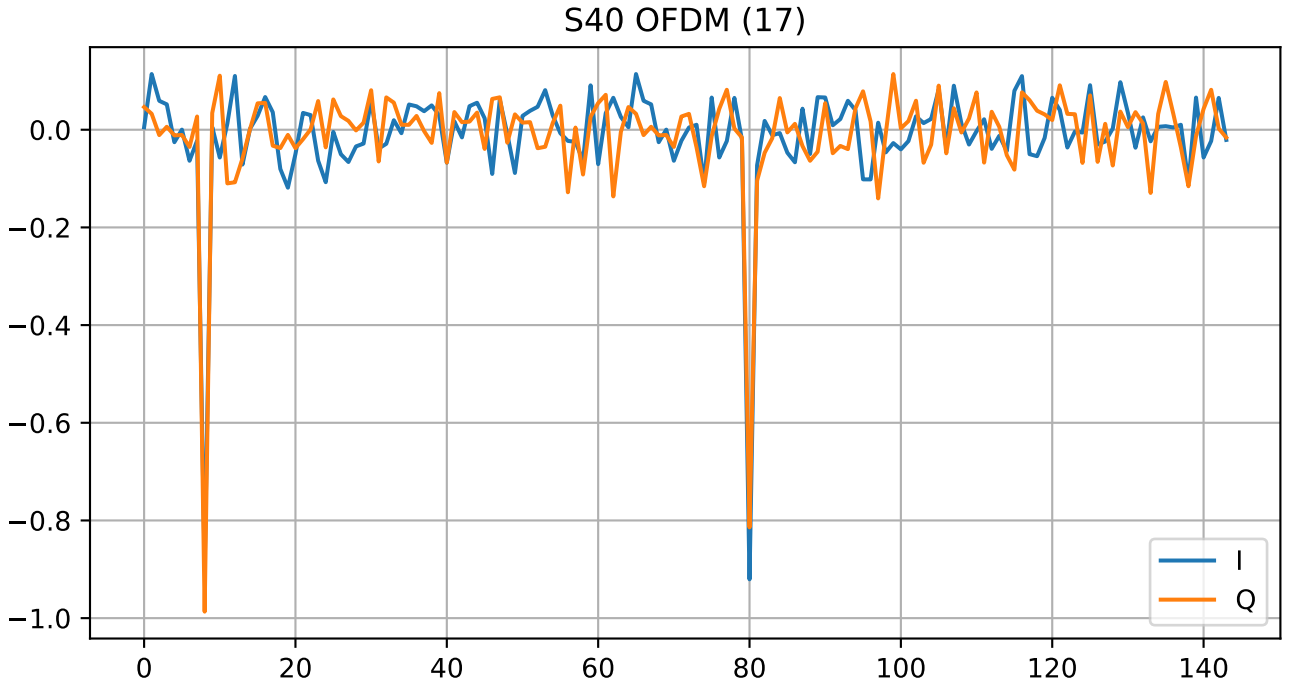

S40 Sensing (15)
Power
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0
10
20
30
40
50
60
70

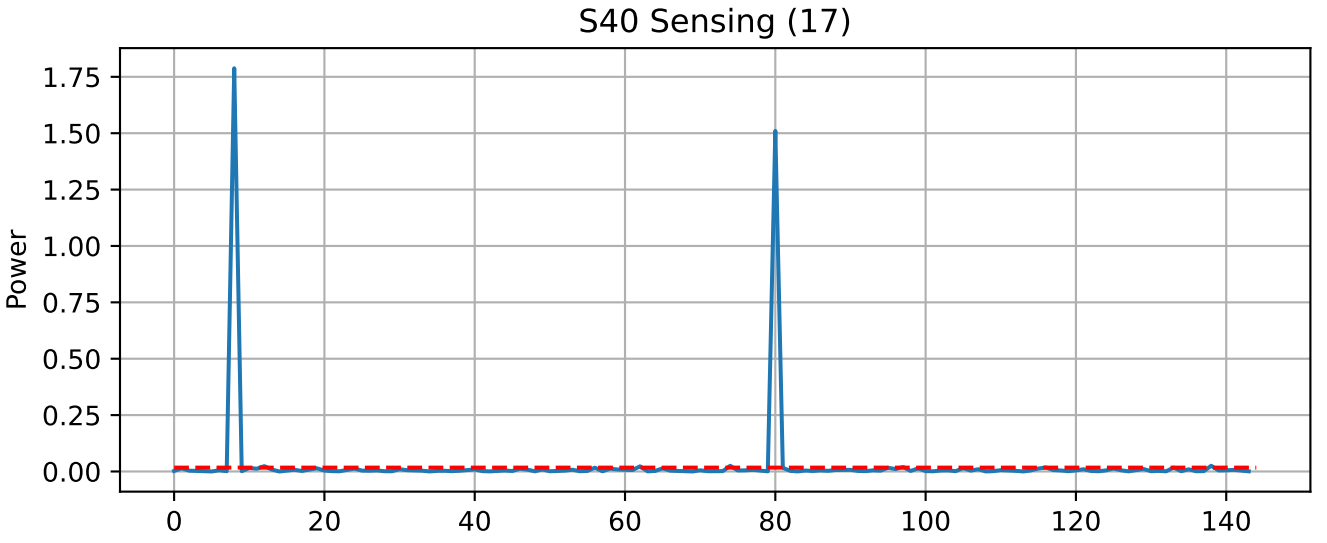

S40 Constellation (16)
Msg: "hellloooo??..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

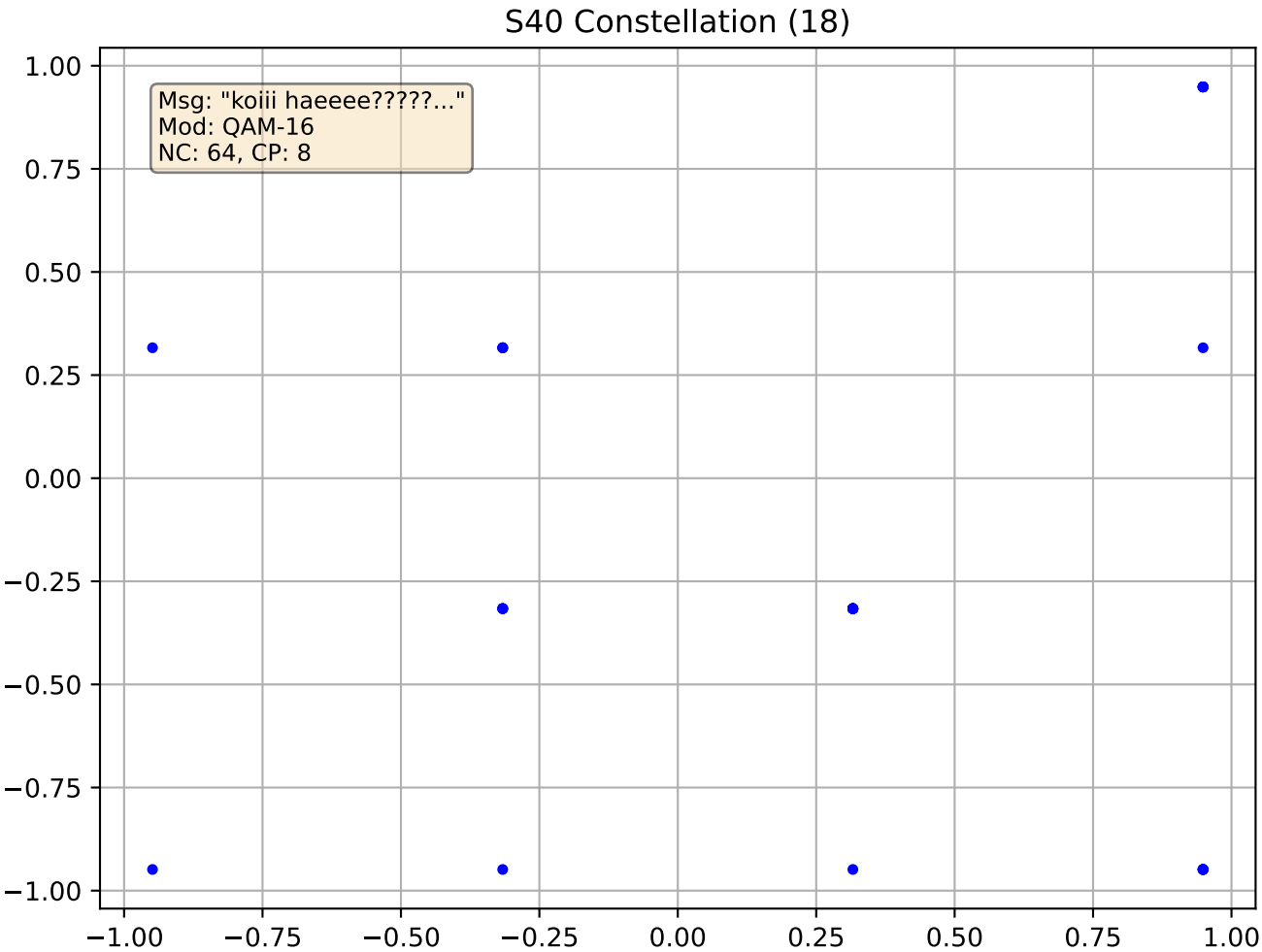

S40 OFDM (16)
I
Q
0.2
0.1
0.0
−0.1
−0.2
0
10
20
30
40
50
60
70

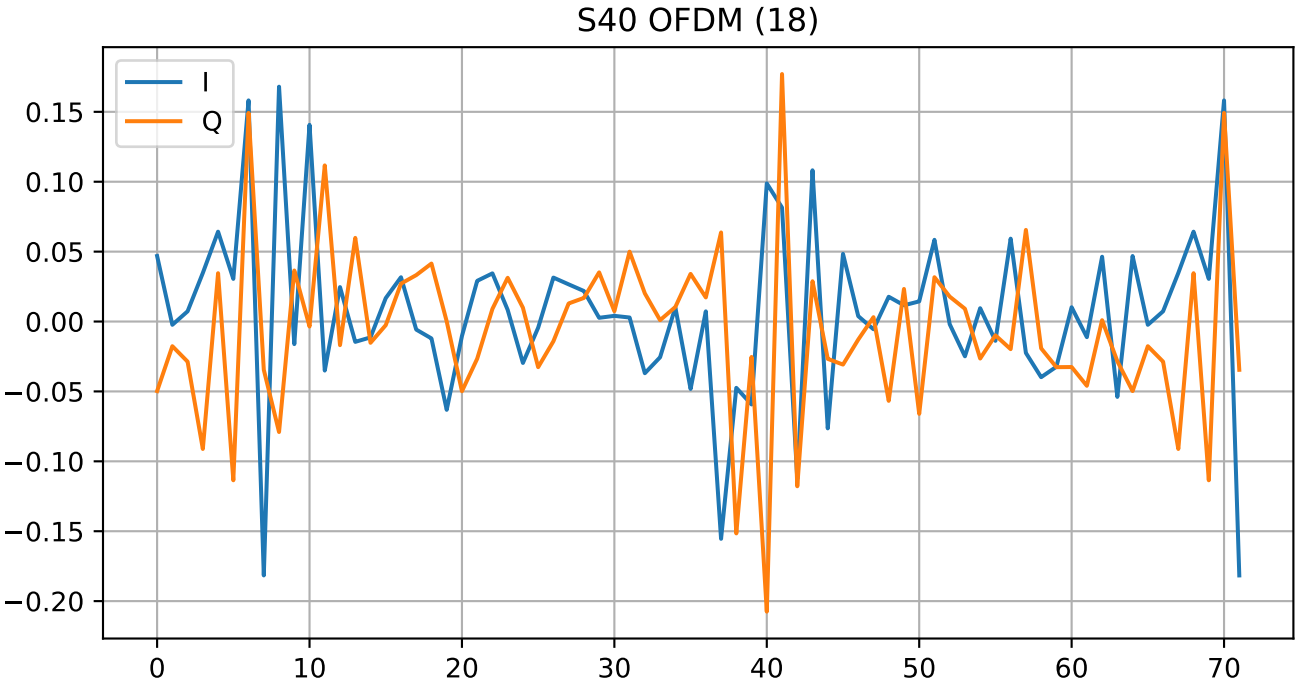

S40 Sensing (16)
Power
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0
10
20
30
40
50
60
70

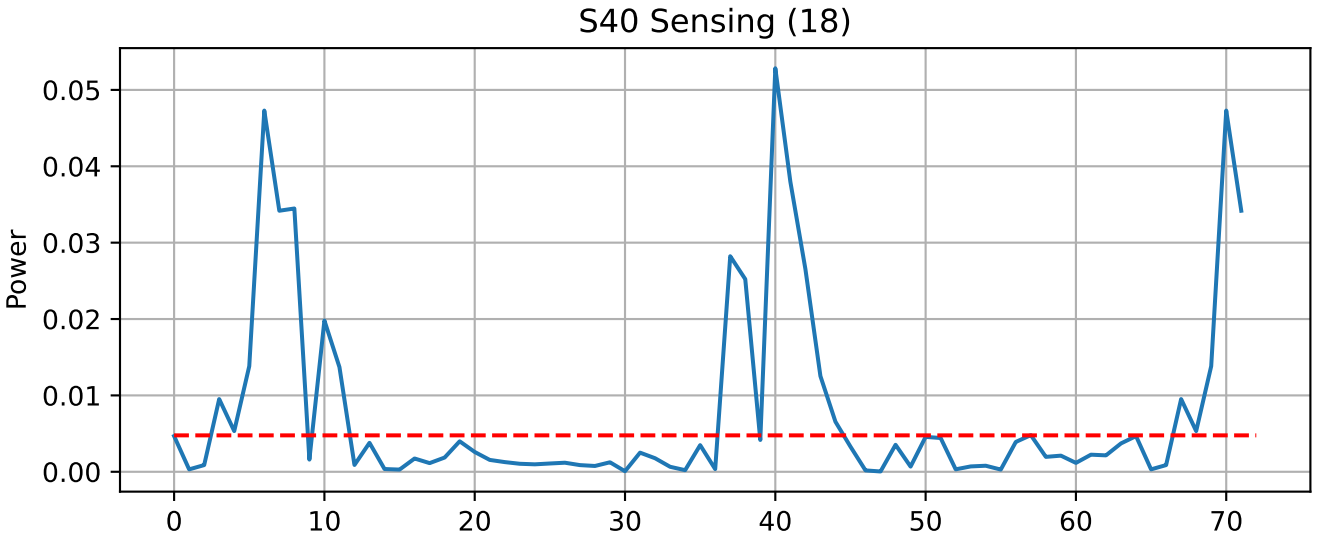

S40 Constellation (17)
Msg: "[JAMMED] CORRUPTED DATA..."
Mod: QAM-16
NC: 64, CP: 8
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1.75
−1.50
−1.25
−1.00
−0.75
−0.50
−0.25
0.00

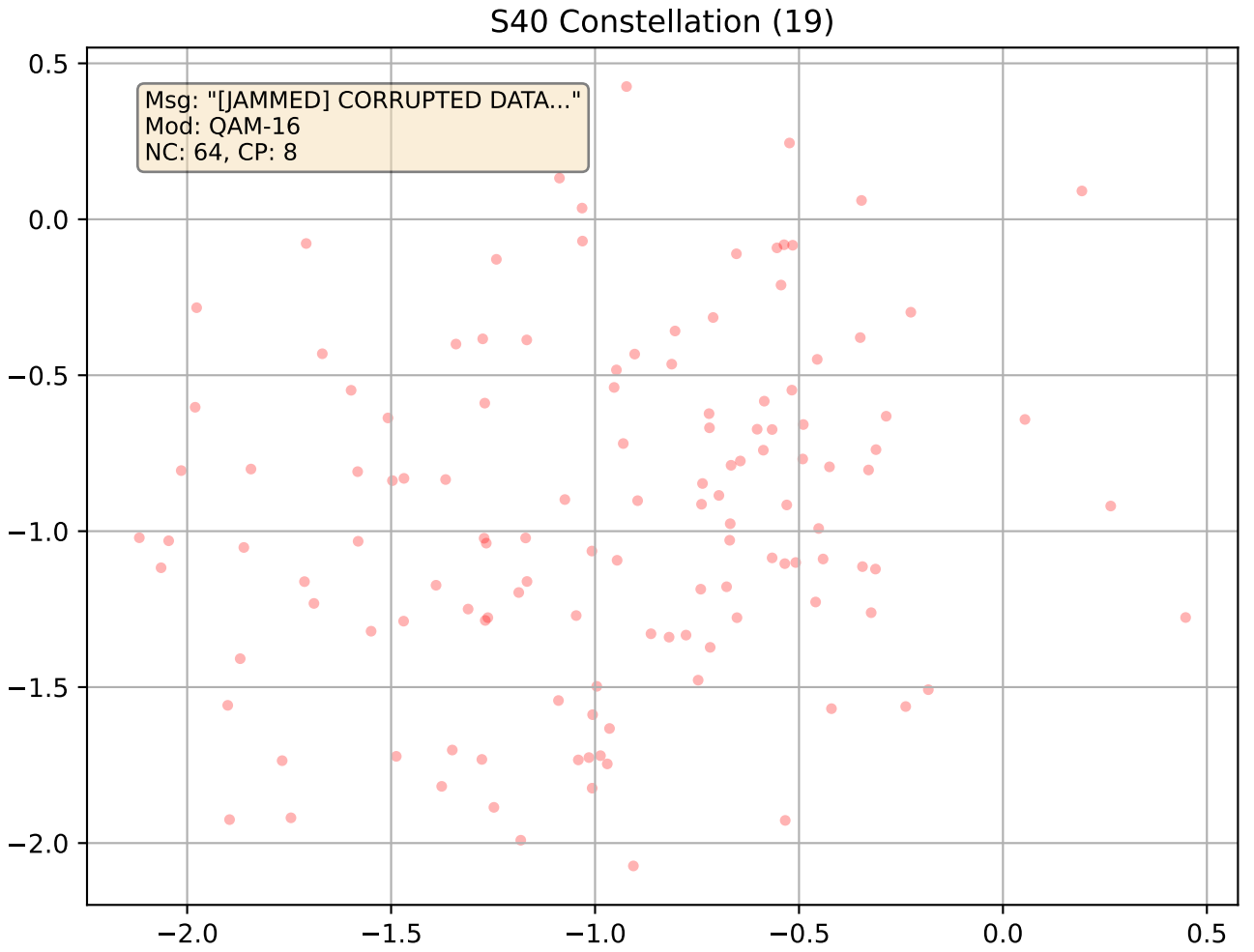

S40 OFDM (17)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
20
40
60
80
100
120
140
I
Q

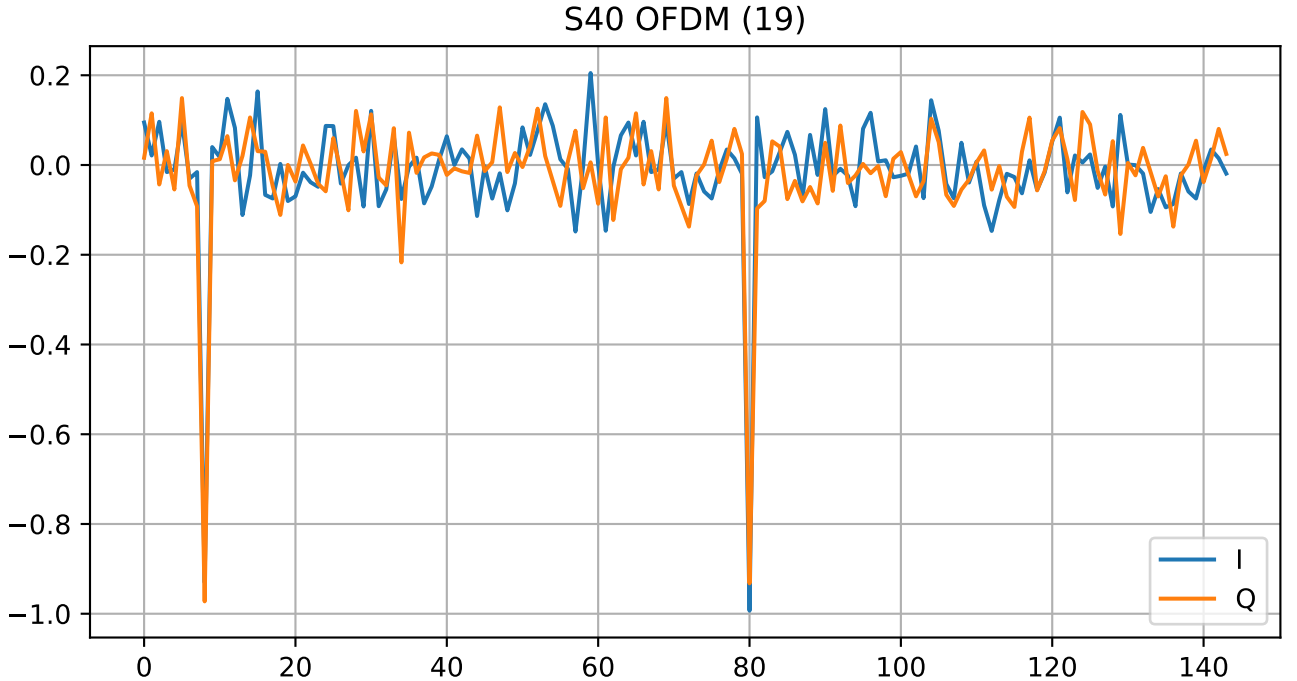

S40 Sensing (17)
Power
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
20
40
60
80
100
120
140

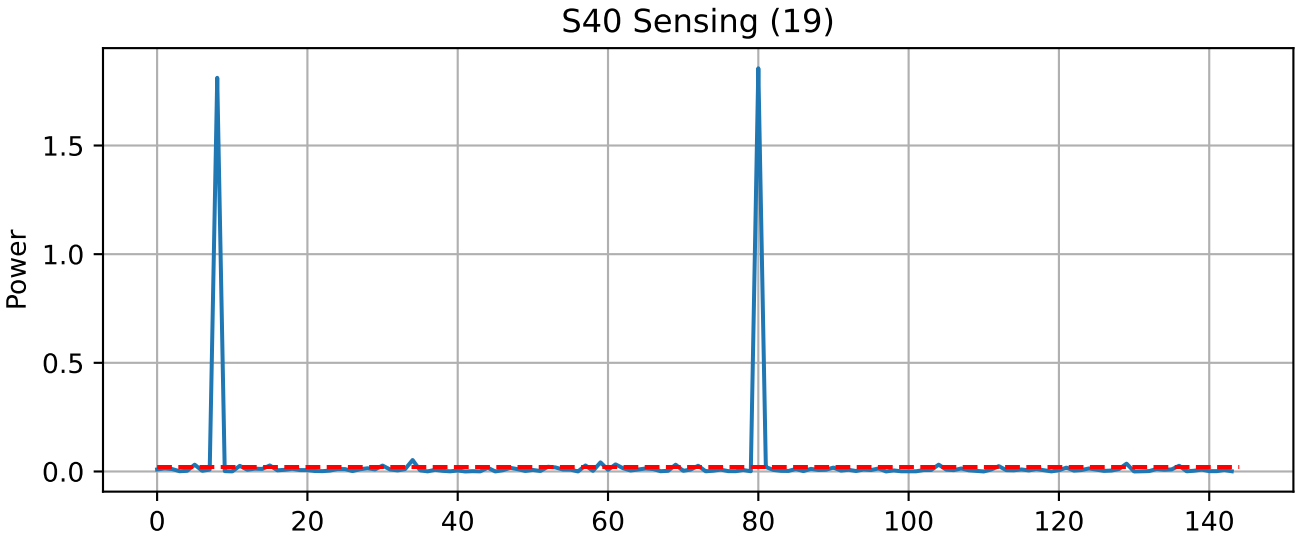

S40 Constellation (18)
Msg: "koiii haeeee?????..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

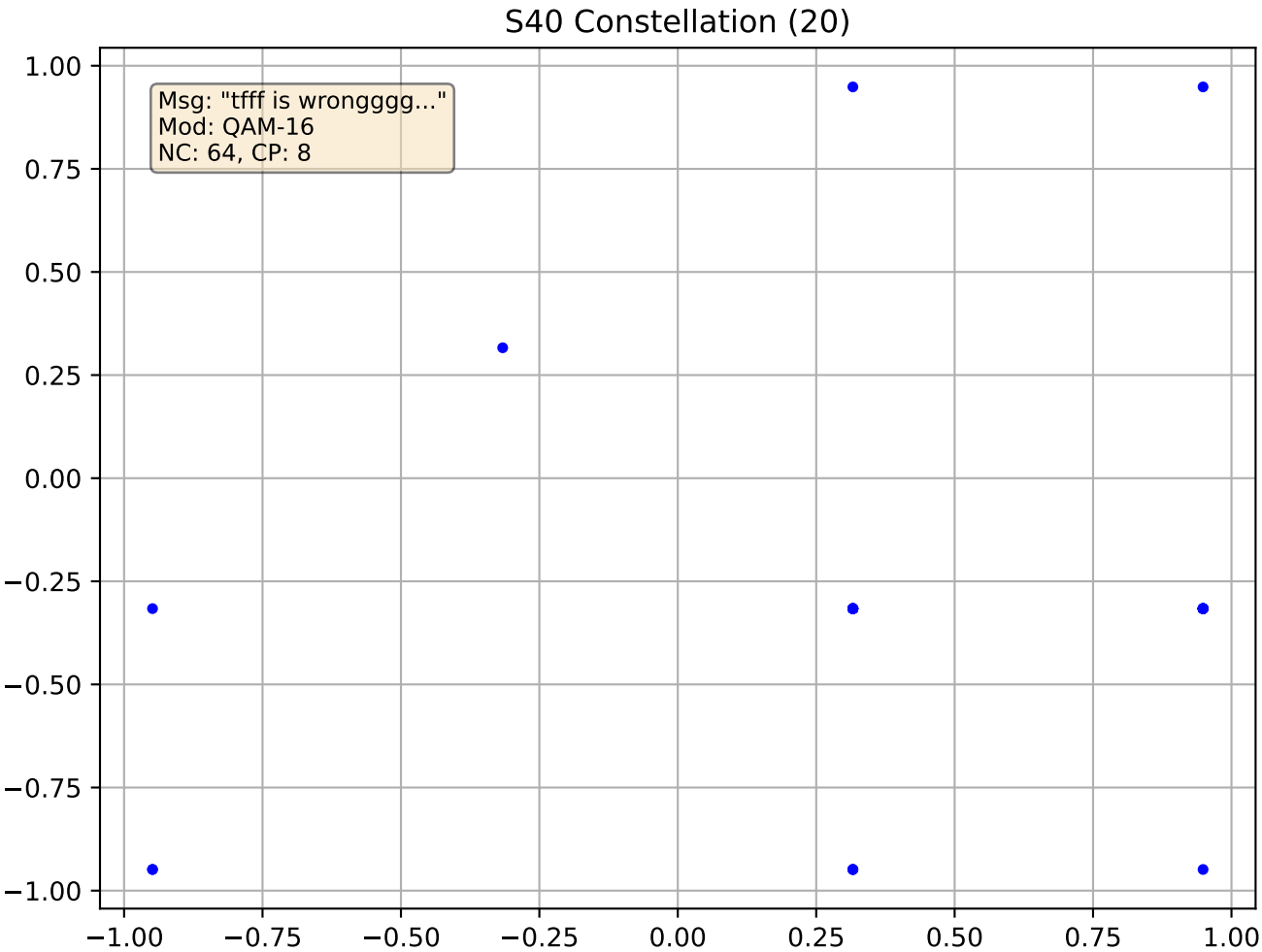

S40 OFDM (18)
I
Q
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
0
10
20
30
40
50
60
70

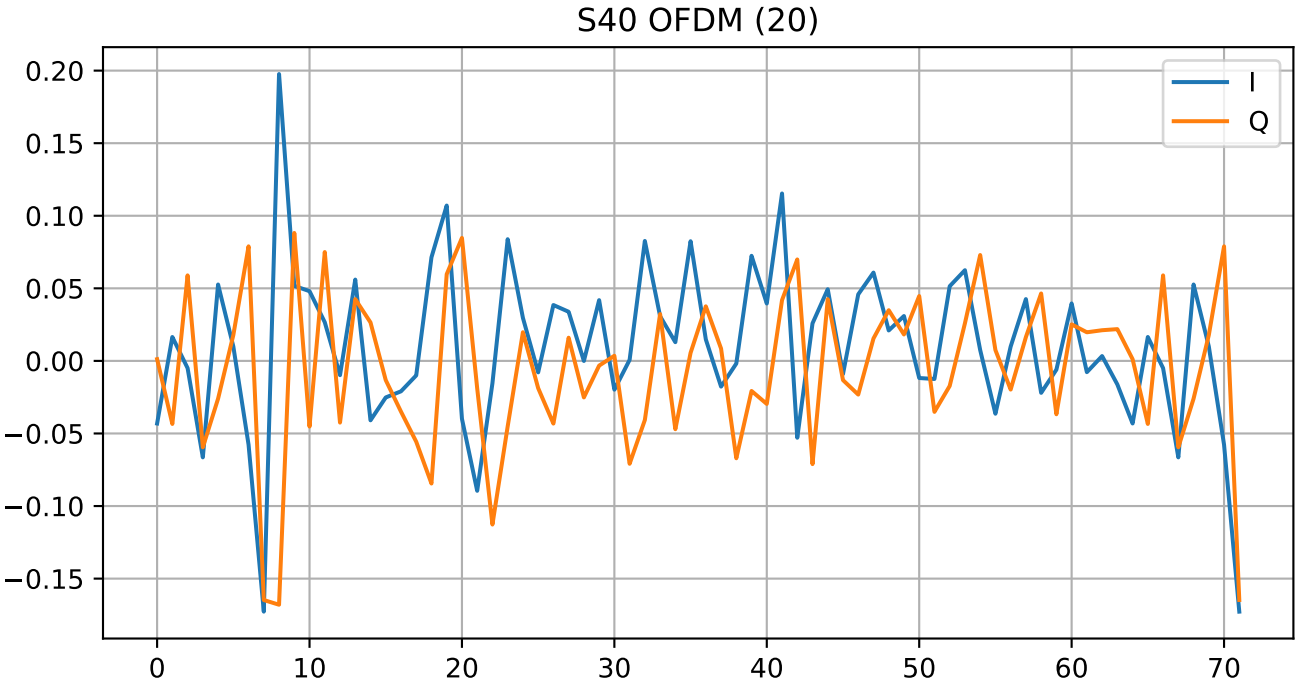

S40 Sensing (18)
Power
0.05
0.04
0.03
0.02
0.01
0.00
0
10
20
30
40
50
60
70

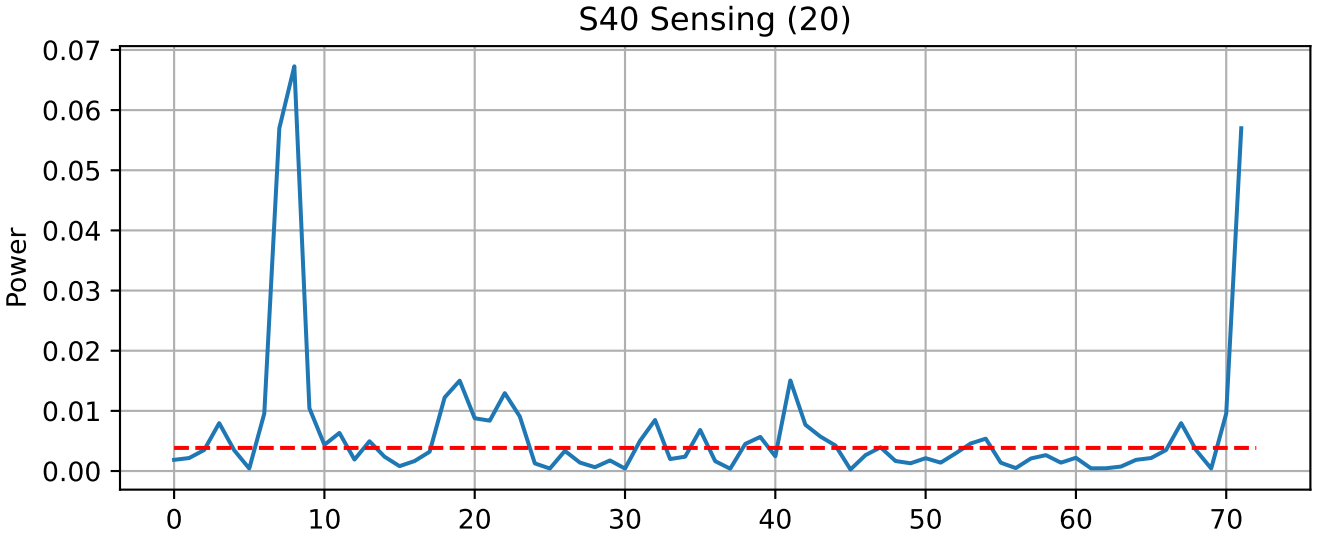

S40 Constellation (19)
Msg: "[JAMMED] CORRUPTED DATA..."
Mod: QAM-16
NC: 64, CP: 8
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5

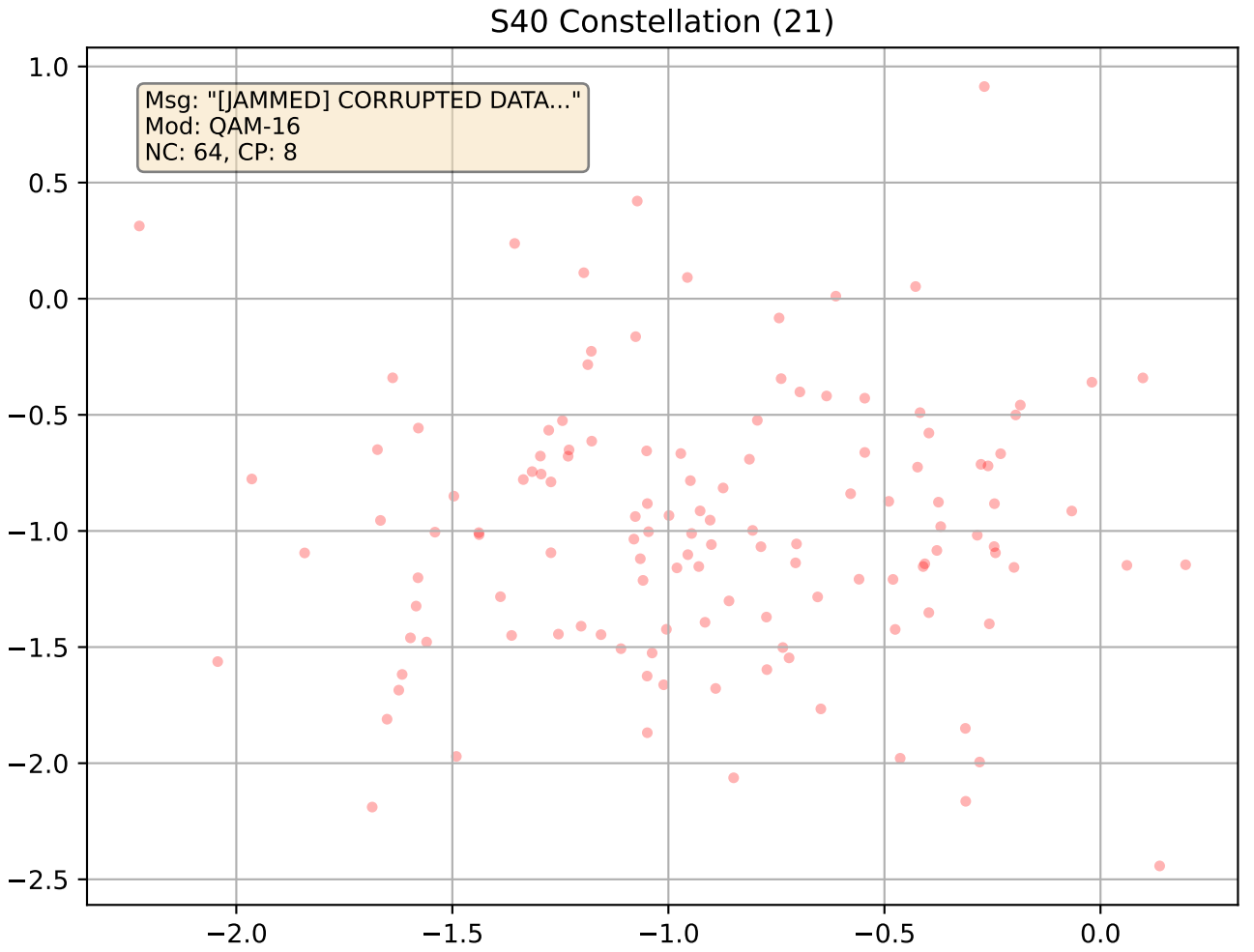

S40 OFDM (19)
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
20
40
60
80
100
120
140
I
Q

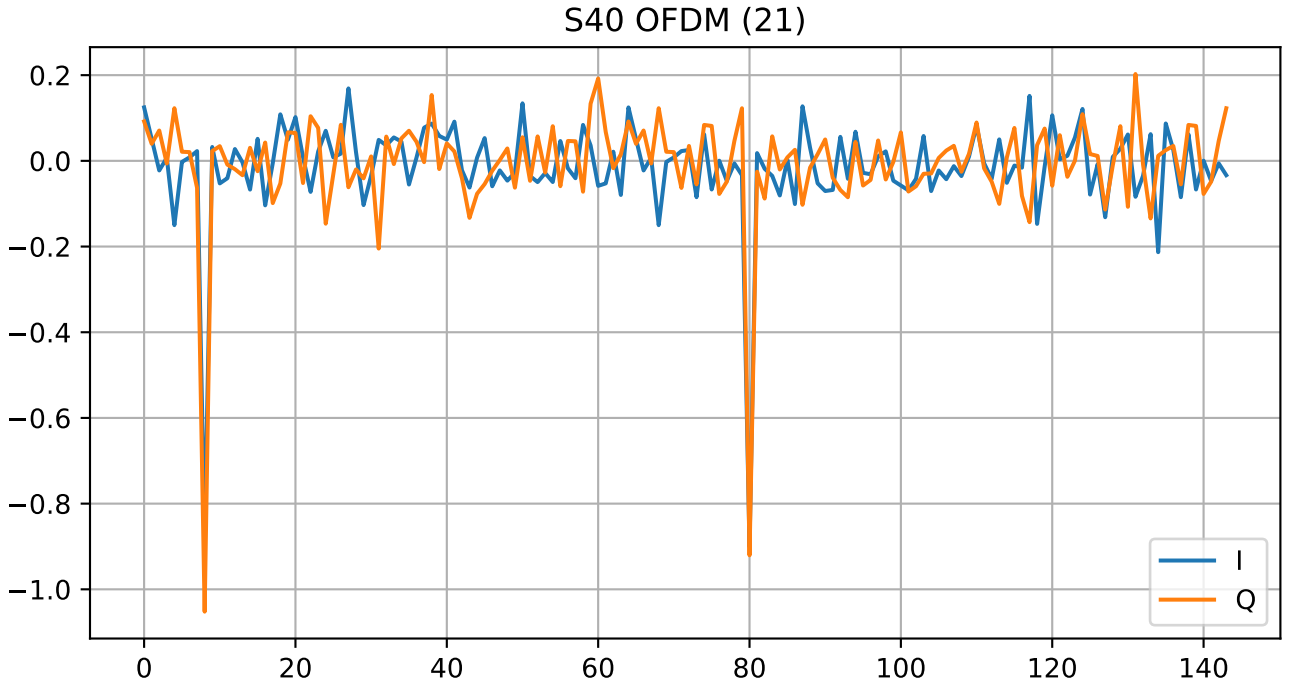

S40 Sensing (19)
Power
1.5
1.0
0.5
0.0
0
20
40
60
80
100
120
140

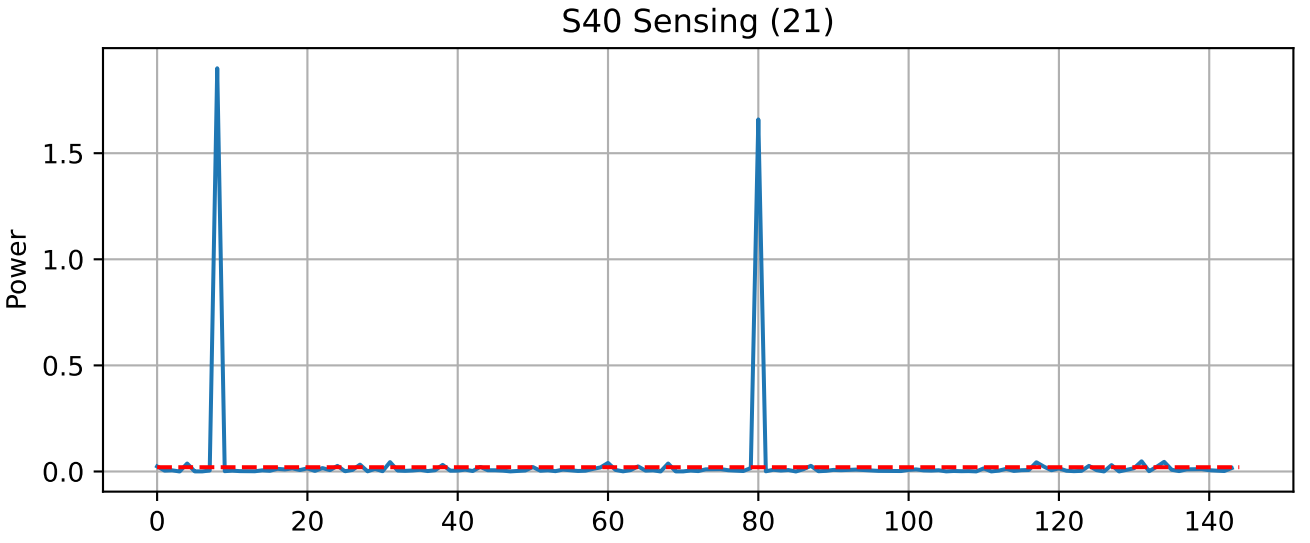

S40 Constellation (20)
Msg: "tfff is wronggg..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

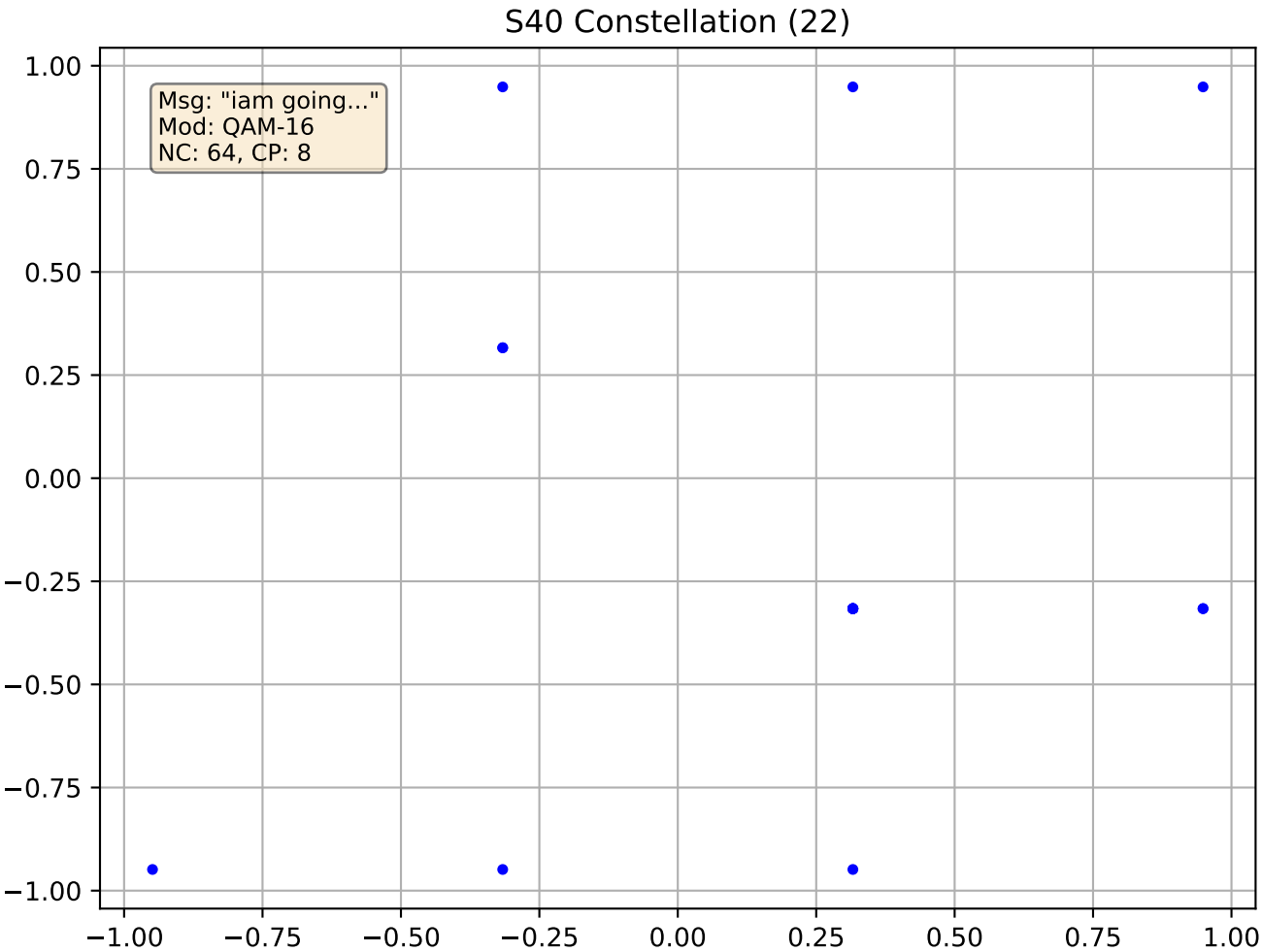

S40 OFDM (20)
I
Q
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
50
60
70

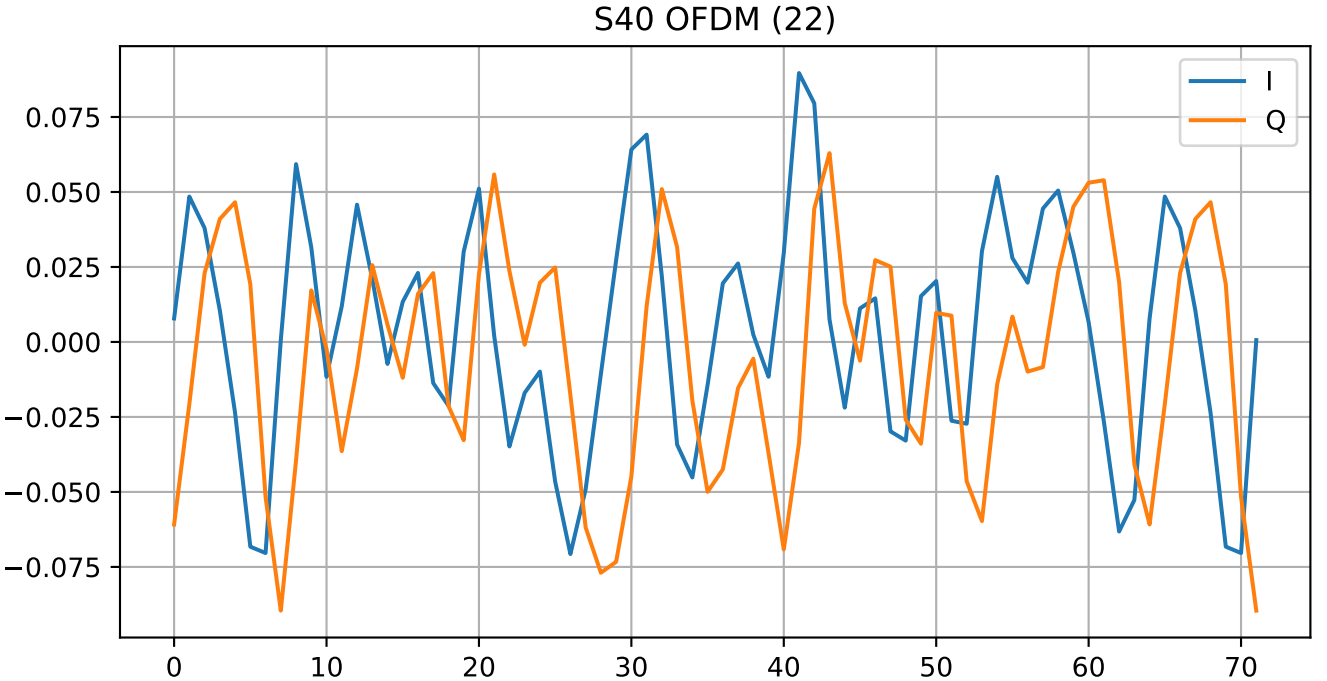

S40 Sensing (20)
Power
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0
10
20
30
40
50
60
70

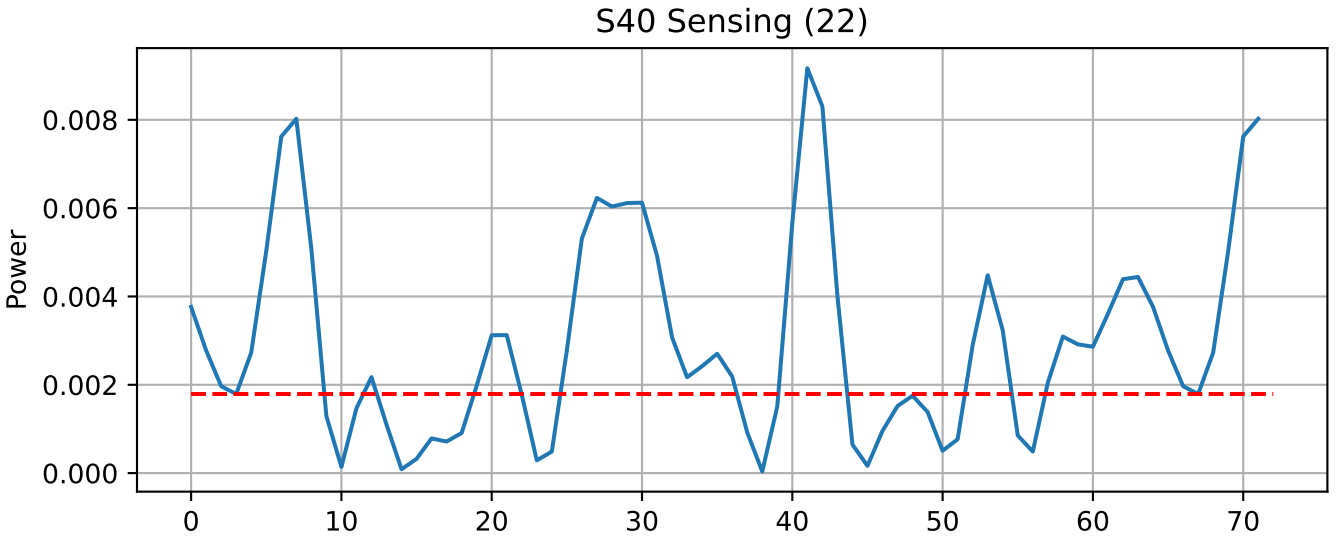

S40 Constellation (21)
Msg: "[JAMMED] CORRUPTED DATA..."
Mod: QAM-16
NC: 64, CP: 8
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0

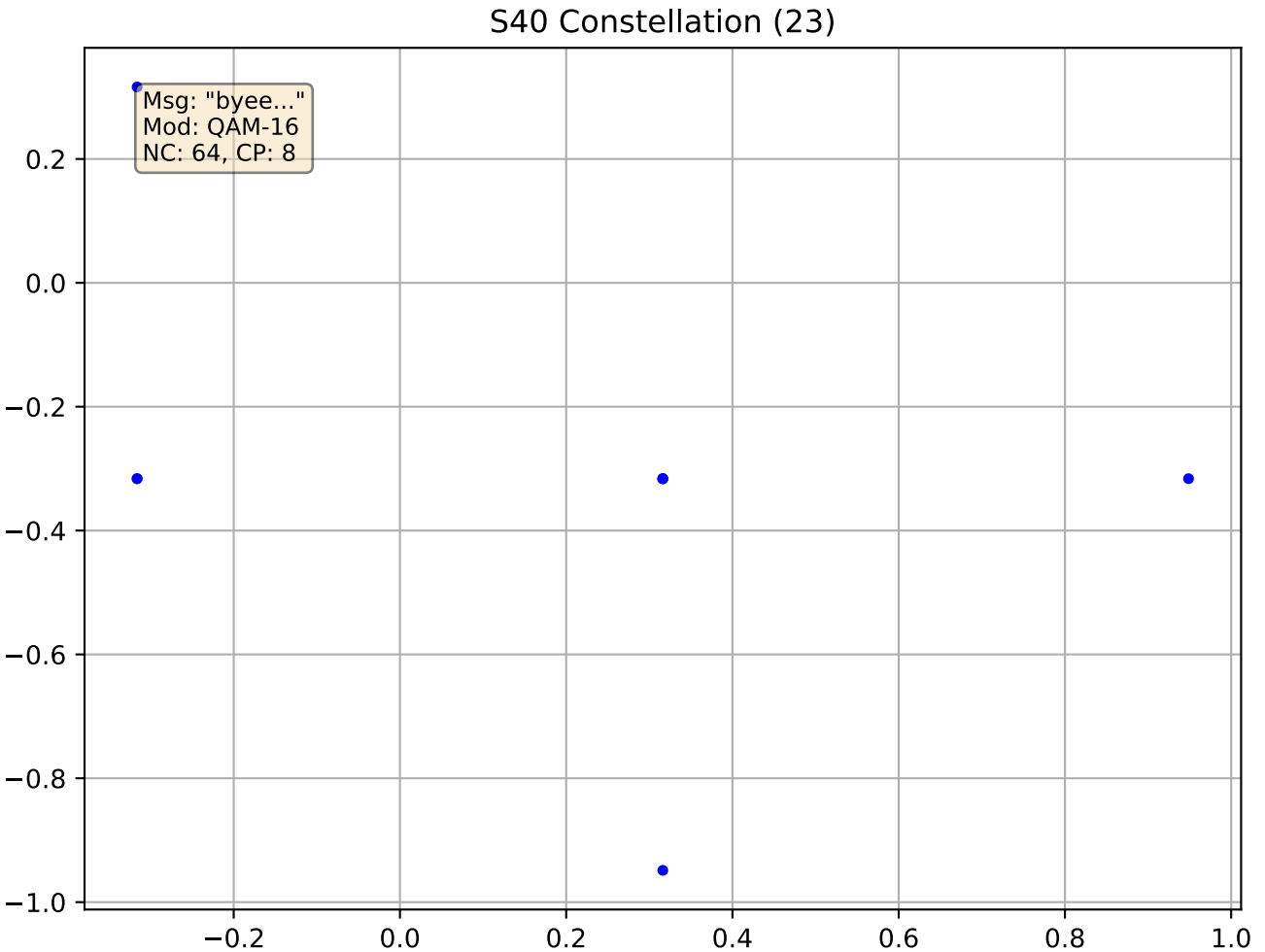

S40 OFDM (21)
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
20
40
60
80
100
120
140
I
Q

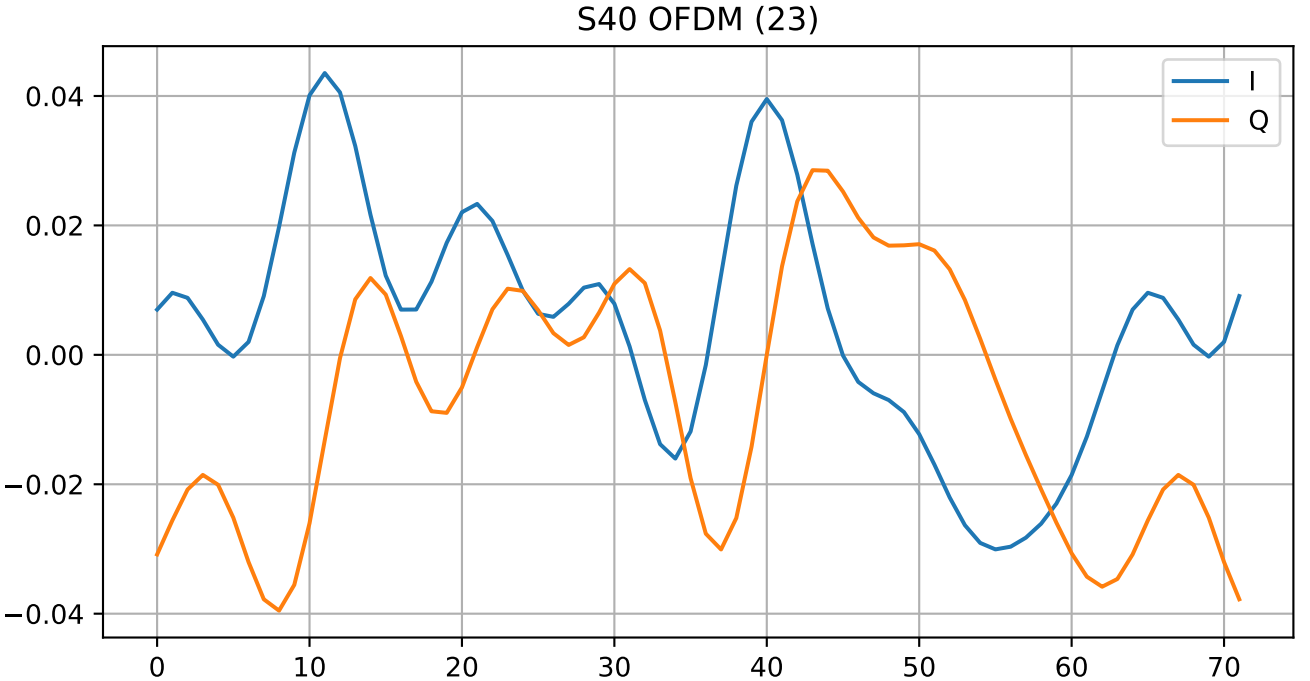

S40 Sensing (21)
Power
1.5
1.0
0.5
0.0
0
20
40
60
80
100
120
140

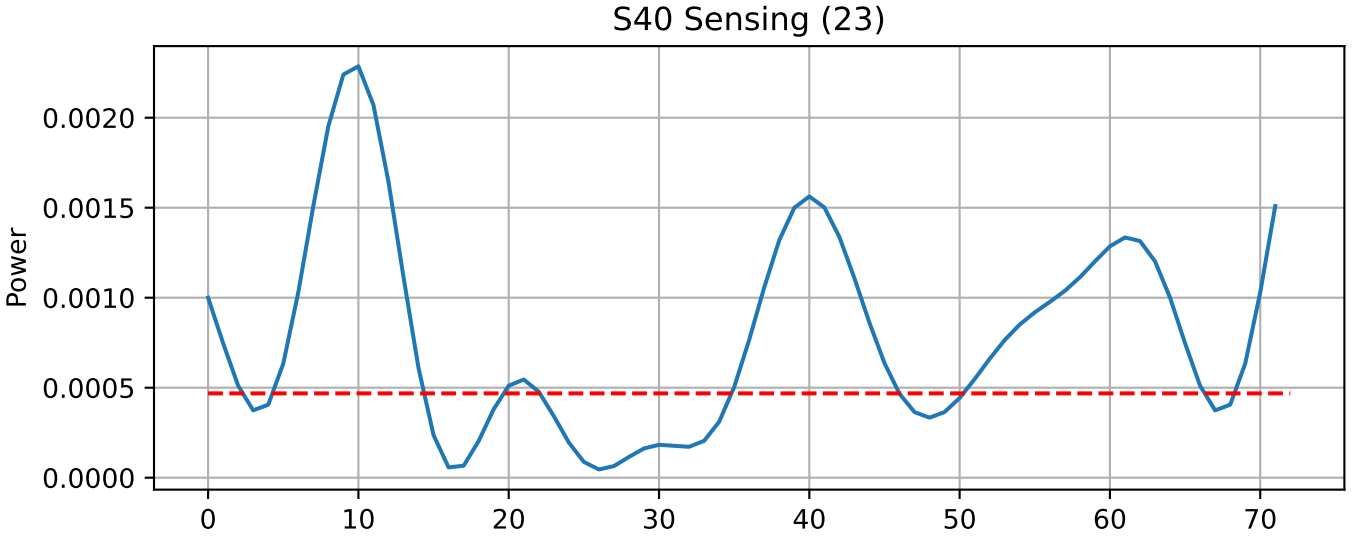

S40 Constellation (22)
Msg: "iam going..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

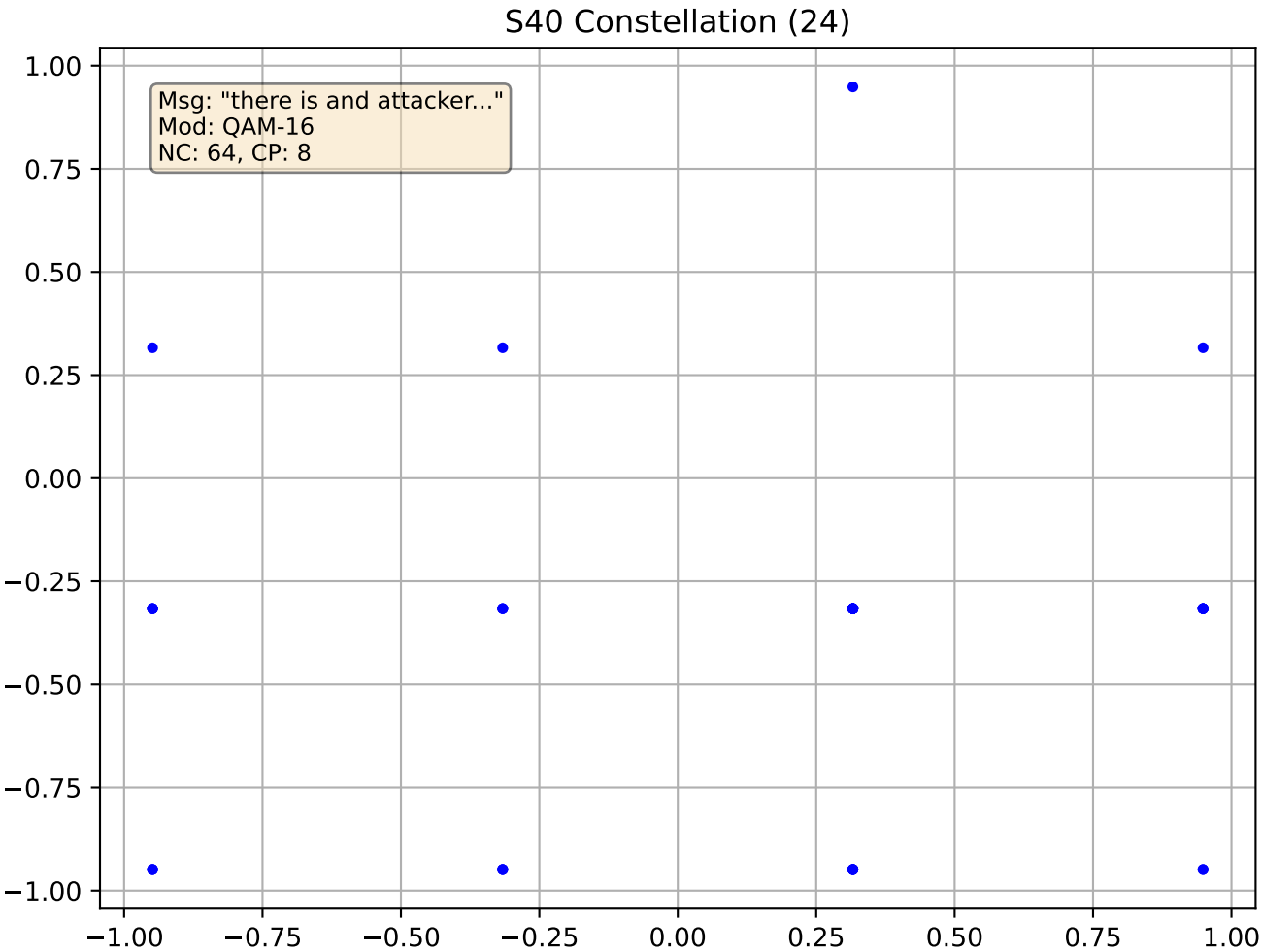

S40 OFDM (22)
I
Q
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
0
10
20
30
40
50
60
70

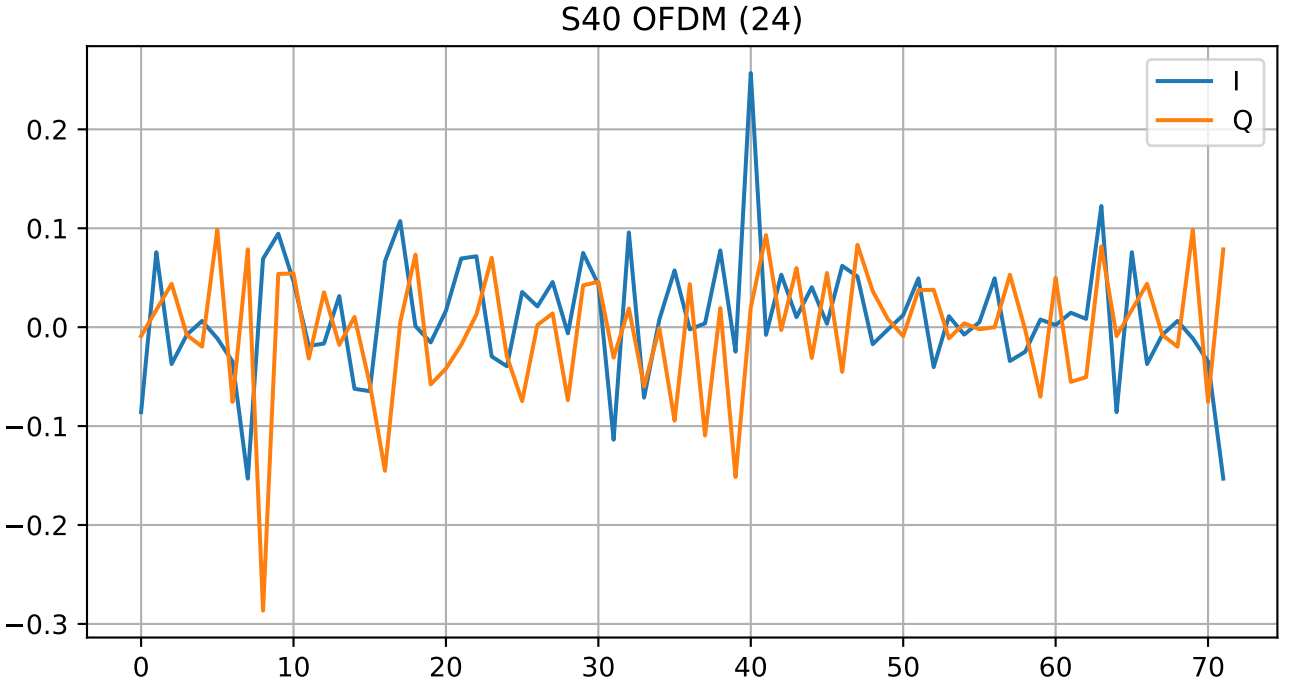

S40 Sensing (22)
Power
0.000
0.002
0.004
0.006
0.008
0
10
20
30
40
50
60
70

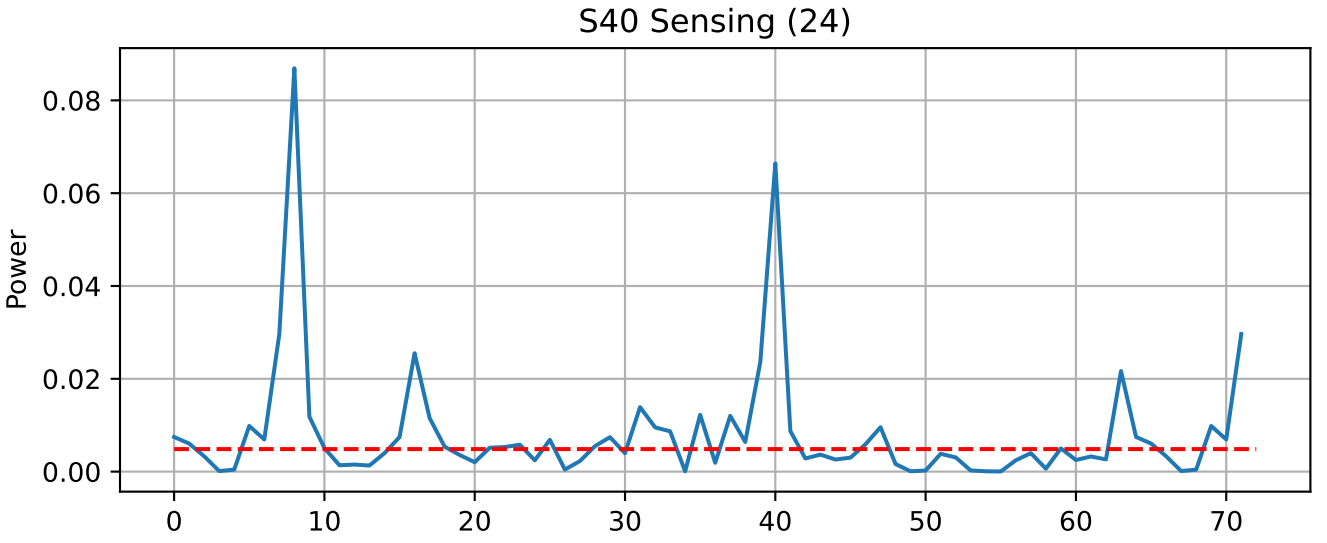

S40 Constellation (23)
Msg: "byee..."
Mod: QAM-16
NC: 64, CP: 8
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0

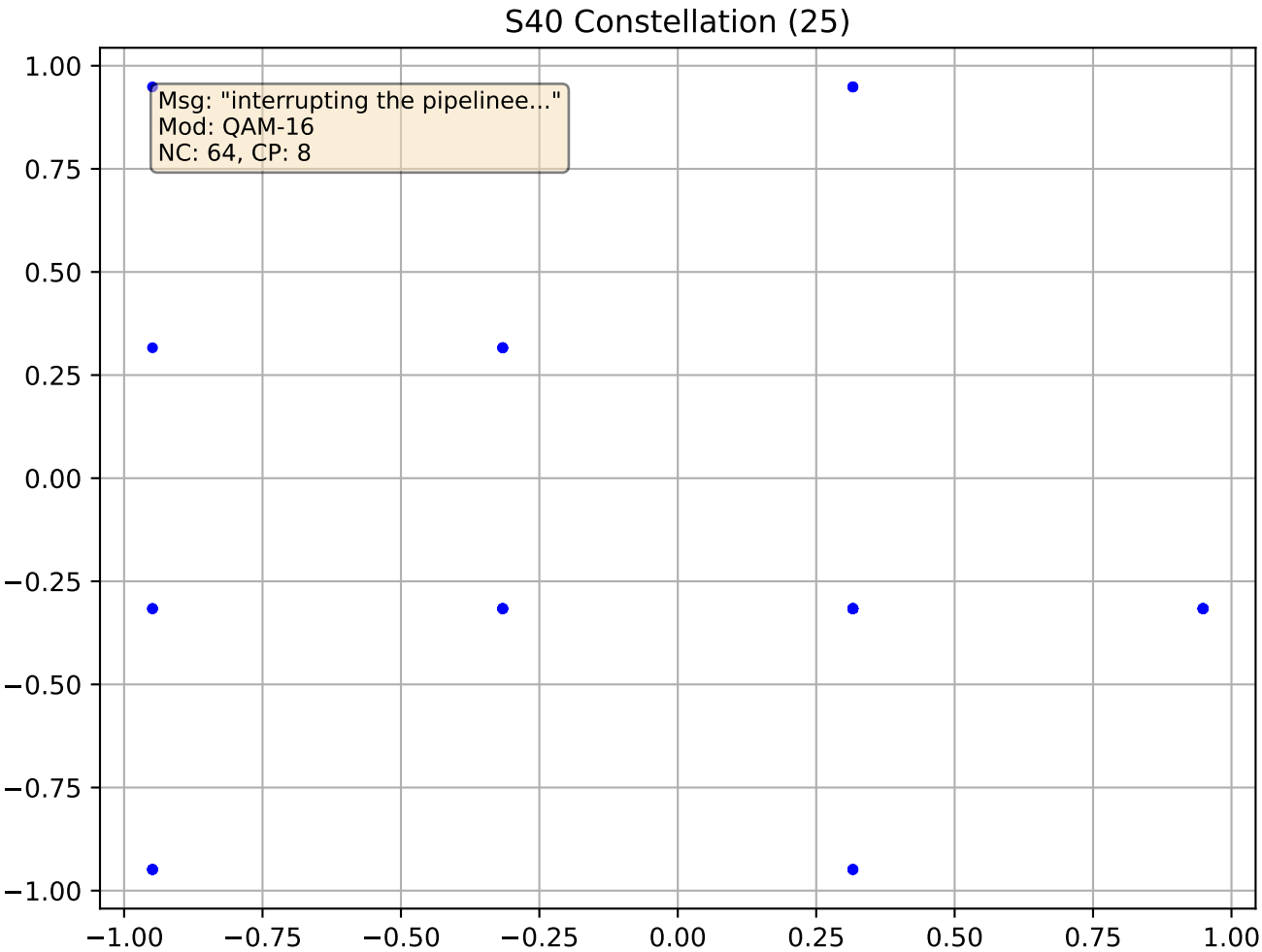

S40 OFDM (23)
I
Q
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
50
60
70

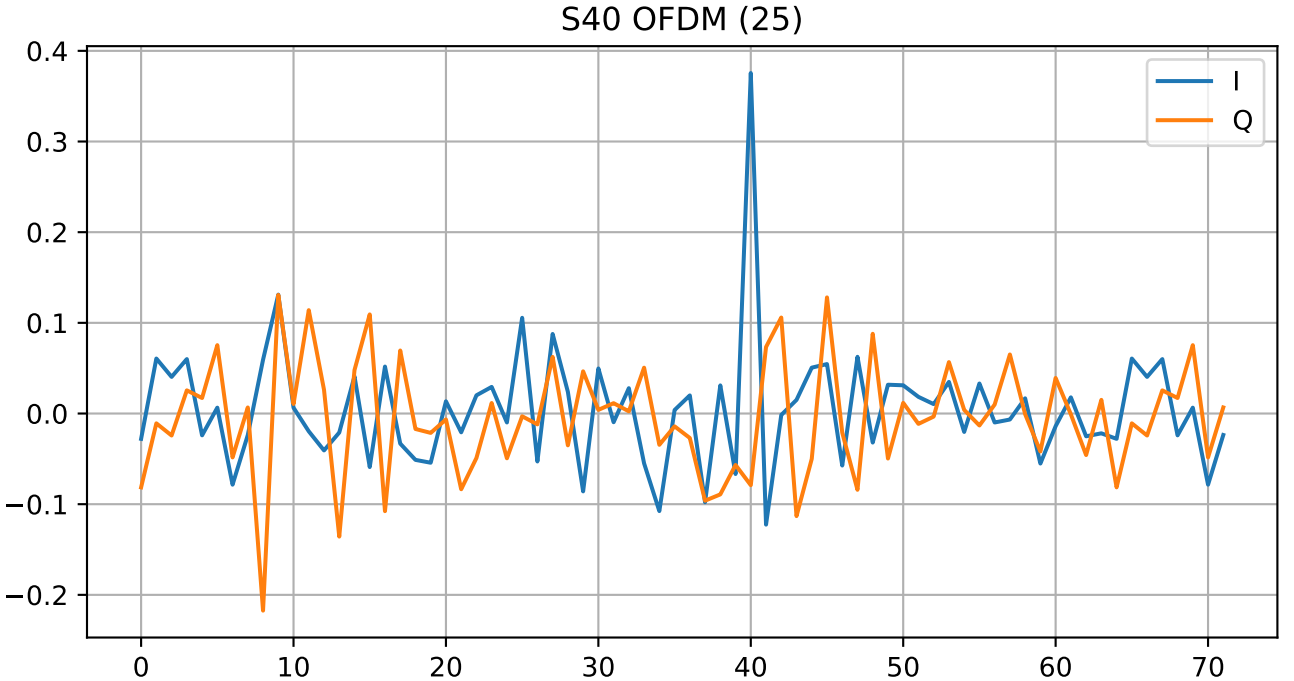

S40 Sensing (23)
Power
0.0020
0.0015
0.0010
0.0005
0.0000
0
10
20
30
40
50
60
70

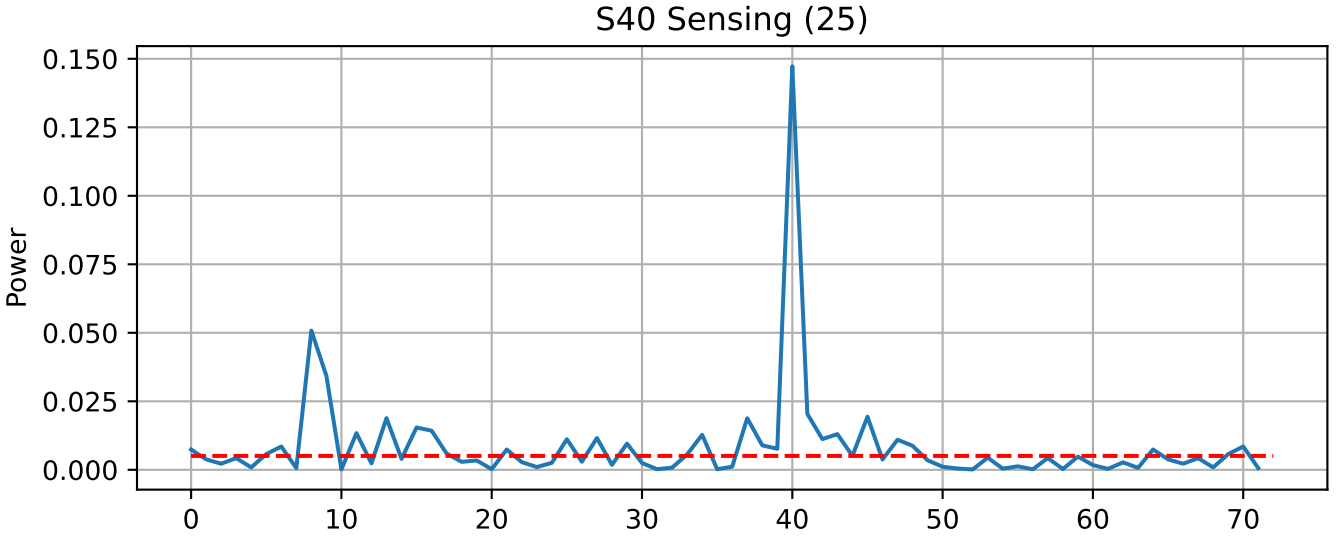

S40 Constellation (24)
Msg: "there is and attacker..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

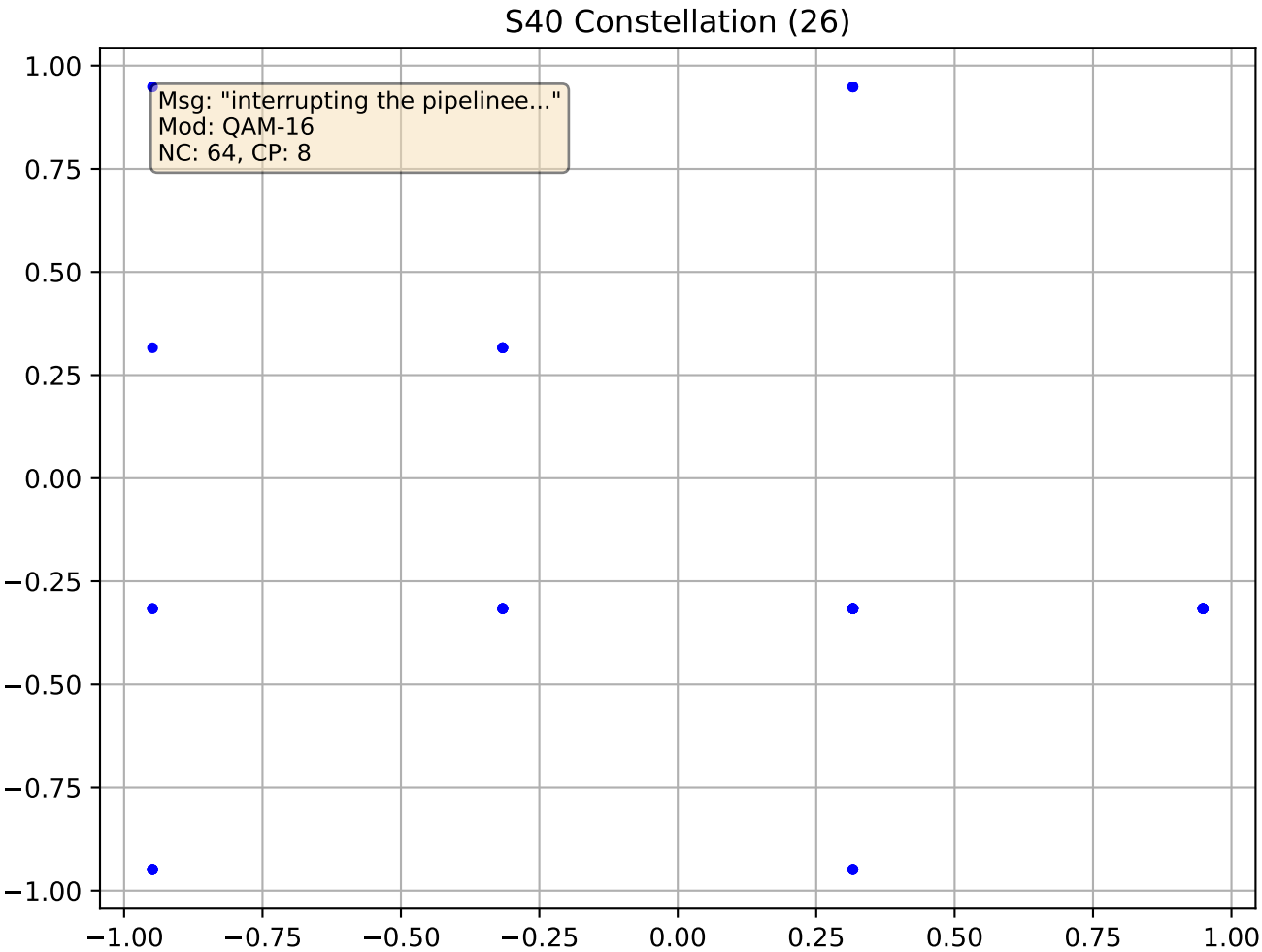

S40 OFDM (24)
I
Q
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
10
20
30
40
50
60
70

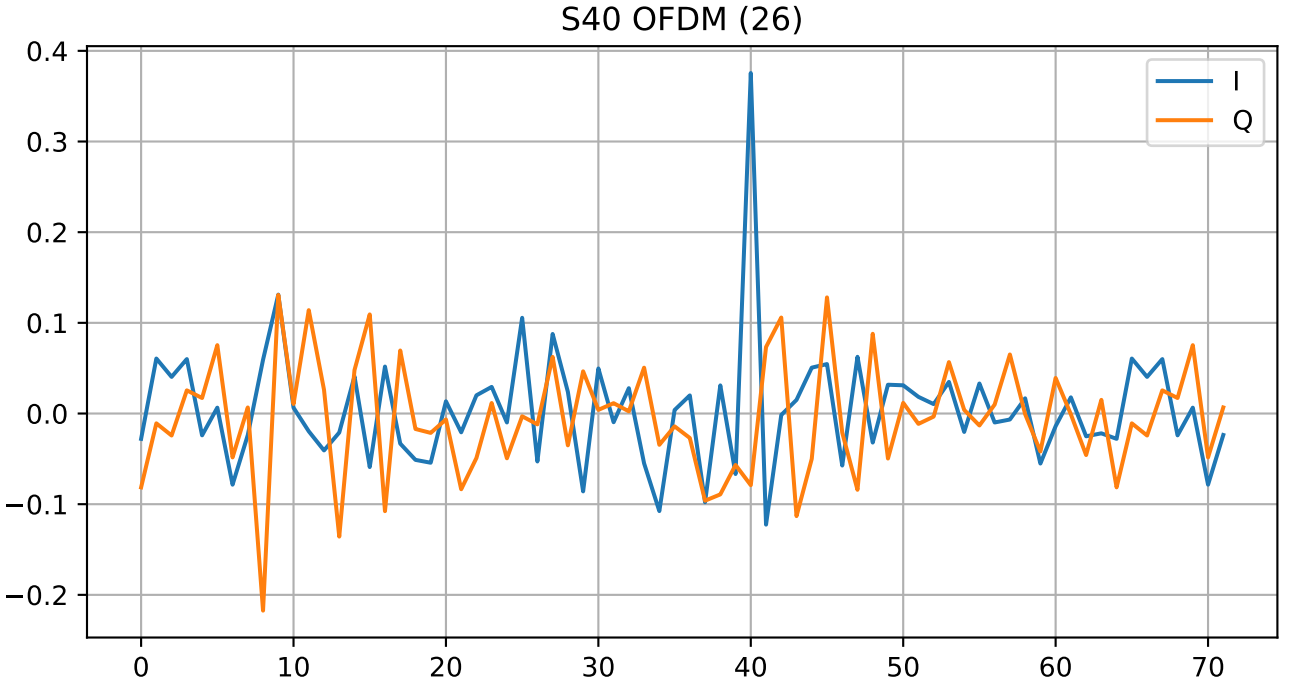

S40 Sensing (24)
Power
0.08
0.06
0.04
0.02
0.00
0
10
20
30
40
50
60
70

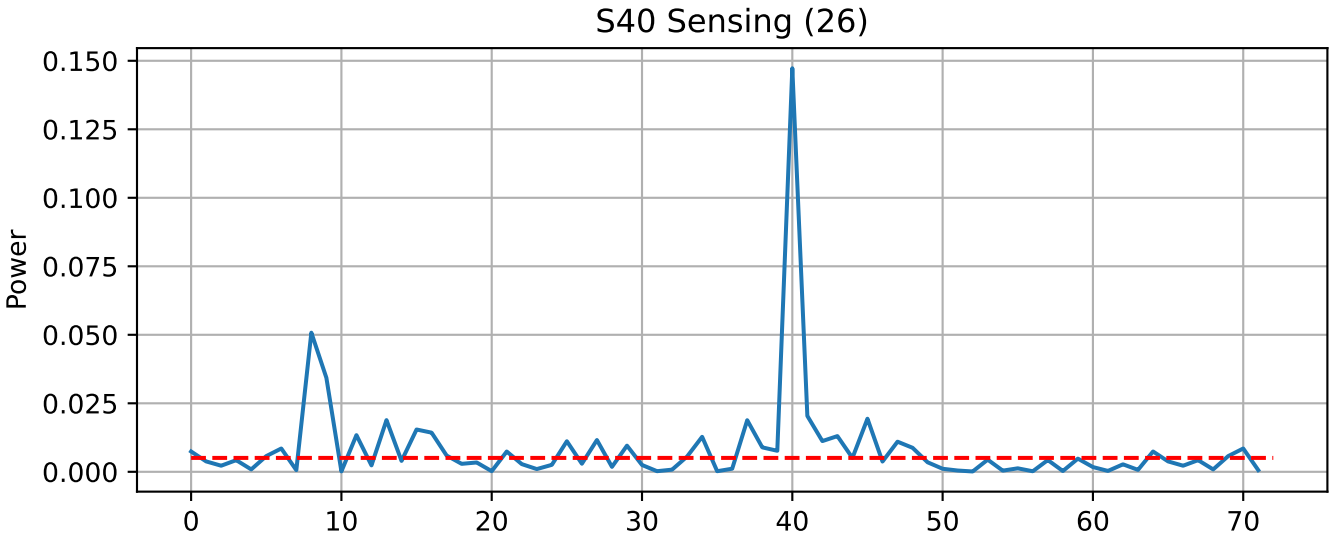

S40 Constellation (25)
Msg: "interrupting the pipelinee..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

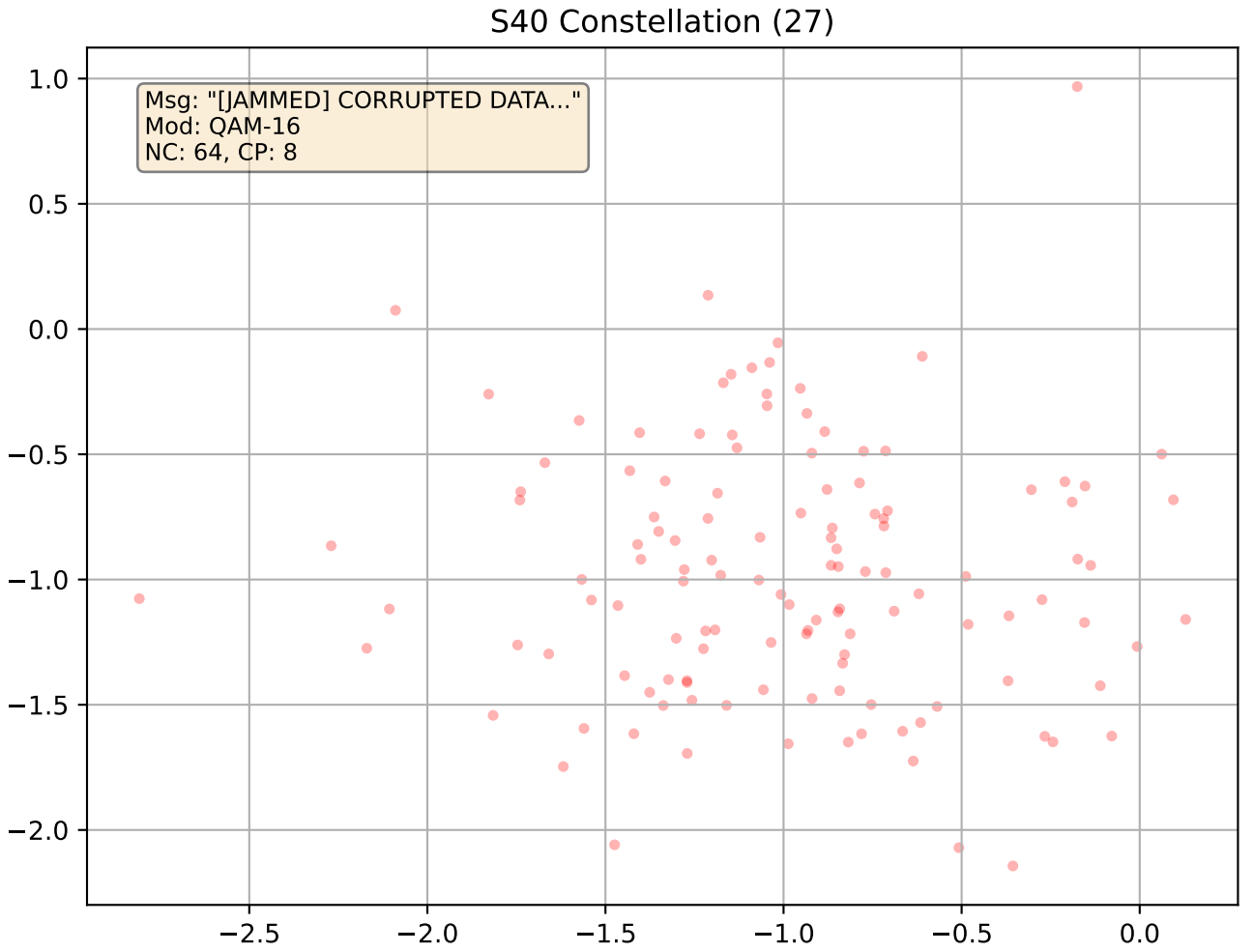

S40 OFDM (25)
I
Q
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
0
10
20
30
40
50
60
70

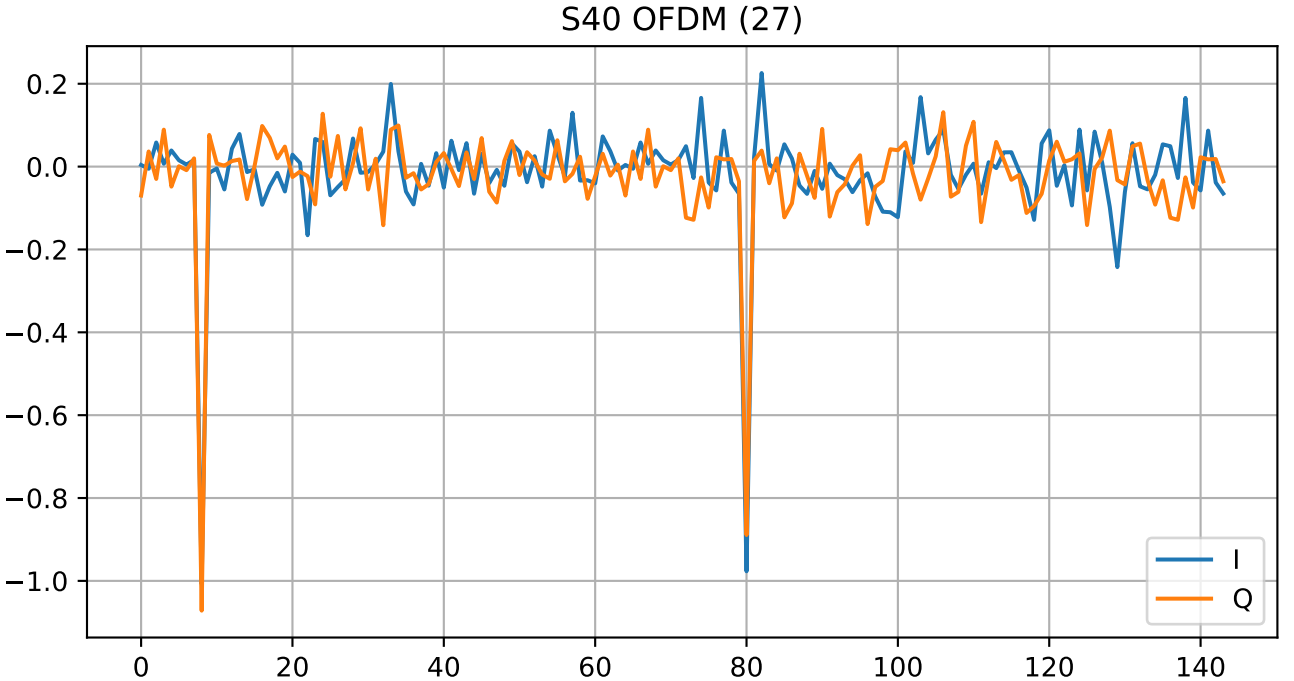

S40 Sensing (25)
Power
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
10
20
30
40
50
60
70

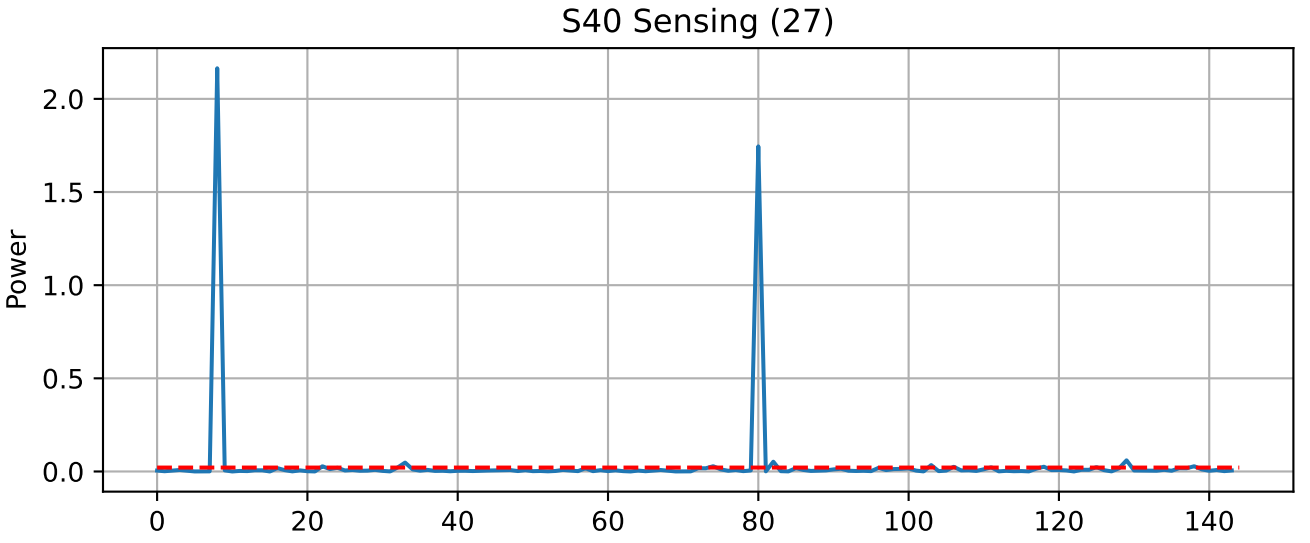

S40 Constellation (26)
Msg: "interrupting the pipelinee..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

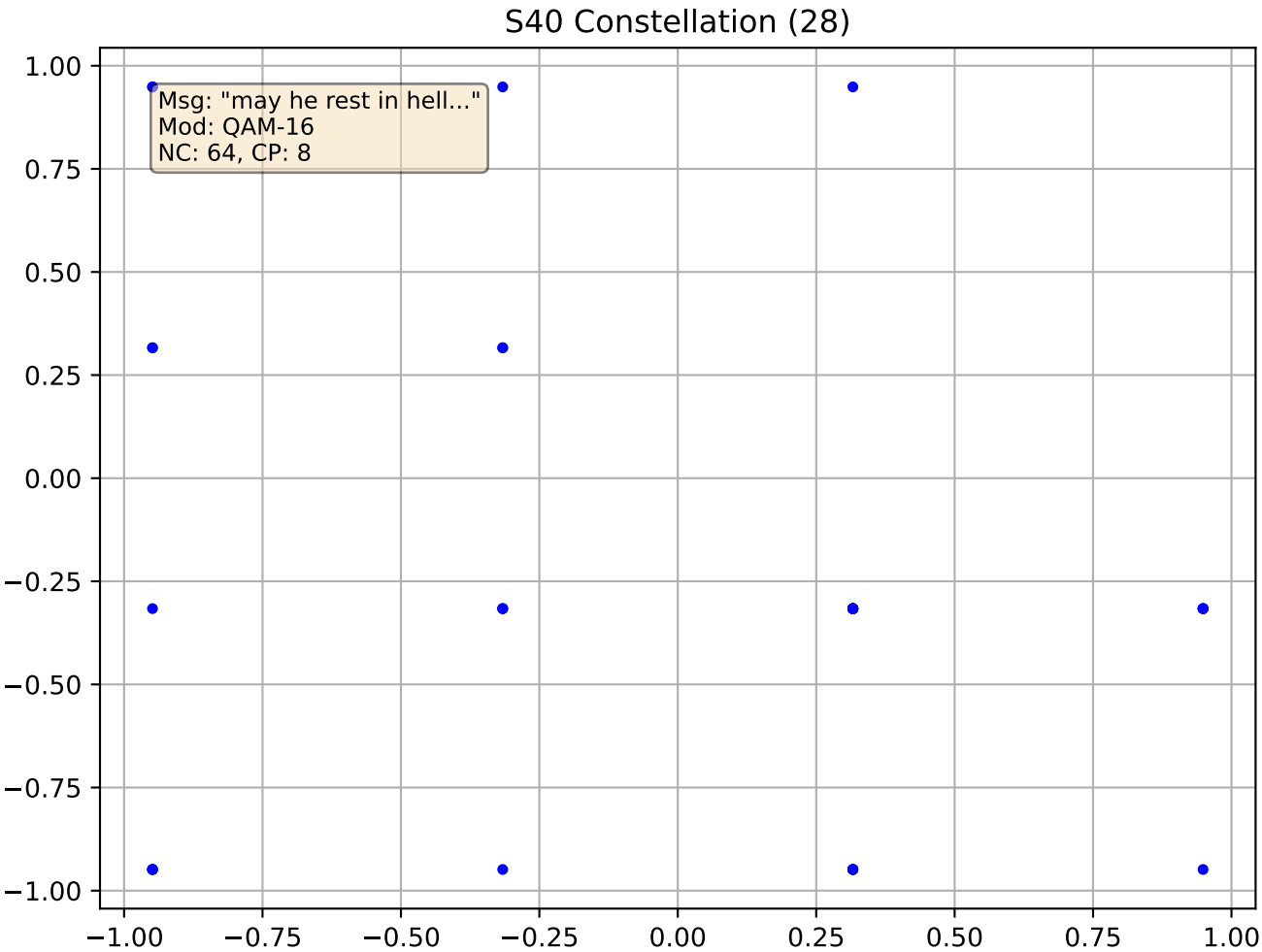

S40 OFDM (26)
I
Q
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
0
10
20
30
40
50
60
70

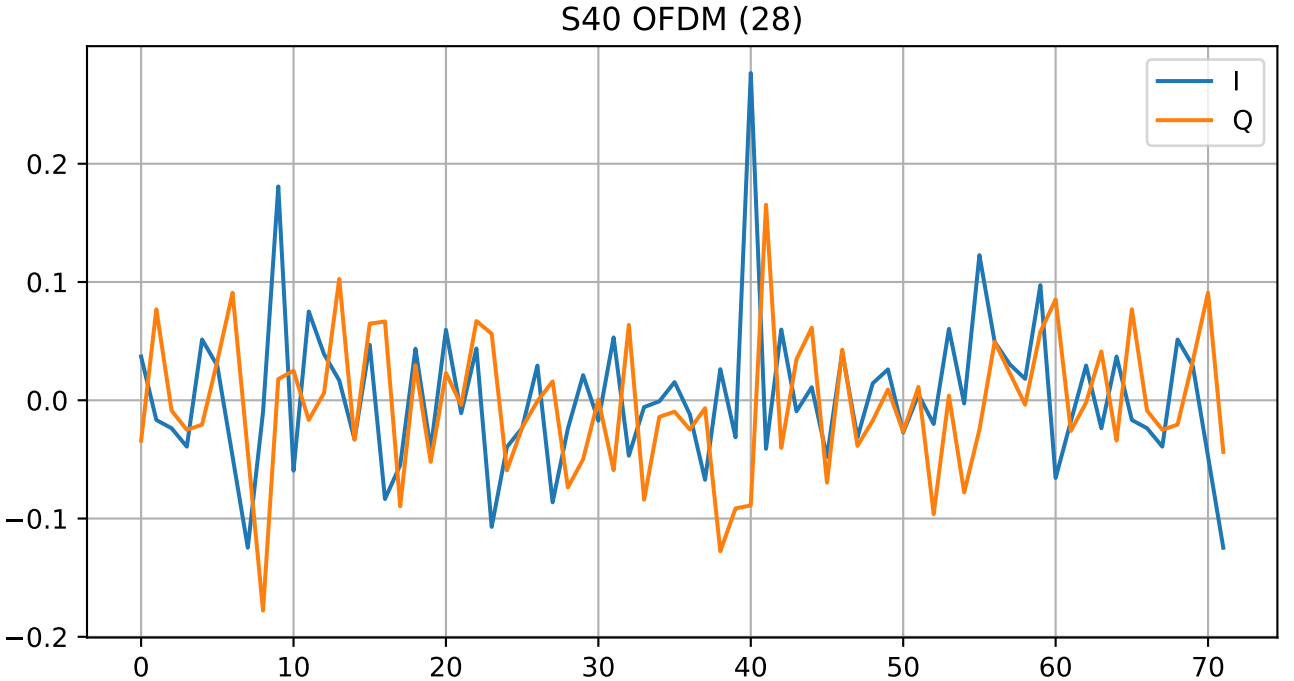

S40 Sensing (26)
Power
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
10
20
30
40
50
60
70

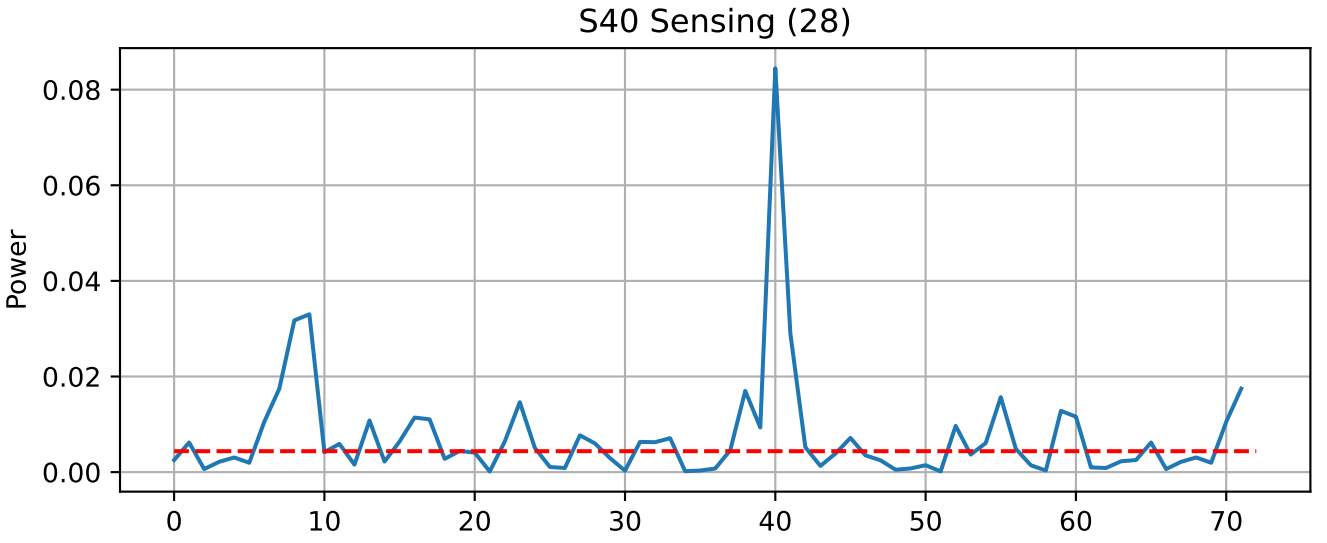

S40 Constellation (27)
Msg: "[JAMMED] CORRUPTED DATA..."
Mod: QAM-16
NC: 64, CP: 8
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0


S40 OFDM (27)
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
20
40
60
80
100
120
140
I
Q


S40 Sensing (27)
Power
2.0
1.5
1.0
0.5
0.0
0
20
40
60
80
100
120
140


S40 Constellation (28)
Msg: "may he rest in hell..."
Mod: QAM-16
NC: 64, CP: 8
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00


S40 OFDM (28)
I
Q
0.2
0.1
0.0
−0.1
−0.2
0
10
20
30
40
50
60
70


S40 Sensing (28)
Power
0.08
0.06
0.04
0.02
0.00
0
10
20
30
40
50
60
70